Design of Operational amplifiers

1

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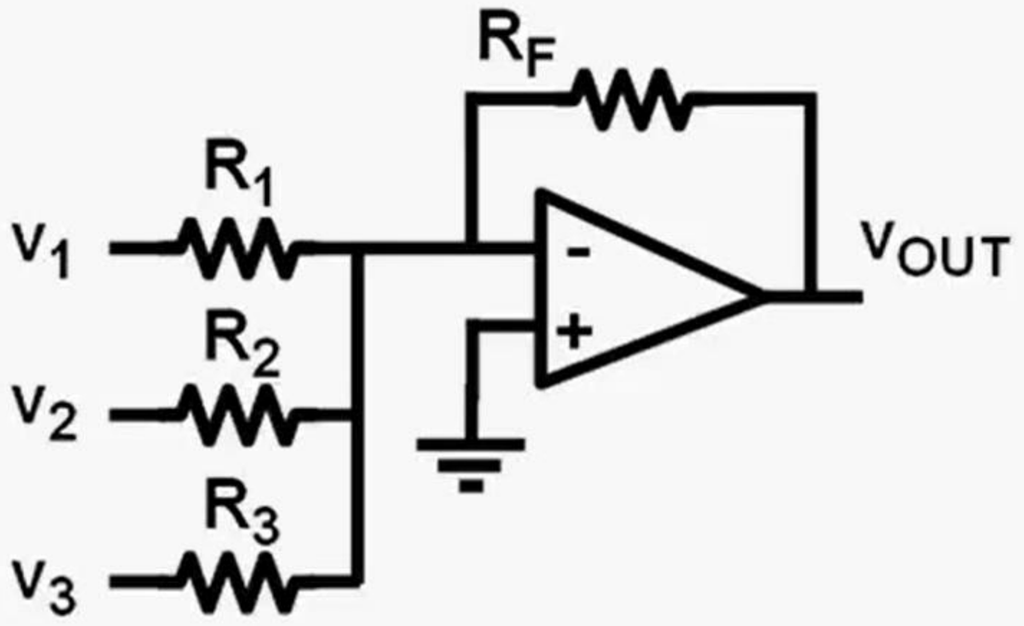


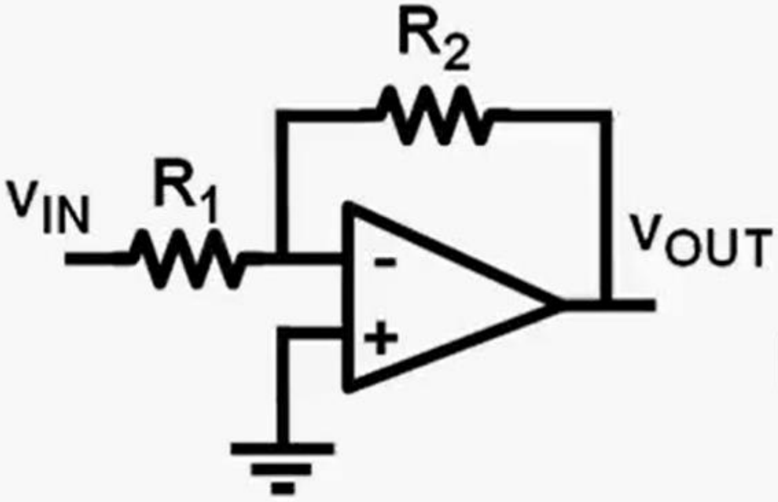
$$-\frac{V_{OUT}}{R_F} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$

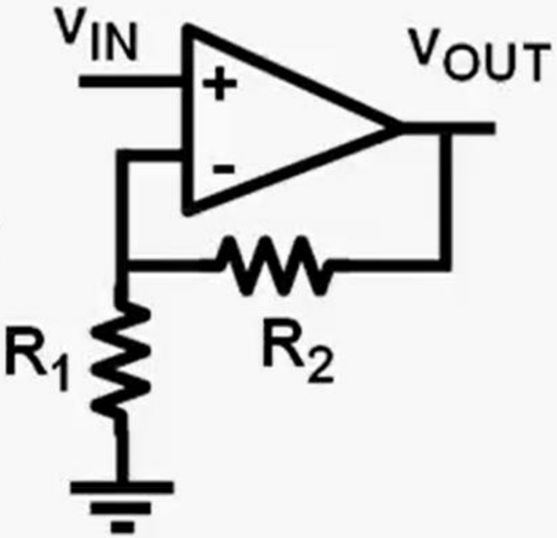
Requires High gain
High speed
Low power

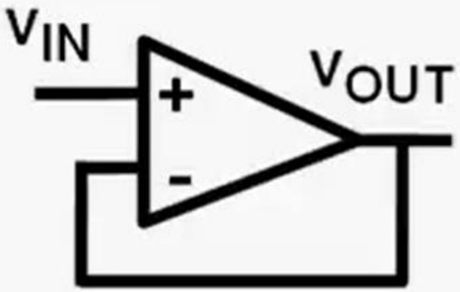
Opamp specs :

Voltage Gain	= High
Input current	= 0
Differential input voltage	= 0
BW	= High
GBW	= Very High

Basics of opamp







$$A_V = -\frac{R_2}{R_1}$$

$$R_{IN} = R_1$$

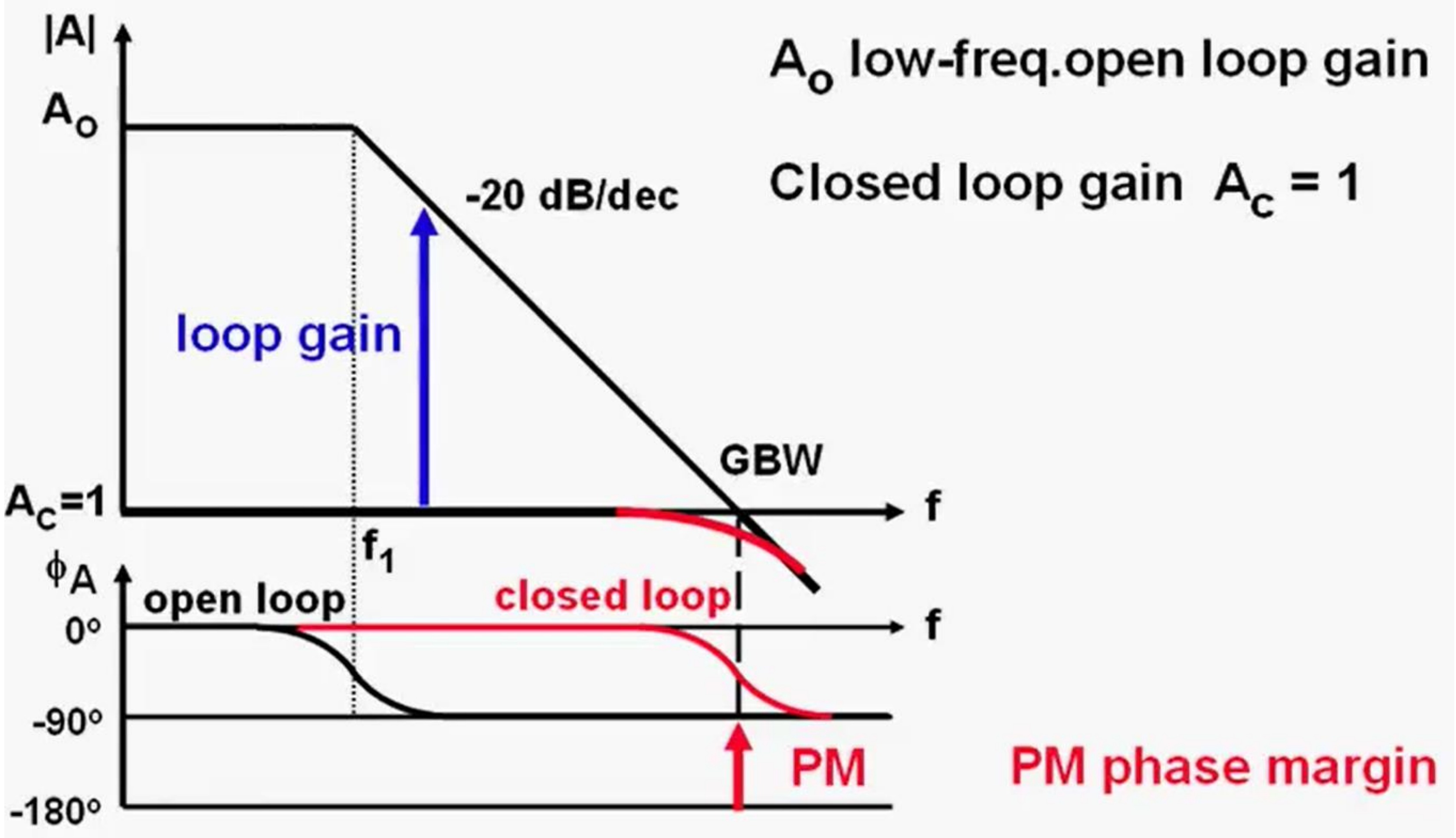
$$A_V = 1 + \frac{R_2}{R_1}$$

$$R_{IN} = \infty$$

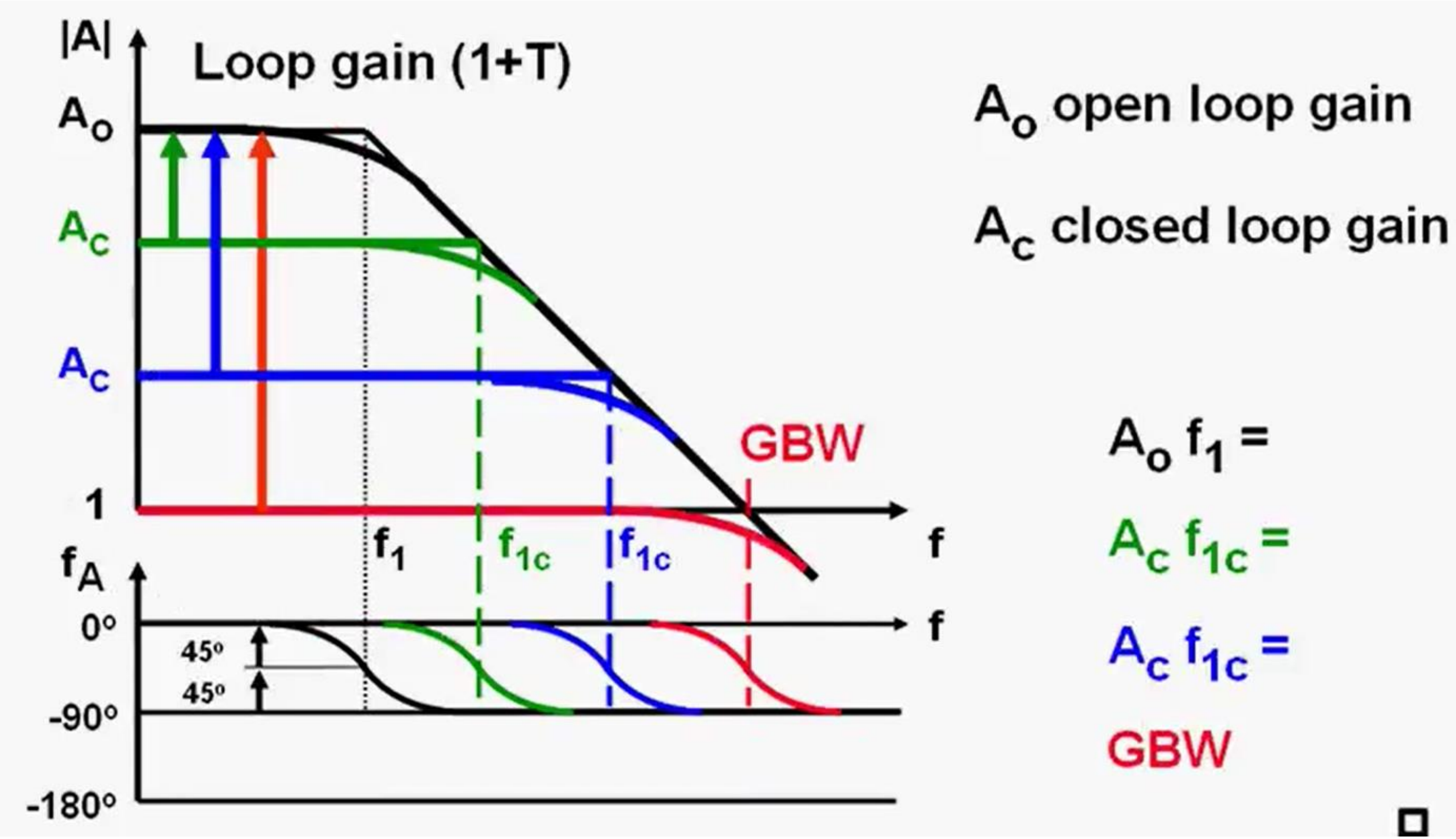
$$A_V = 1$$

$$R_{IN} = \infty$$

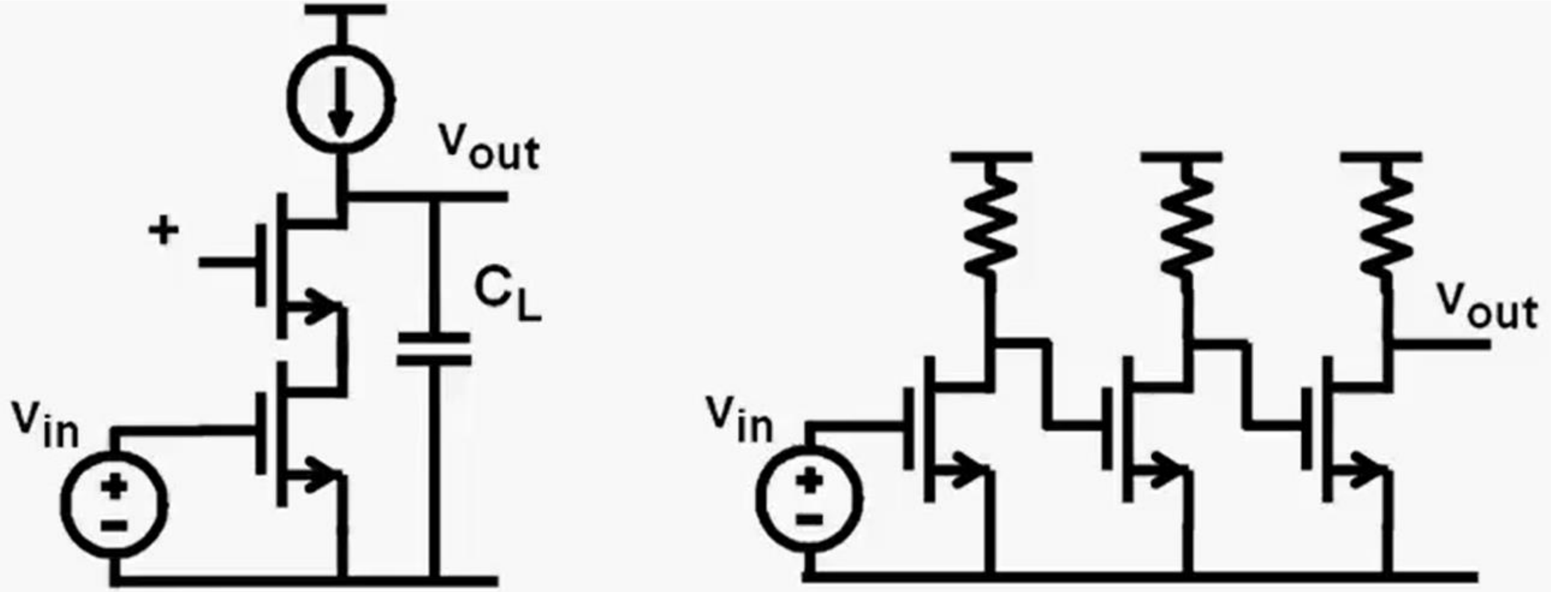
Basics of opamp



Basics of opamp



Basics of opamp

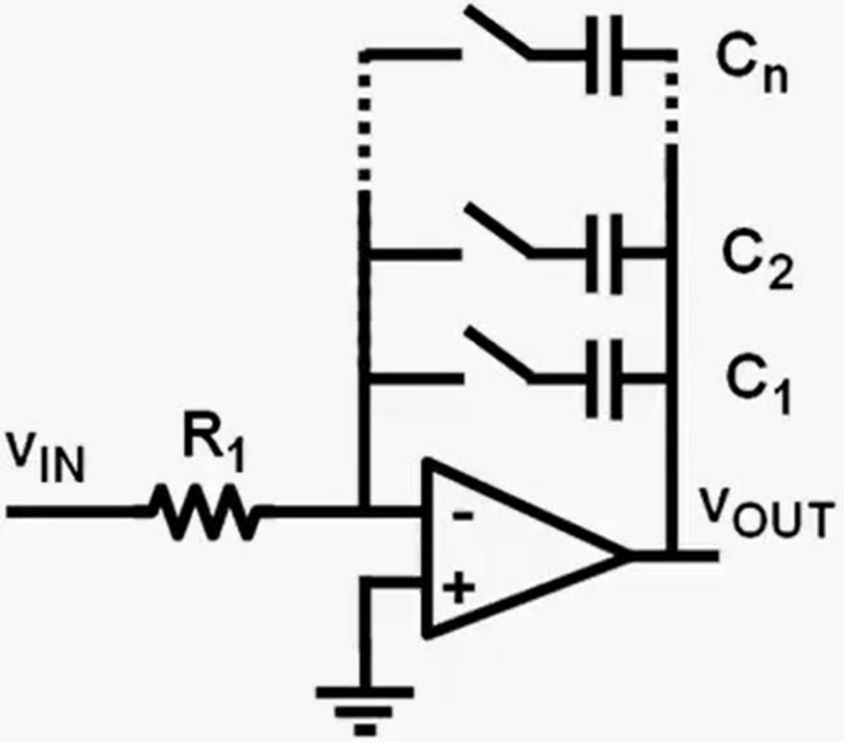


Operational amplifier :
Single-pole amplifier
High impedance = high gain
Exchange Gain-Bandwidth
Stable for all gain values

Wideband amplifier :
Multiple-pole amplifier
Low impedances at nodes
Wide Bandwidth
Stable for one gain only

Basics of opamp

Opamps and passive components (R, C)



Advantages :

S/N up to 100 dB

THD very low < - 90 dB

Disadvantages :

Opamps :

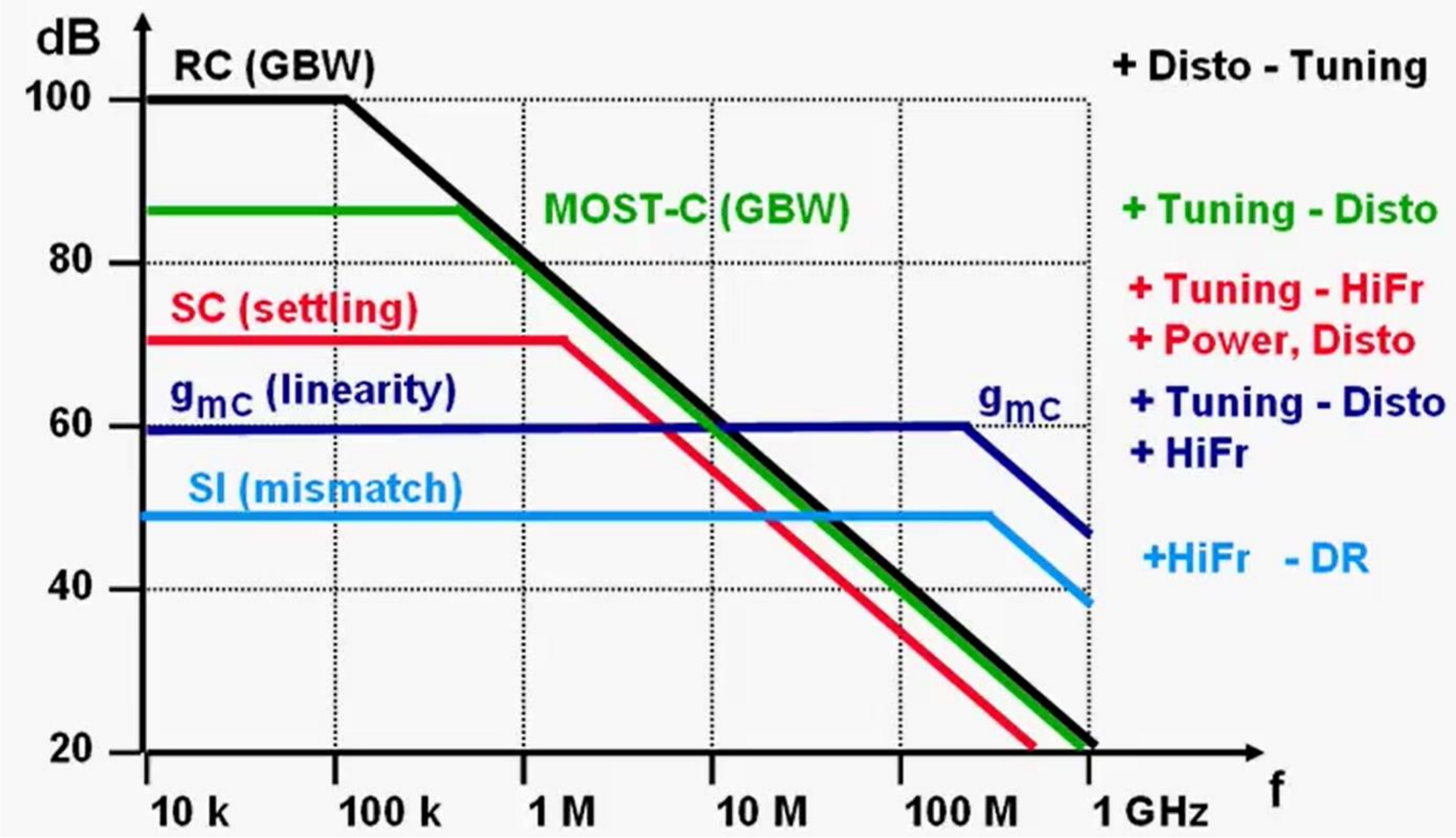
only for low frequencies

Errors on R, C ≈ 20 %

>> tune C's

Ref. Hollman, JSSC July 01, pp.1148-1153

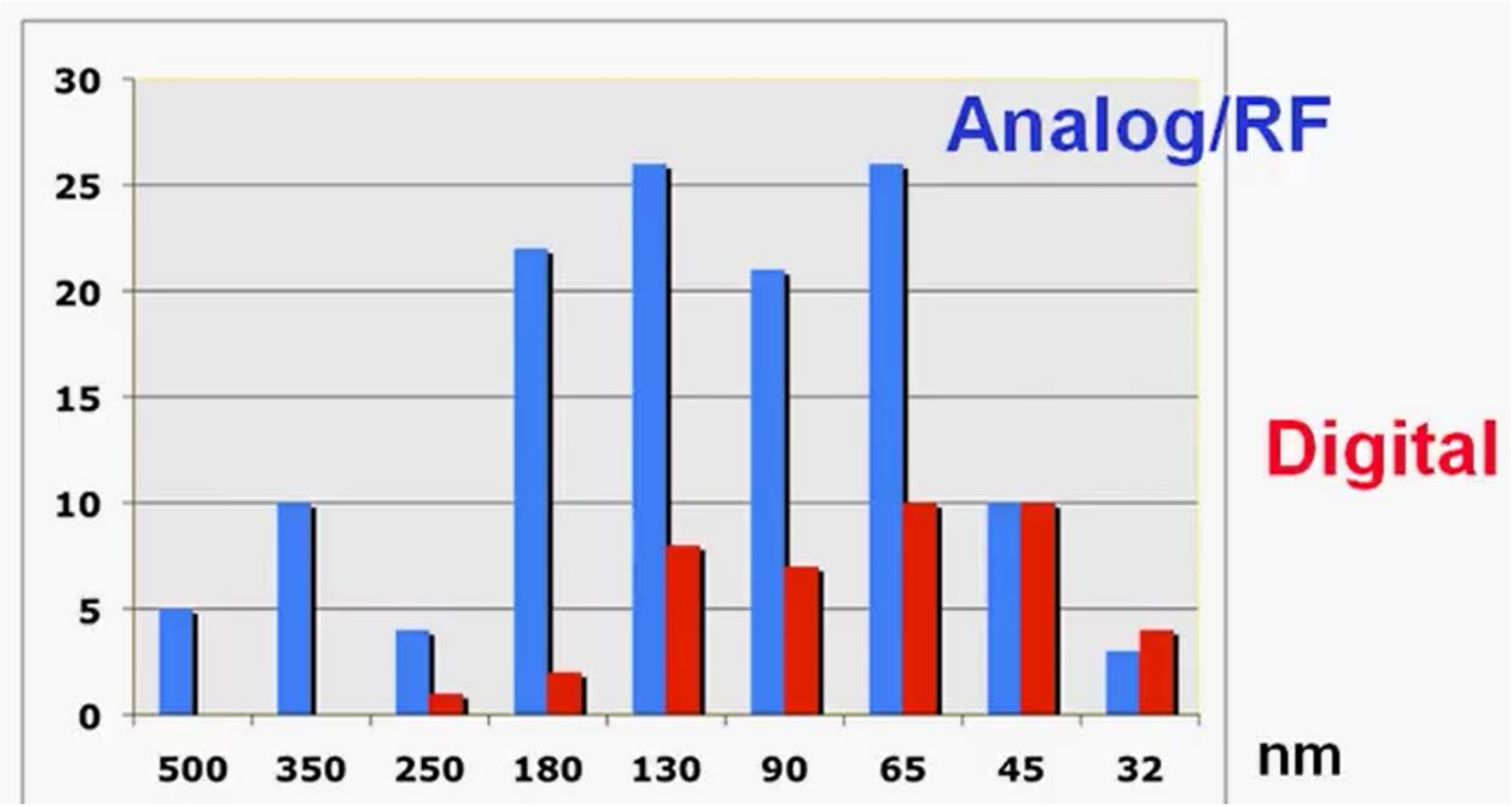
Basics of opamp



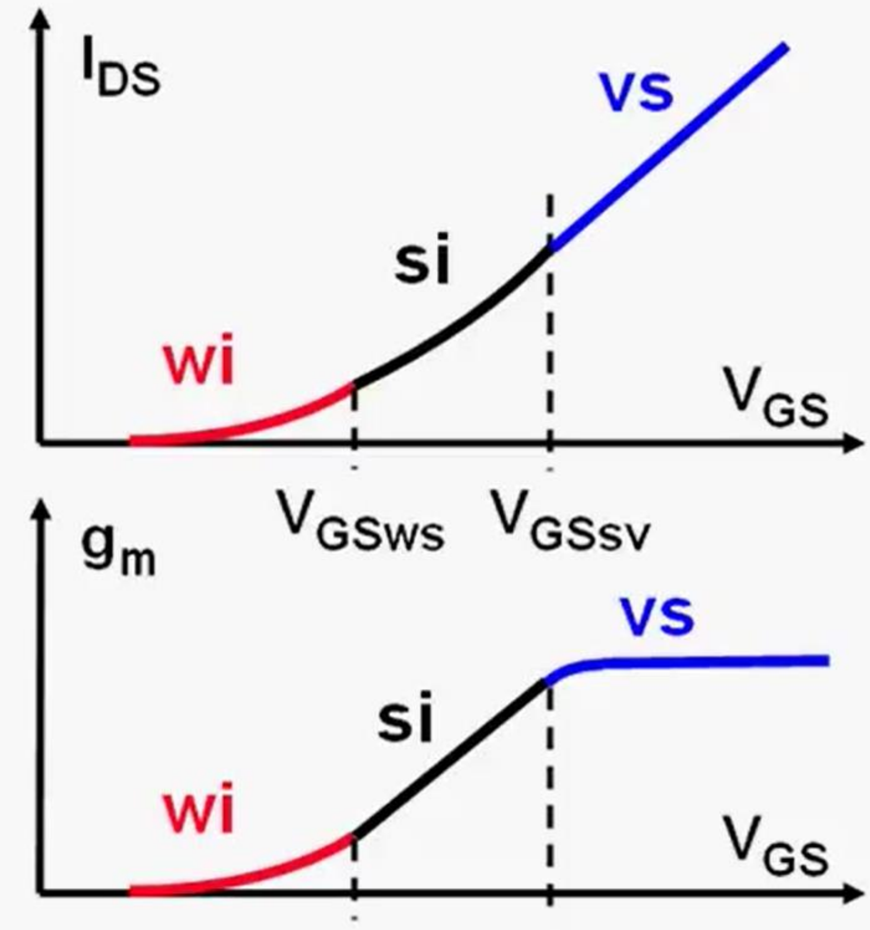
CMOS tecnologies Vs opamp

Nanometer CMOS technologies are leading to smaller channel lengths along with supply voltage **below or equal to 1V** !!!!

CMOS tecnologies Vs opamp



CMOS tecnologies Vs opamp



$$I_{DS} = K'_n \frac{W}{L} (V_{GS}-V_T)^2$$

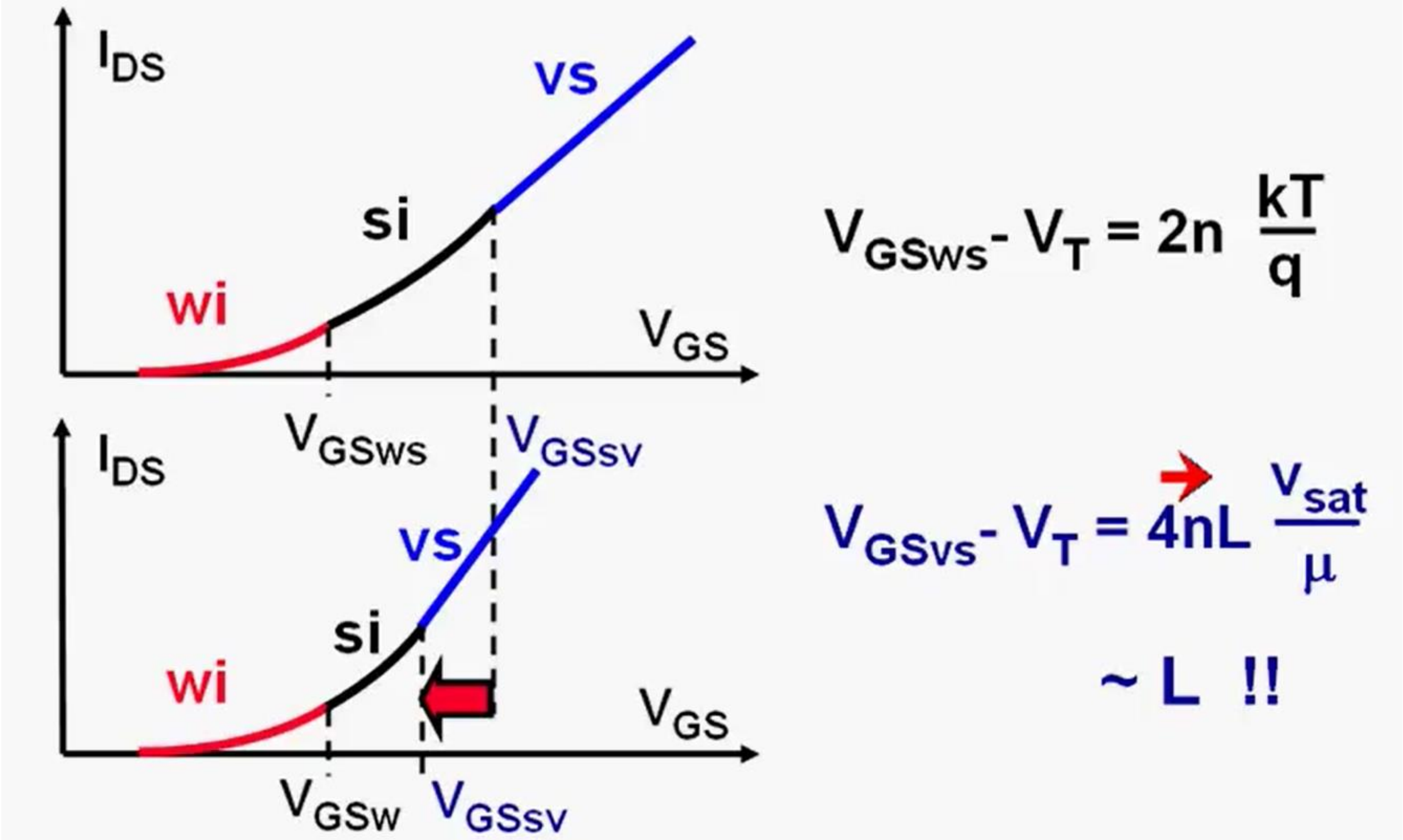
$$K' \approx \mu C_{ox}$$

$$I_{DS} = K''_n W (V_{GS}-V_T)$$

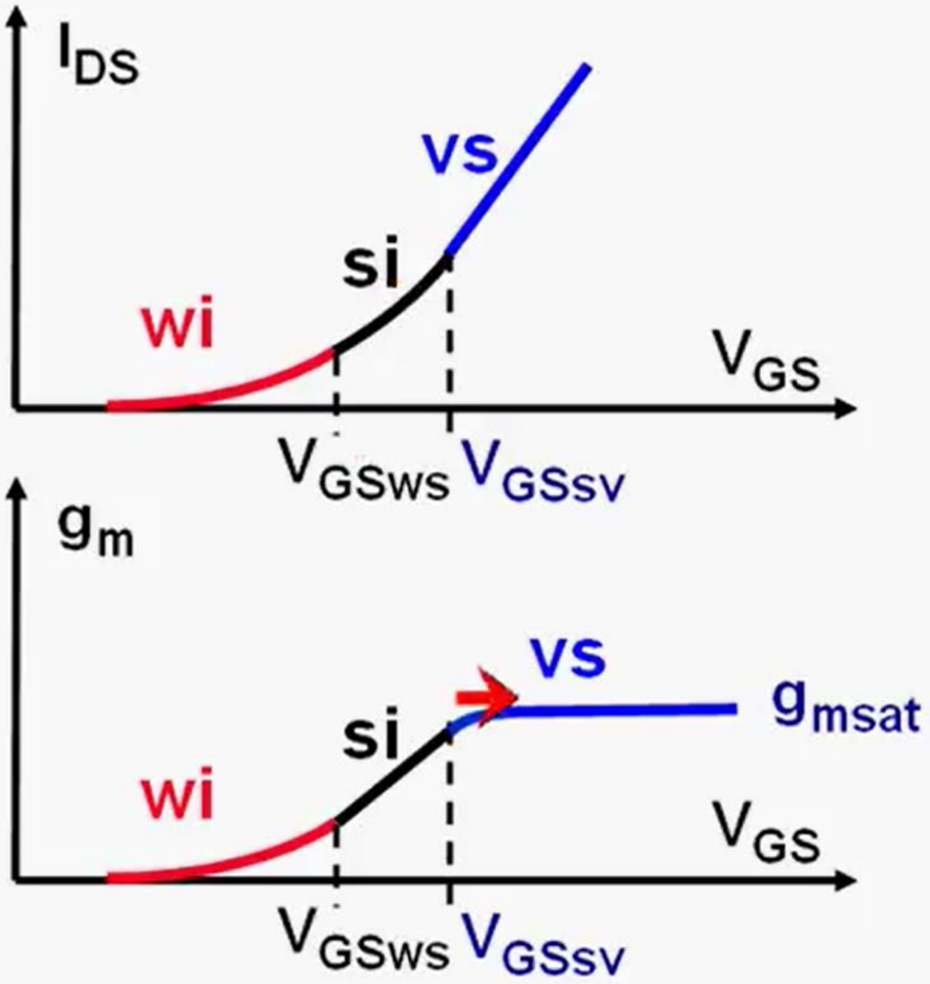
$$K''_n \approx v_{sat} C_{ox}$$

10^7 cm/s

CMOS tecnologies Vs opamp



CMOS tecnologies Vs opamp



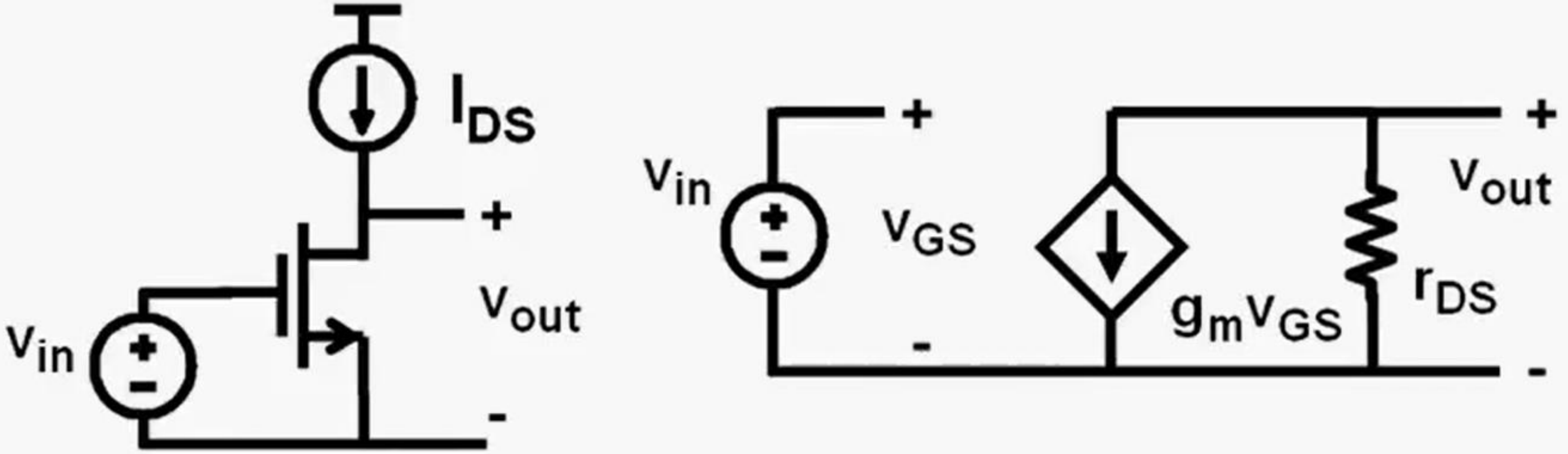
$g_{msat} = WC_{ox}v_{sat}$

$W \sim L$ & $C_{ox} \sim \frac{1}{L}$

$g_{msat} \approx$ constant

g_{msat} slowly increases with smaller L !!

CMOS tecnologies Vs opamp



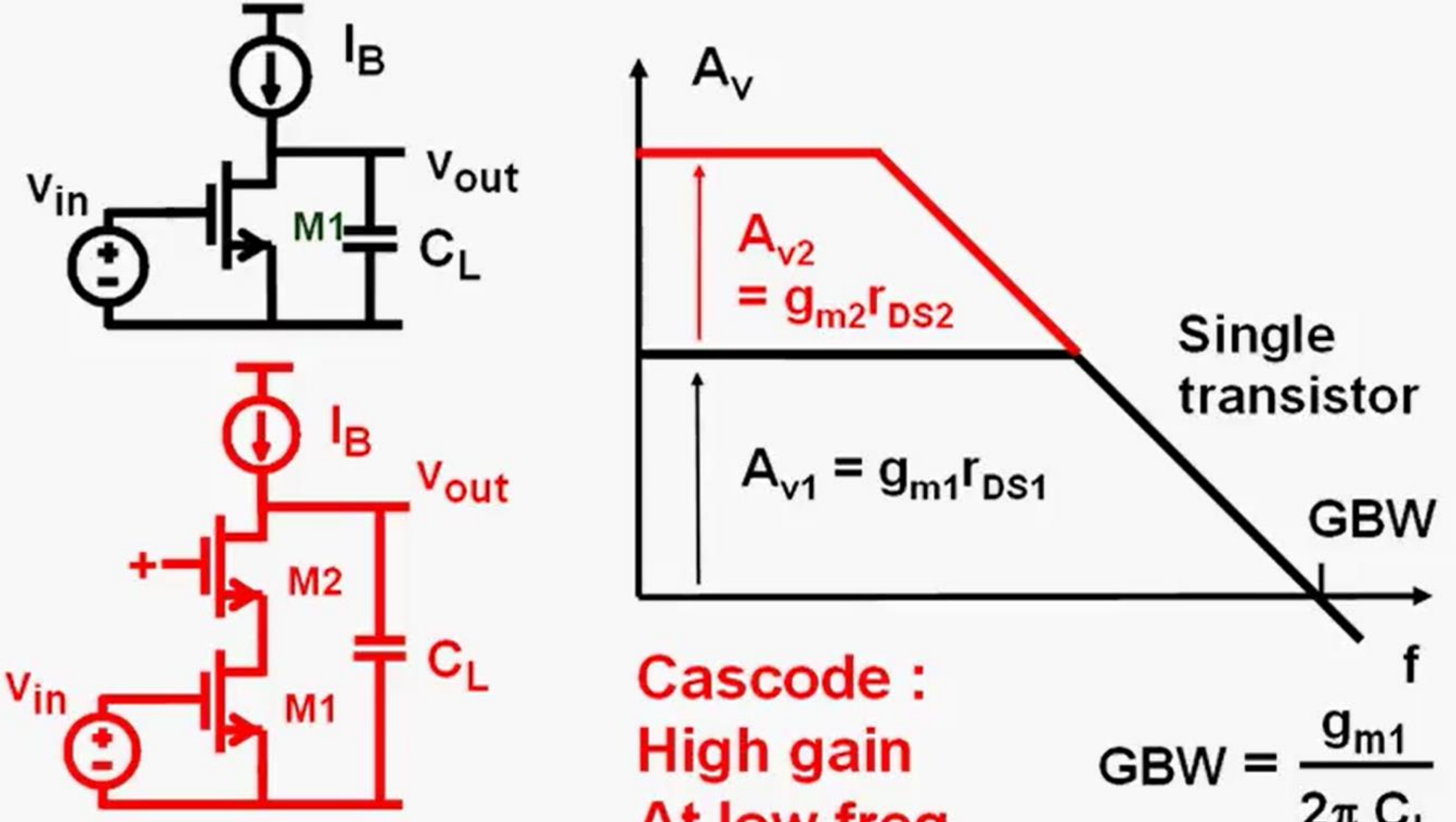
$$A_v = g_m r_{DS} = \frac{2\,I_{DS}}{V_{GS}-V_T}\,\frac{V_E L}{I_{DS}} = \frac{2\,V_E L}{V_{GS}-V_T}$$

For L = 65 nm ($V_E \approx 5$ V/µm & $V_{GS}-V_T \approx 0.2$ V) : $A_V \approx 3.2$!!!

As CMOS tecnology node reduces opamp gain reduces significantly

Sol : Gain Boosting !!!!

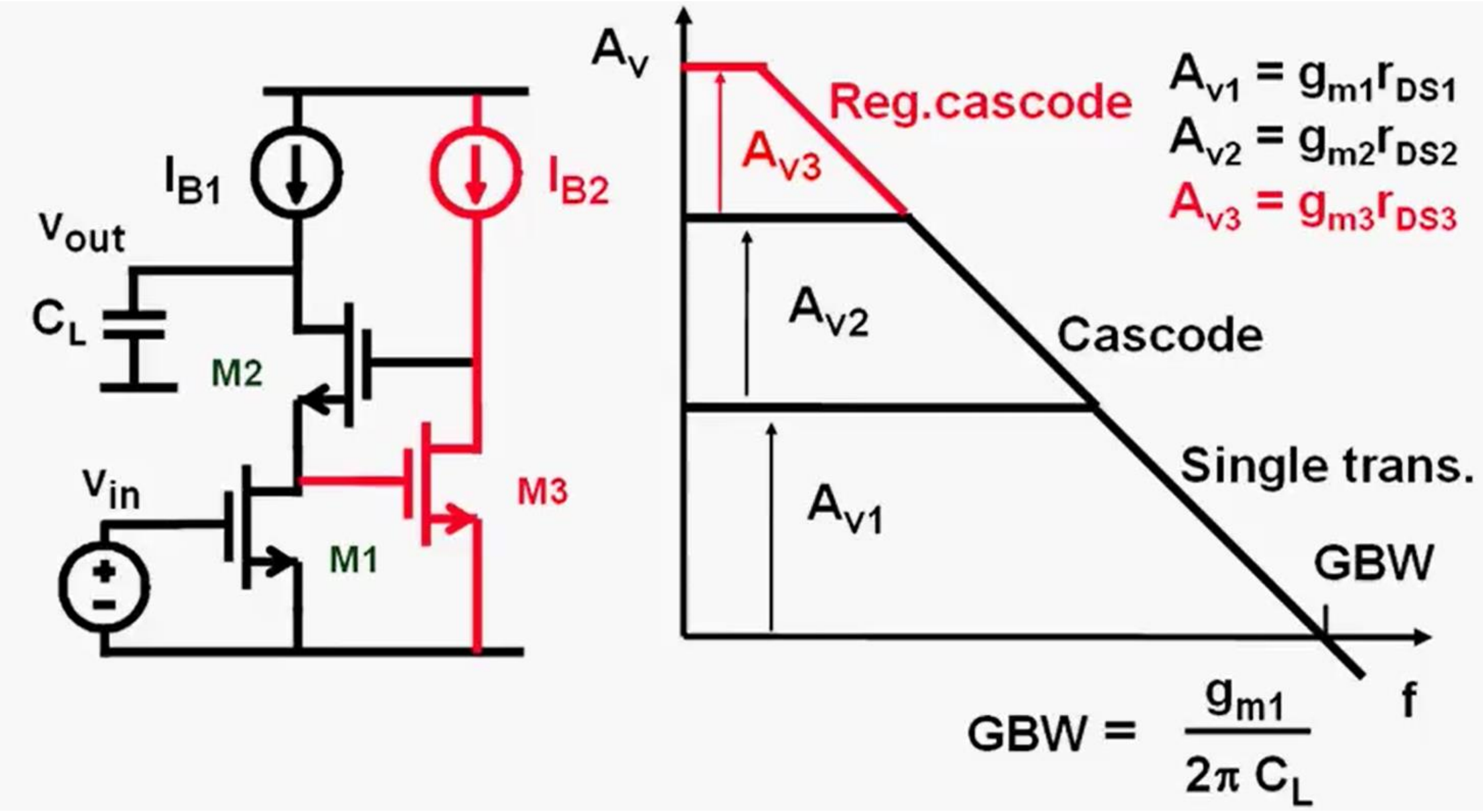
Idea 1



Cascode :
High gain
At low freq.

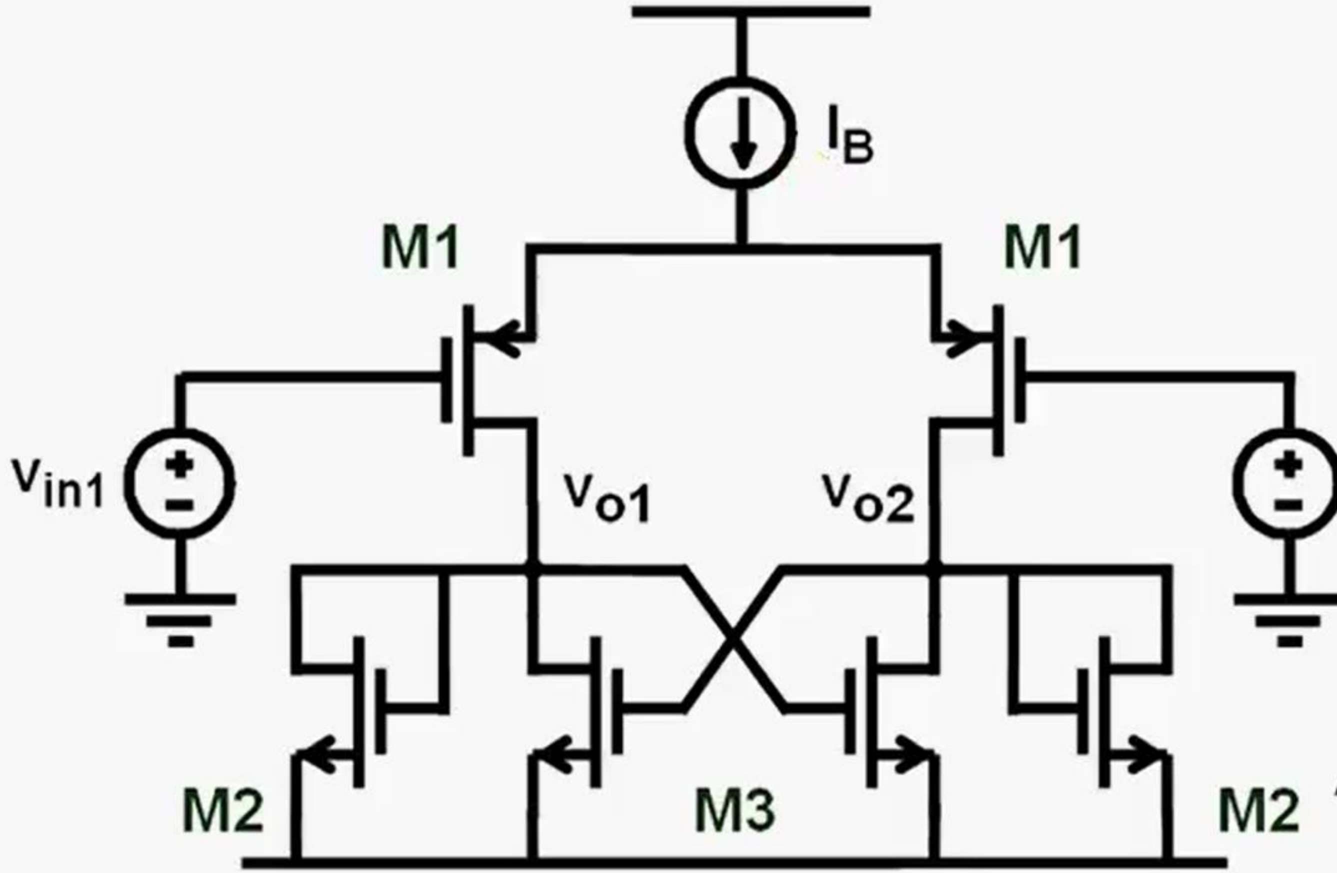
$$GBW = \frac{g_{m1}}{2\pi\, C_L}$$

Idea 2



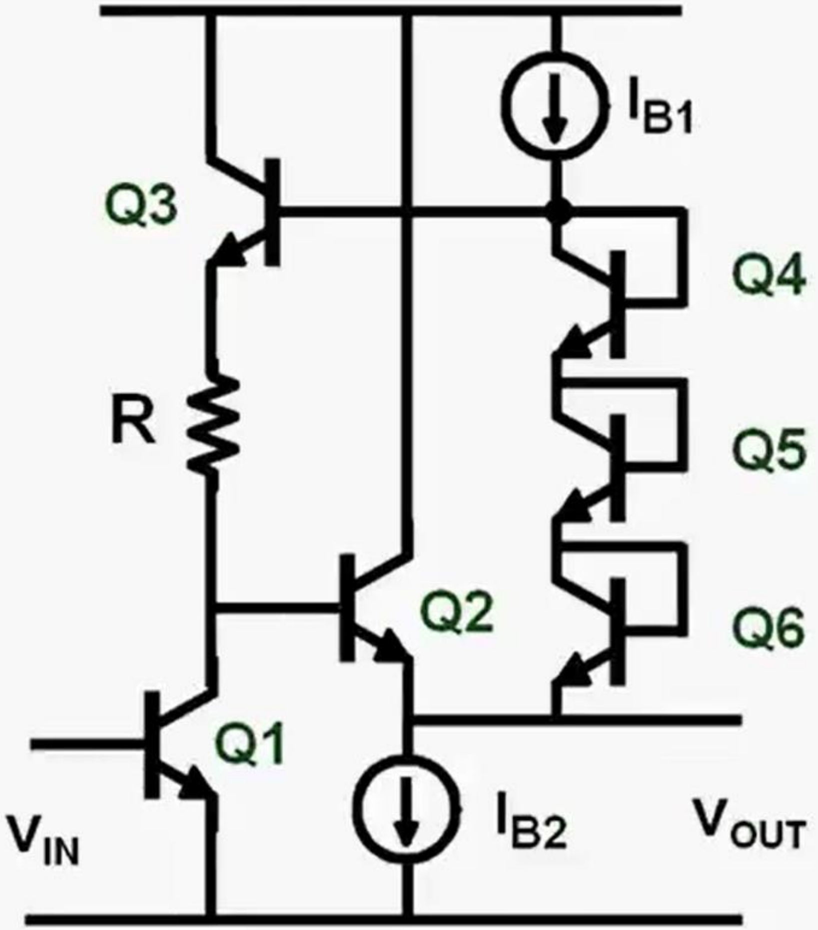
I_{B1}
I_{B2}
V_{out}
C_L
M2
M1
M3
V_{in}
A_V
Reg.cascode
A_{V3}
A_{V2}
A_{V1}
Cascode
Single trans.
GBW
f
$A_{v1} = g_{m1} r_{DS1}$
$A_{v2} = g_{m2} r_{DS2}$
$A_{v3} = g_{m3} r_{DS3}$
$GBW = \frac{g_{m1}}{2\pi C_L}$

Idea 3



$$A_v = \frac{g_{m1}}{g_{m2} - g_{m3}}$$

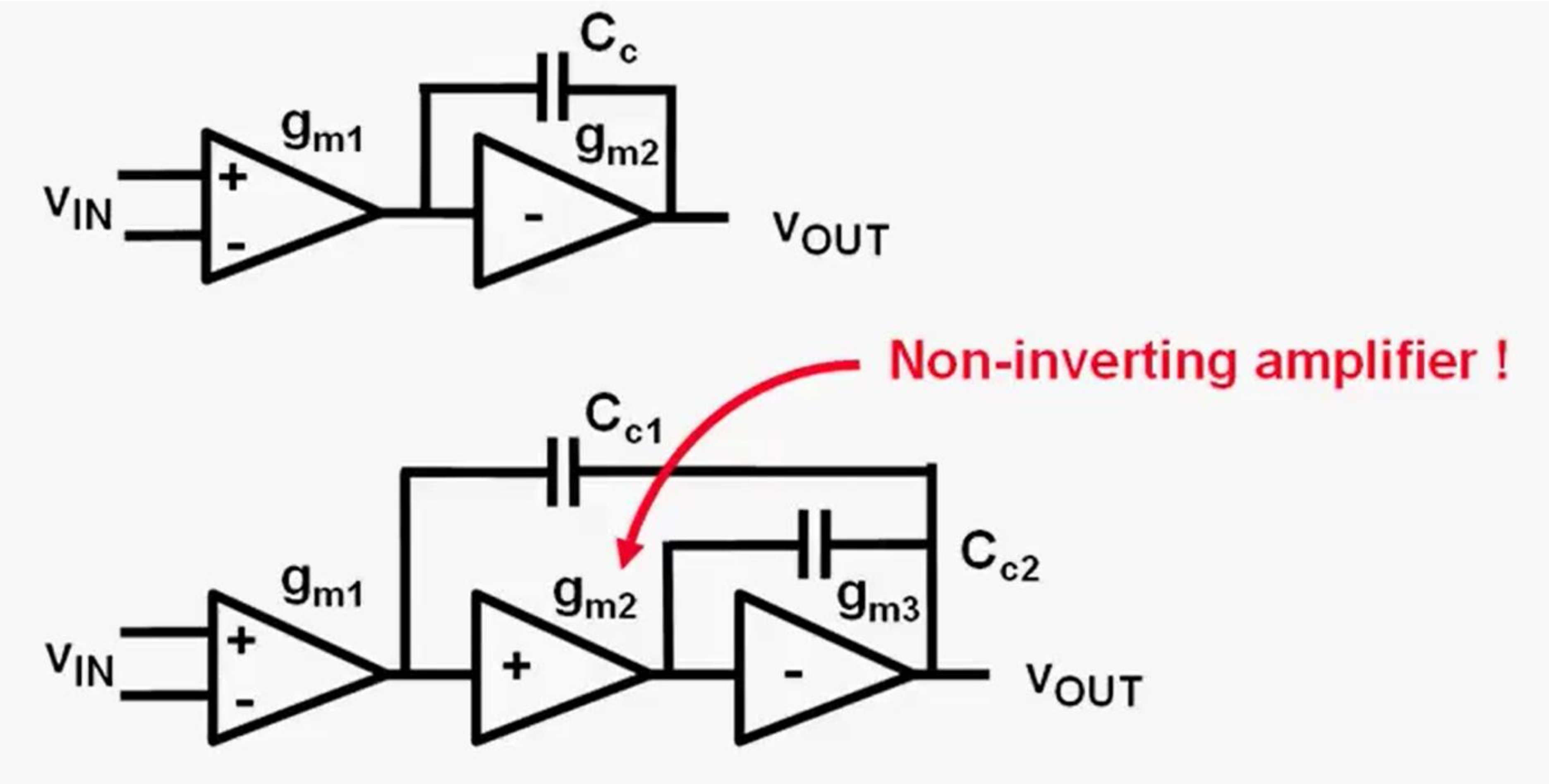
Idea 4



R is
bootstrapped out :
Very high gain !

Ref.: Nordholt
JSSC June 85, 688-696

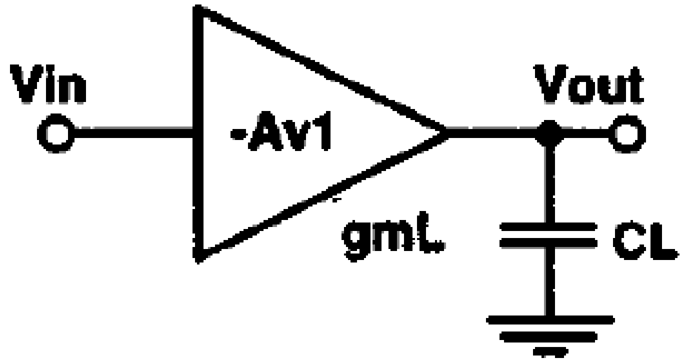
Idea 5



C_c
g_{m1}
g_{m2}
v_{IN}
+
-
-
v_{OUT}
Non-inverting amplifier !
C_{c1}
C_{c2}
g_{m1}
g_{m2}
g_{m3}
v_{IN}
+
-
+
-
v_{OUT}

Speed, Noise and Power !!!

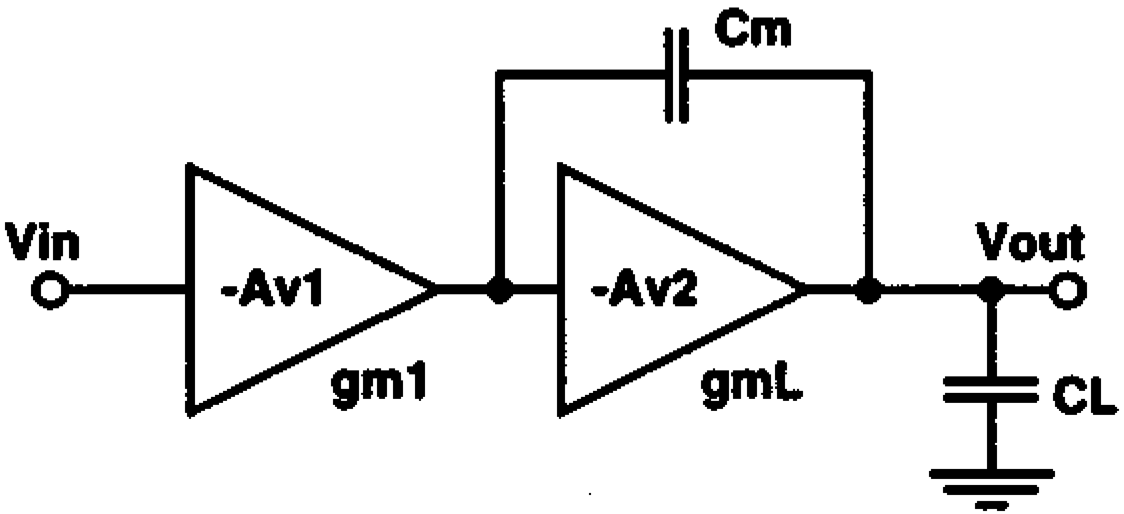
Quick Revison



$$A_{v_{\text{single}}}(s) = \frac{g_{mL} R_L}{1 + sC_L R_L}$$

$$\text{GBW} = \frac{g_{mL}}{C_L}$$

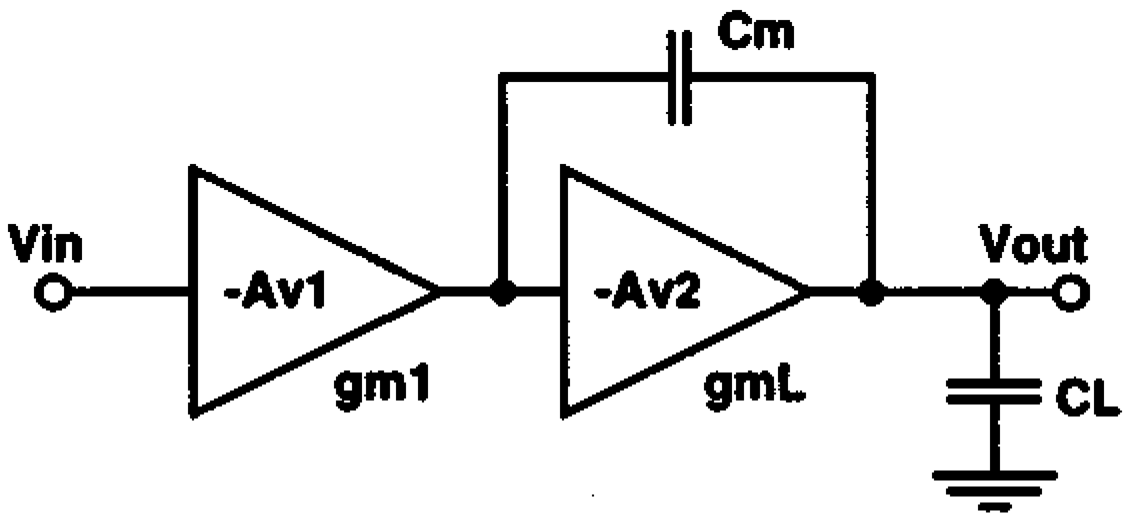
Quick Revison



$$A_{v_{(\mathrm{SMC})}}(s) = \frac{g_{m1} g_{mL} R_{o1} R_L \left(1 - s\frac{C_m}{g_{mL}}\right)}{(1 + s C_m g_{mL} R_{o1} R_L)\left(1 + s\frac{C_L}{g_{mL}}\right)}$$

$$\mathrm{GBW} = g_{m1}/C_m$$

Quick Revison



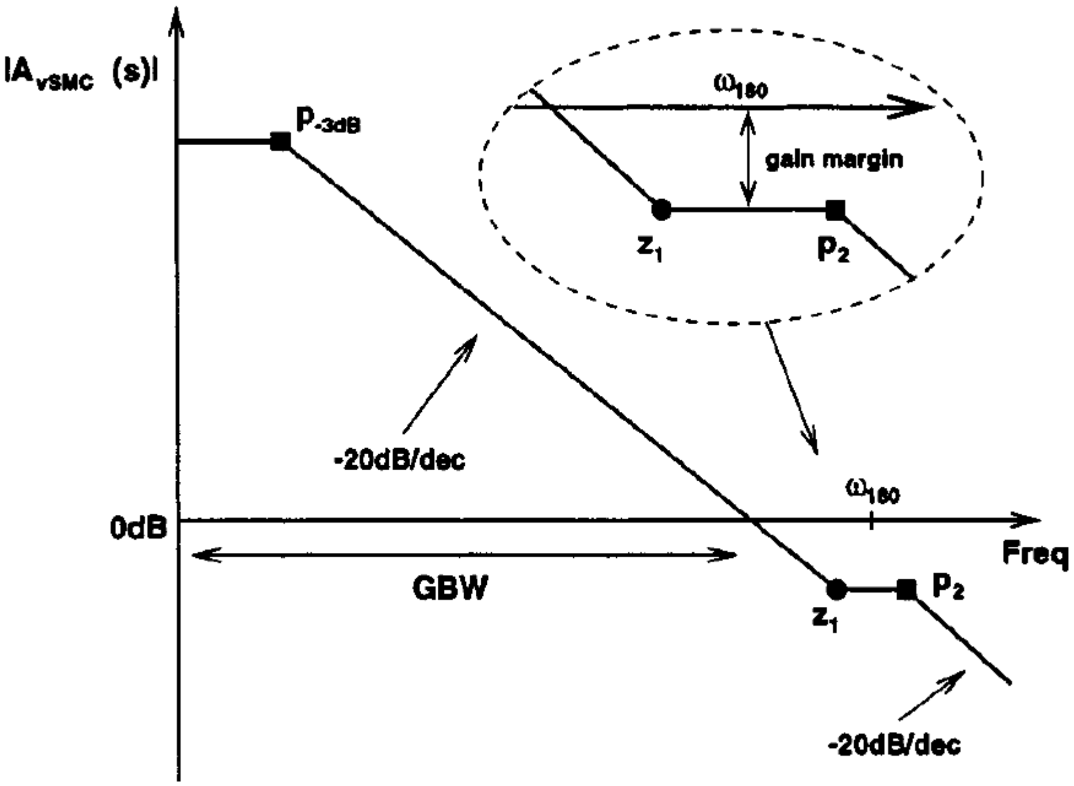
$$\mathrm{GBW} = g_{m1}/C_m$$

$$p_{-3\text{ dB}} = 1/C_m g_{mL} R_{o1} R_L$$

$$p_2 = g_{mL}/C_L$$

$$z_1 = -g_{mL}/C_m$$

Quick Revison



GBW is generally set to
be half of to obtain a good PM !!!

$$g_{mL}/C_L = 2g_{m1}/C_m$$

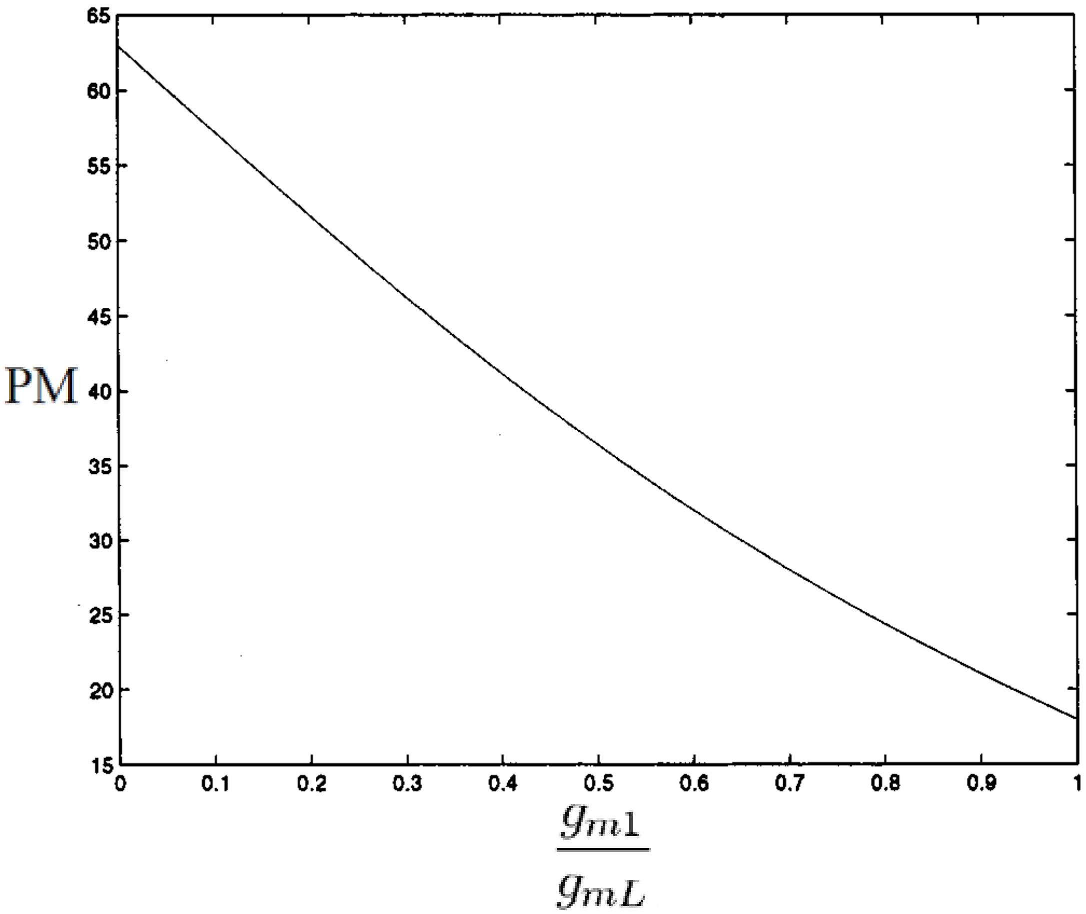
$$C_m = 2\left(\frac{g_{m1}}{g_{mL}}\right)C_L$$

As Z1 locates before P2, the gain margin is small and the amplifier may be unstable !!!

Quick Revison

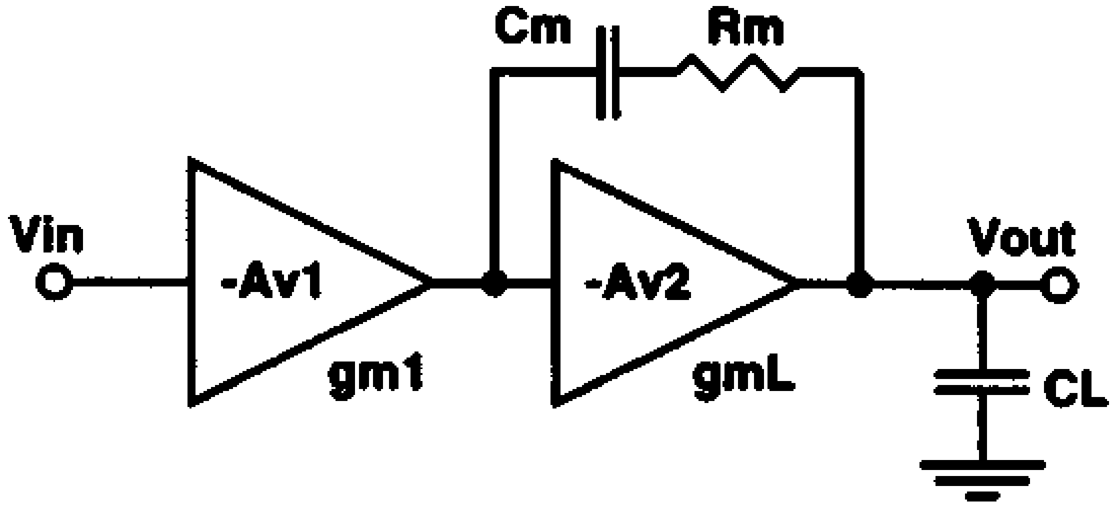
$$\mathrm{PM} = 180^\circ - \tan^{-1}\left(\frac{\mathrm{GBW}}{p_{-3\ \mathrm{dB}}}\right) - \tan^{-1}\left(\frac{\mathrm{GBW}}{p_2}\right) - \tan^{-1}\left(\frac{\mathrm{GBW}}{|z_1|}\right)$$

$$\approx \quad 63^\circ - \tan^{-1}\left(\frac{g_{m1}}{g_{mL}}\right)$$



(Leung et al, TCAS1, 2001)

Quick Revison

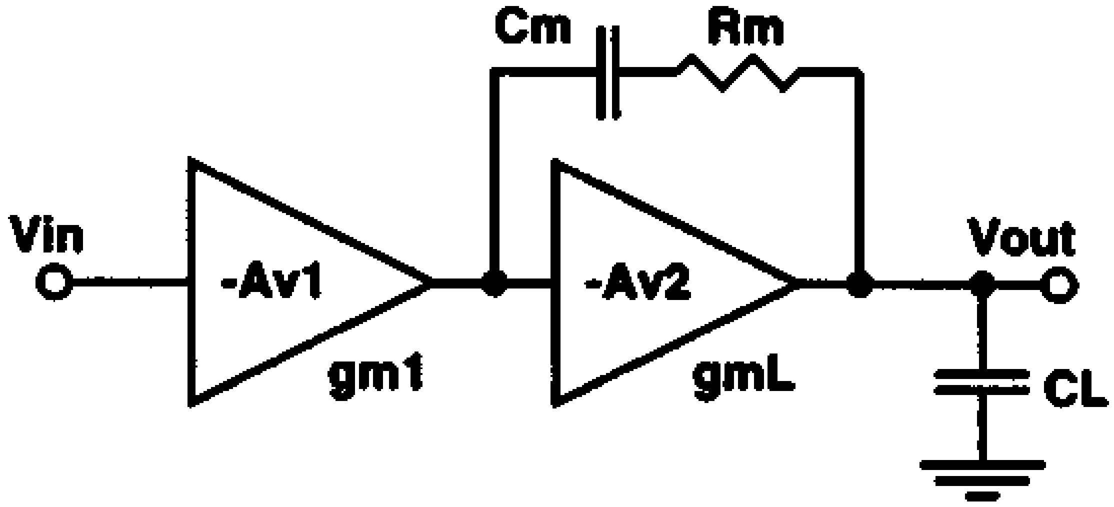


$$z_1 = 1/C_m\left(R_m - 1/g_{mL}\right)$$

$$\mathrm{GBW} = g_{m1}/C_m$$

$$A_{v(\mathrm{SMCNR})}(s) = \frac{g_{m1}g_{mL}R_{o1}R_L\left[1 + sC_m\left(R_m - \frac{1}{g_{mL}}\right)\right]}{\left[1 + sC_m(R_m + g_{mL}R_{o1}R_L)\right]\left[1 + s\frac{C_L(R_{o1}+R_m)R_L}{R_m+g_{mL}R_{o1}R_L}\right]}$$

Quick Revison

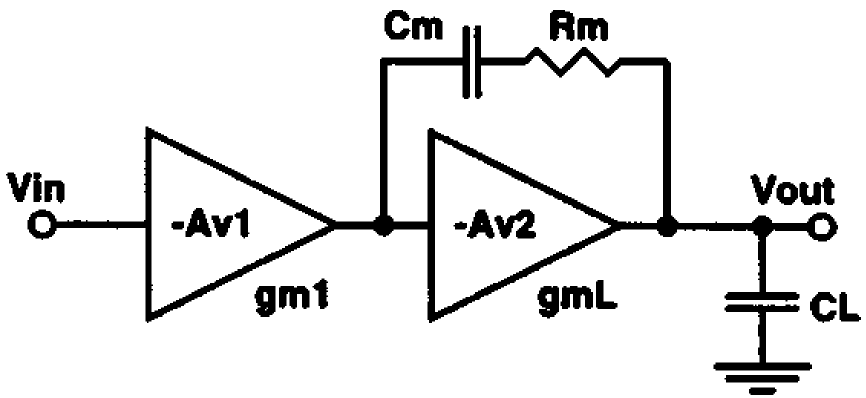


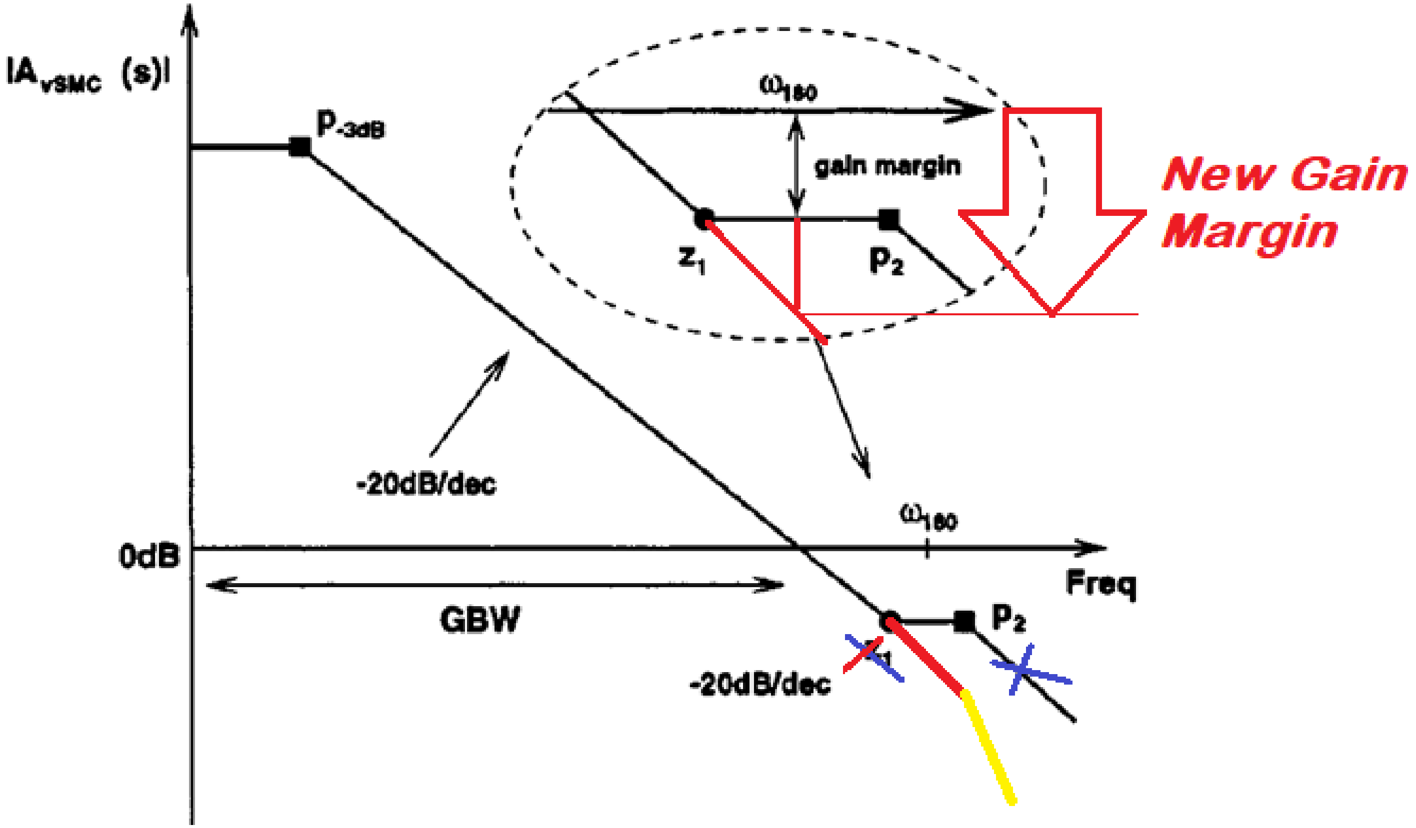
$$z_1 = 1/C_m\,(R_m - 1/g_{mL})$$

$$\mathrm{GBW} = g_{m1}/C_m$$

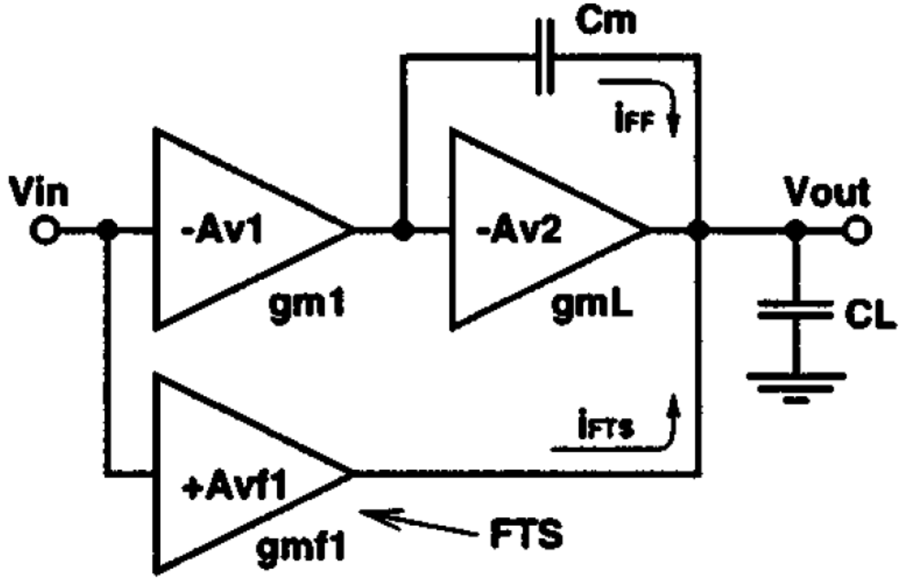
$$A_{v(\mathrm{SMCNR})}(s) = \frac{g_{m1}g_{mL}R_{o1}R_L\left[1 + sC_m\left(R_m - \frac{1}{g_{mL}}\right)\right]}{\left[1 + sC_m(R_m + g_{mL}R_{o1}R_L)\right]\left[1 + s\frac{C_L(R_{o1}+R_m)R_L}{R_m+g_{mL}R_{o1}R_L}\right]}$$

Quick Revison





Quick Revison



$$A_{v_{(MZC)}}(s) = \frac{g_{m1} g_{mL} R_{o1} R_L \left[1 + s\frac{C_m (g_{mf1} - g_{m1})}{g_{m1} g_{mL}}\right]}{(1 + sC_m g_{mL} R_{o1} R_L)\left(1 + s\frac{C_L}{g_{mL}}\right)}$$

Speed, Noise and Power !!!

Speed, Noise and Power

Thermal noise :

no v_{sat}

$$\overline{dv_{ieq}}^2 = 4kT\left(\frac{2/3}{g_m}\right) df \sim \frac{L}{W}$$

Increases for smaller W/L !

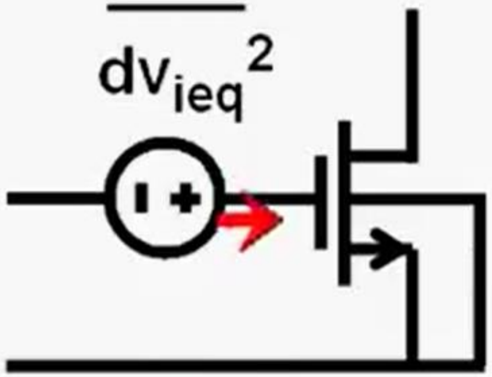
1/f noise

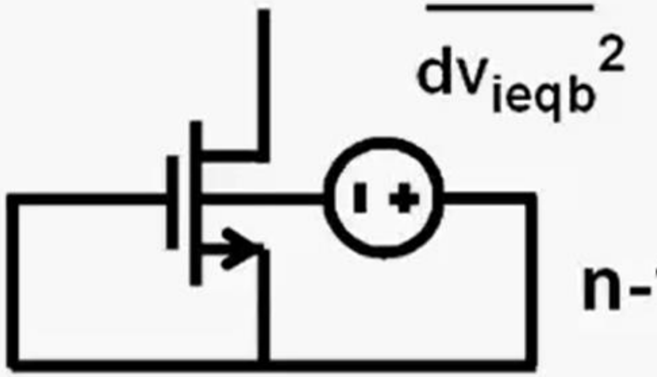
$$\overline{dv_{ieqf}}^2 = \frac{KF_F}{WL\,C_{ox}^2}\frac{df}{f} \sim \frac{1}{WLC_{ox}^2}$$

Increases for smaller W and L ?

SNR decreases for smaller W and L !

Speed, Noise and Power



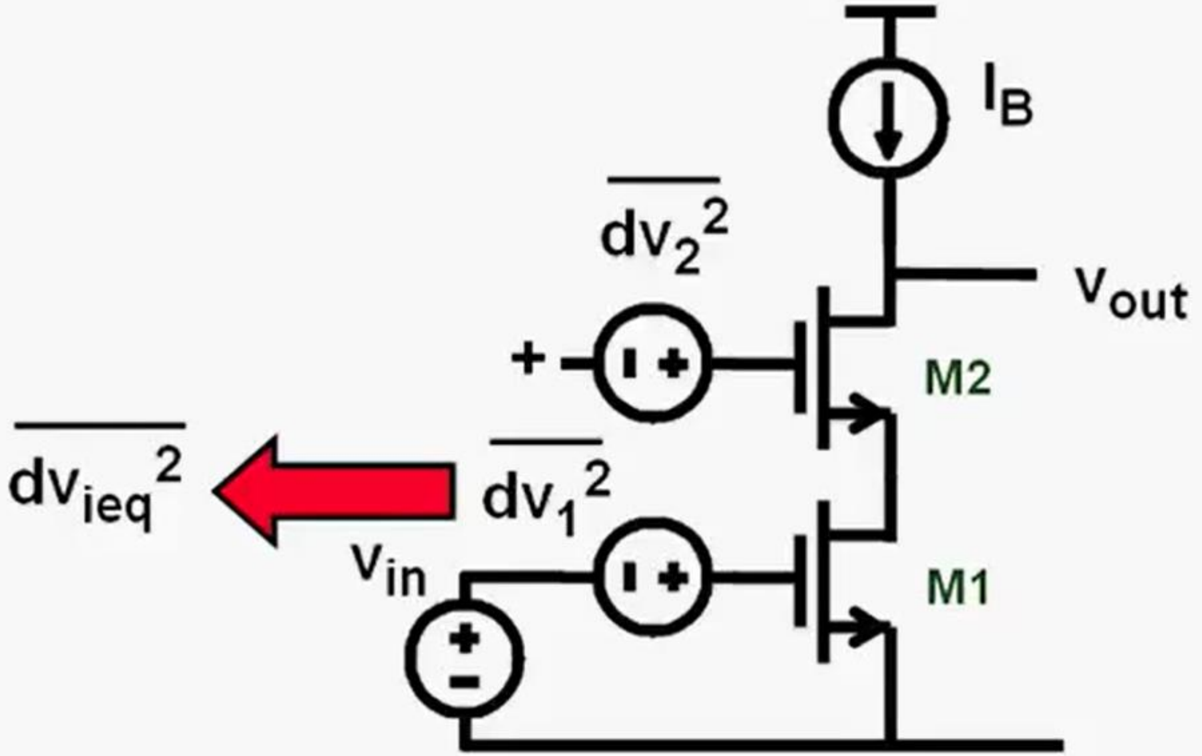


$$n-1 = \frac{g_{mb}}{g_m}$$

$$\overline{dv_{ieq}^2} = 4kT\left(\frac{2/3}{g_m}\right) df \qquad \overline{dv_{ieqb}^2} = 4kT\left(\frac{2/3\ g_m}{g_{mb}^2}\right) df$$

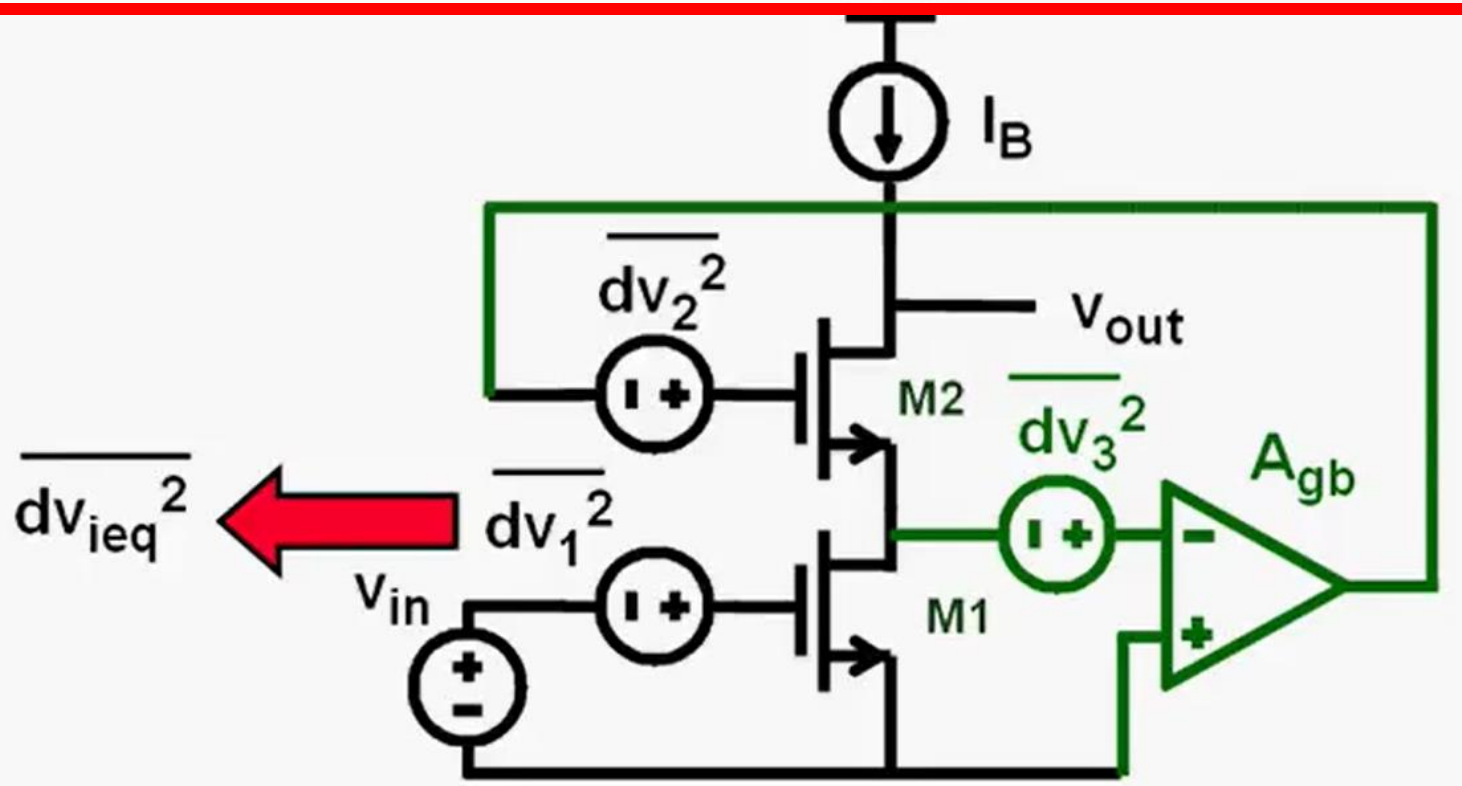
$$\overline{dv_{ieqf}^2} = \frac{KF_F}{WL\ C_{ox}^2}\ \frac{df}{f} \qquad \overline{dv_{ieqfb}^2} = \frac{KF_F}{WL\ C_{ox}^2}\ \frac{g_m^2}{g_{mb}^2}\ \frac{df}{f}$$

Speed, Noise and Power



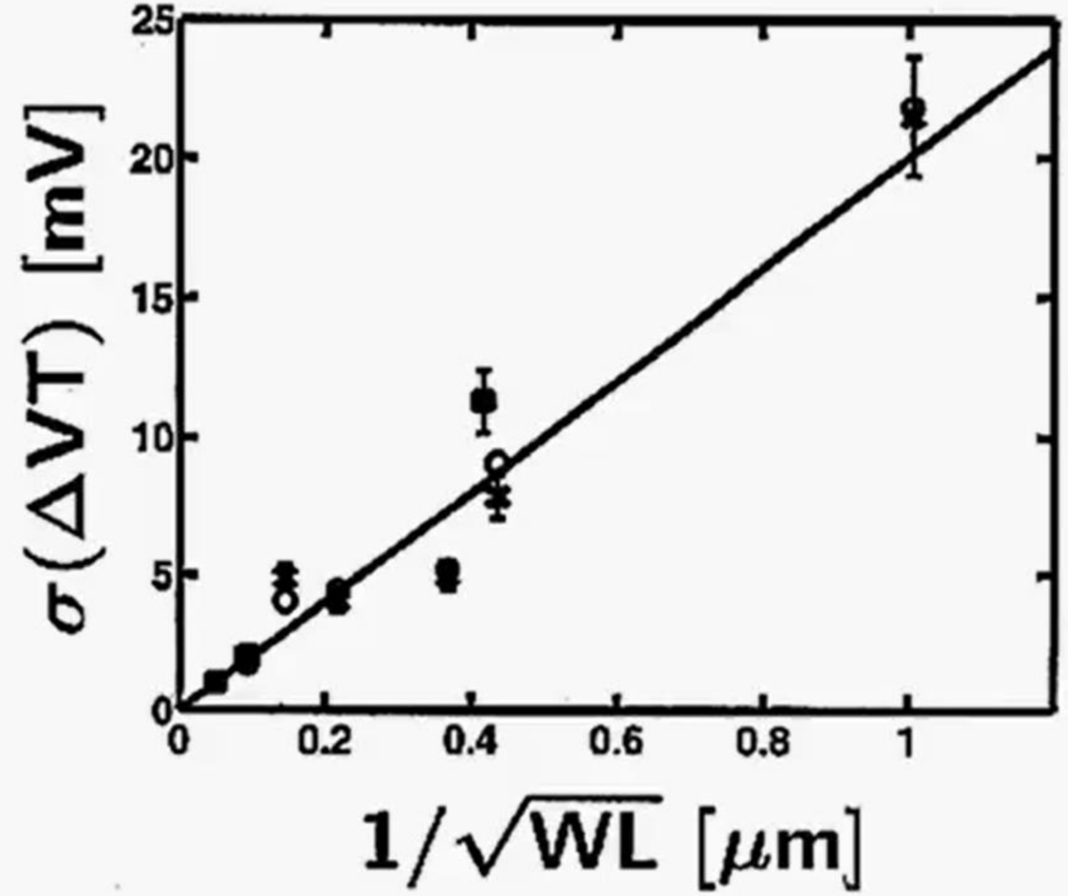
$$\overline{dv_{ieq}^2} = \overline{dv_1^2} + \overline{dv_2^2}\ \frac{1}{(g_{m1}\, r_{o1})^2} \approx \overline{dv_1^2}$$

Speed, Noise and Power



$$\overline{dv_{ieq}^2} = \overline{dv_1^2} + \overline{dv_2^2}\,\frac{1}{(g_{m1}\,r_{o1})^2} + \overline{dv_3^2}\,\frac{1}{(g_{m1}\,r_{o1})^2} \approx \overline{dv_1^2}$$

Speed, Noise and Power

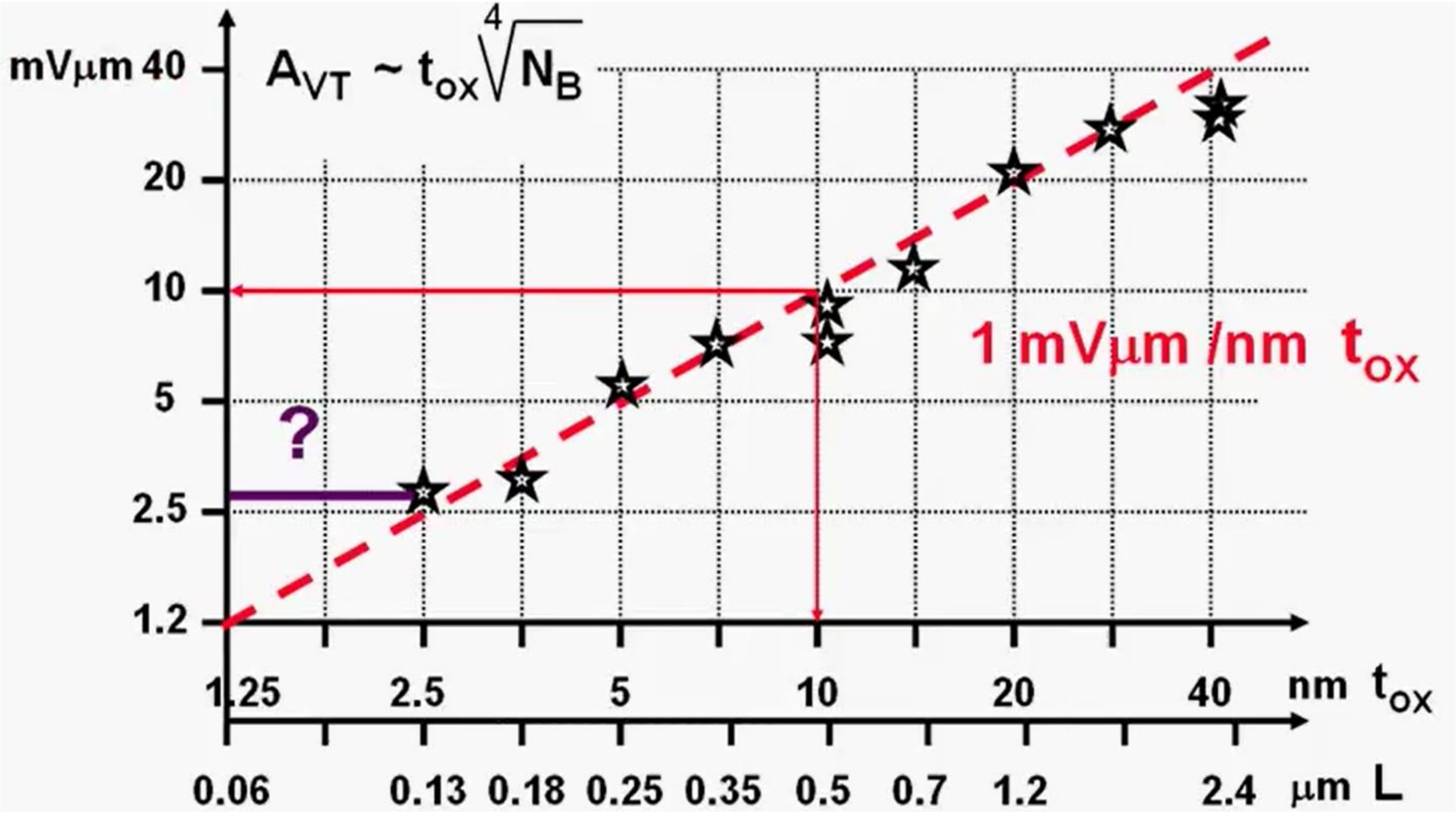


$$\sigma^2(\Delta V_T) \approx \frac{A_{VT}^{\;2}}{WL}$$

$$A_{VT} \sim t_{ox} \sim L$$

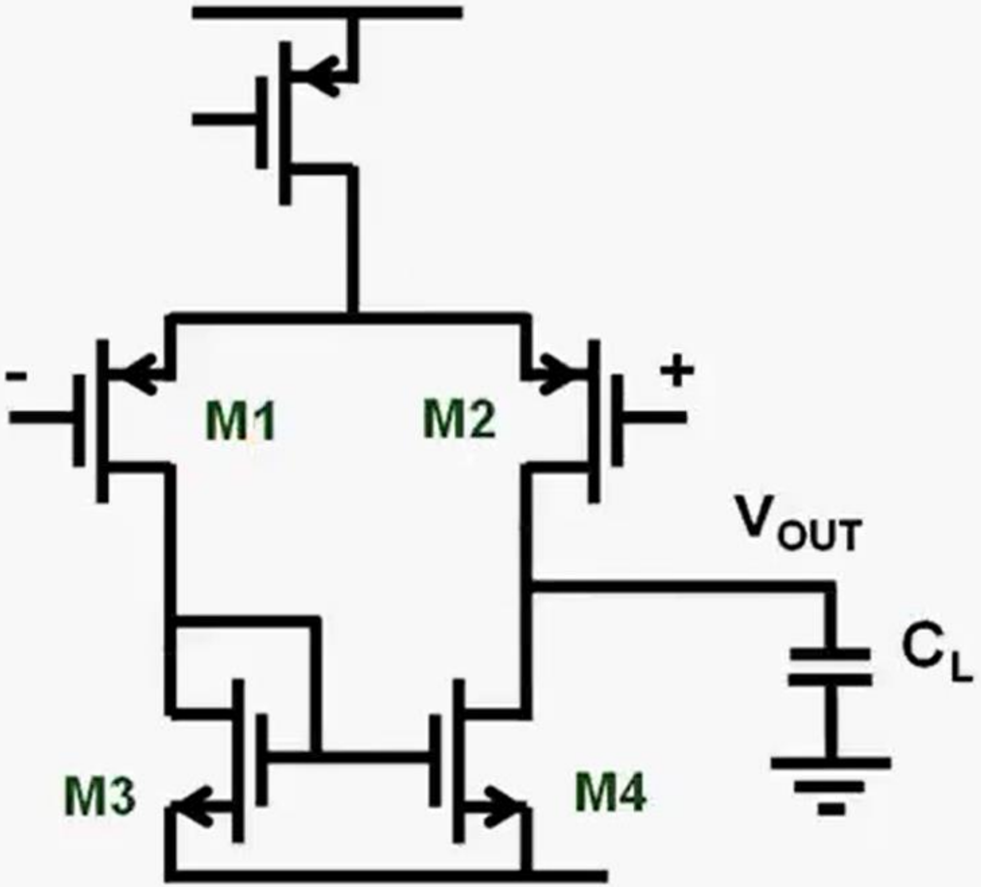
$A_{VT} \approx 10$ mVμm
in 0.5 μm CMOS
$\sigma(\Delta V_T) \approx 3.2$ mV
for W = 10 & L = 1 μm

Speed, Noise and Power



Comparison of few fundamantal AMP Configurations

Speed and fT for simple Diff-amp



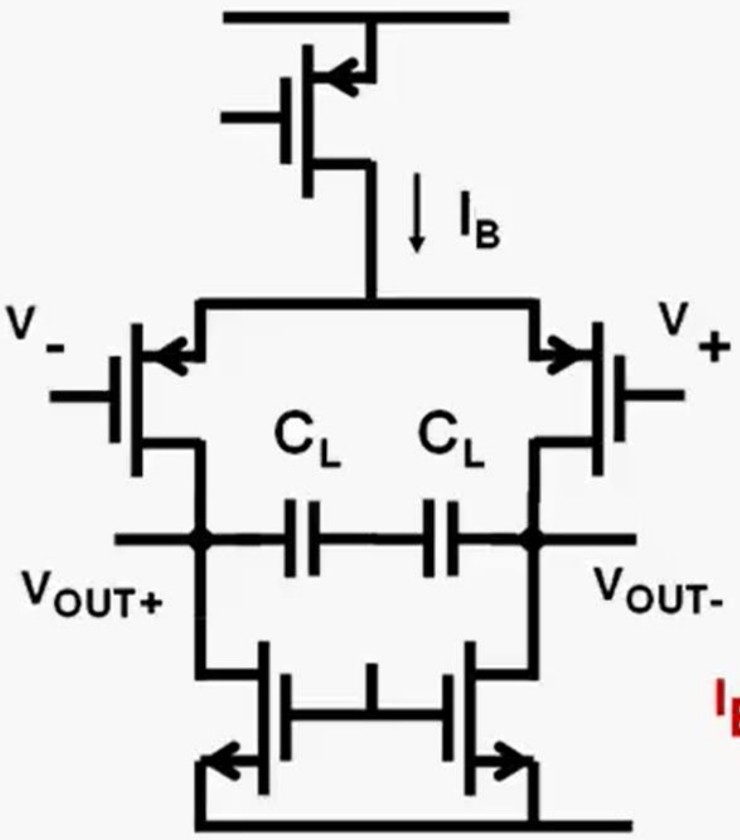
$$GBW = \frac{g_{m1}}{2\pi C_L}$$

$$GBW_{max} \approx \frac{f_{nd}}{2}$$

$$f_{nd} = \frac{g_{m3}}{4\pi C_{GS3}} \approx \frac{f_{T3}}{4}$$

$$GBW_{max} \approx \frac{f_{T3}}{8}$$

Maximum GBW and FOM



$$GBW = \frac{g_{m1}}{2\pi\, C_L} \qquad g_{m1} = \frac{I_B}{V_{GS1}-V_T}$$

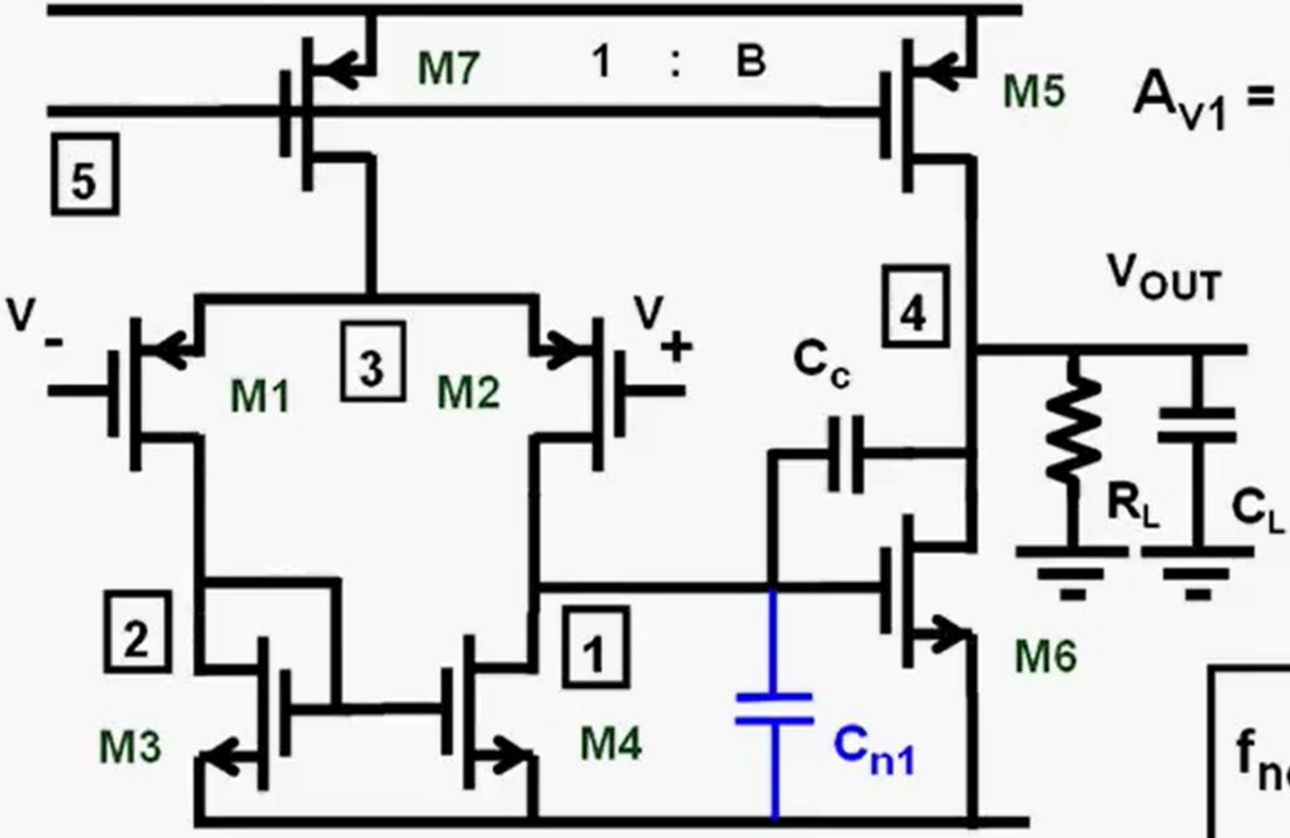
$$\boxed{GBW_{max} = \frac{I_B}{V_{GS1}-V_T}\,\frac{1}{2\pi\, C_L}}$$

$I_B = 10\ \mu A$ $C_L = 1\ pF$ $GBW_{max} \approx 10\ MHz$

$$FOM = \frac{GBW.C_L}{I_B} = 1000$$

MHzpF/mA

Two Stage AMP



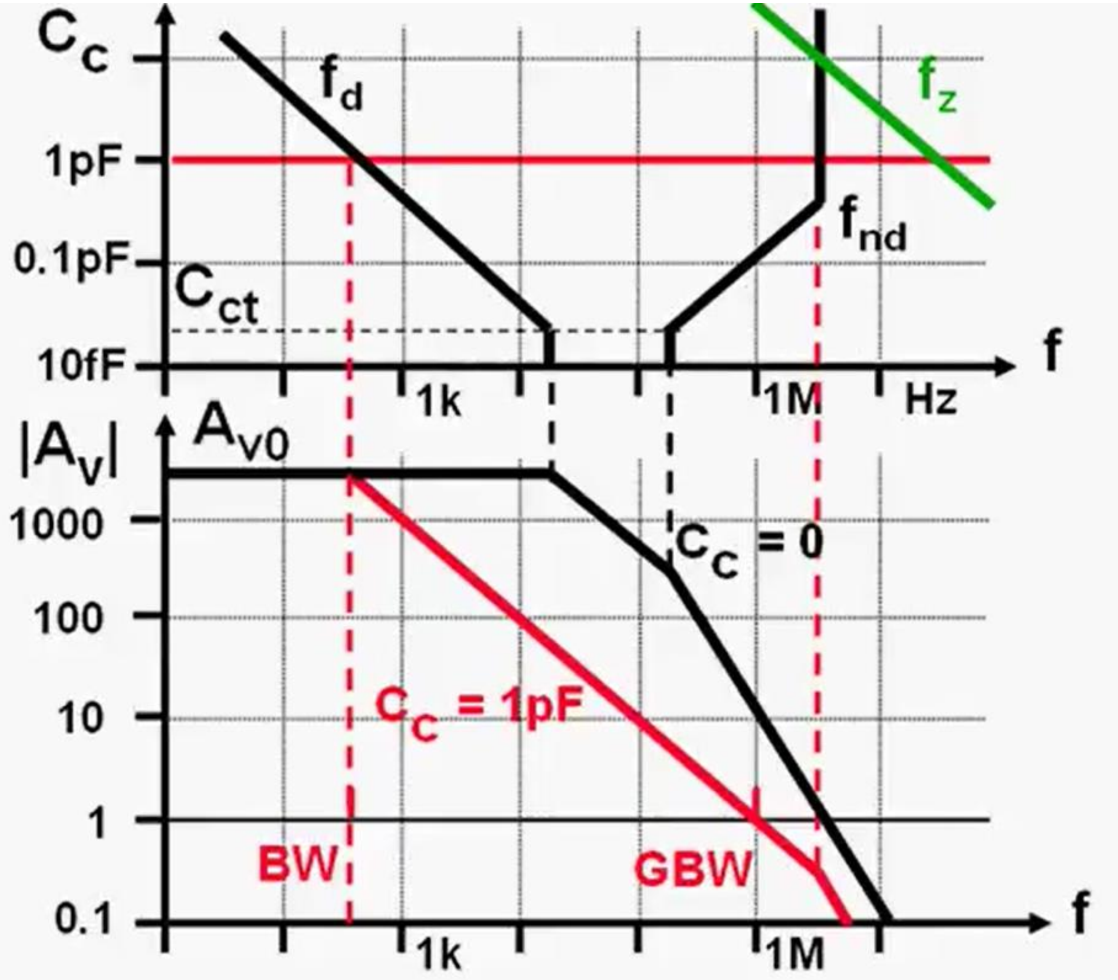
$$A_{V1} = \frac{g_{m1}}{g_{o24}} \quad A_{V2} = \frac{g_{m6}}{g_{Lo6}}$$

$$BW = \frac{g_{o24}}{2\pi A_{V2} C_c}$$

$$\boxed{GBW = \frac{g_{m1}}{2\pi C_c}}$$

$$\boxed{f_{nd} \approx \frac{g_{m6}}{2\pi C_{Ln4}} \frac{1}{1+\frac{C_{n1}}{C_c}}}$$

Two Stage AMP



Pole splitting

is sufficient
for C_c = 1pF

$$f_z = \frac{g_{m6}}{2\pi C_c}$$

Two Stage AMP

$$GBW = \frac{g_{m1}}{2\pi C_c}$$

$$f_{nd} = \frac{g_{m6}}{2\pi C_L} \frac{1}{1 + C_{n1}/C_c}$$

$C_L = \alpha C_c$ $\alpha \approx 2$

$C_c = \beta C_{n1} = \beta C_{GS6}$ $\beta \approx 3$

$f_{nd} = \gamma\, GBW$ $\gamma \approx 2$

$C_{GS} = kW$ $k = 2\ 10^{-11}$ F/cm

$$GBW = \frac{f_{nd}}{\gamma} = \frac{g_{m6}}{2\pi C_L} \frac{1}{\gamma\,(1 + 1/\beta)} \boxed{= \frac{f_{T6}}{\alpha\beta\gamma\,(1 + 1/\beta)}}$$

$$C_L = \alpha C_c = \alpha\beta C_{n1} = \alpha\beta C_{GS6} = \alpha\beta kW_6$$

$W_6 \uparrow$ if $C_L \uparrow$

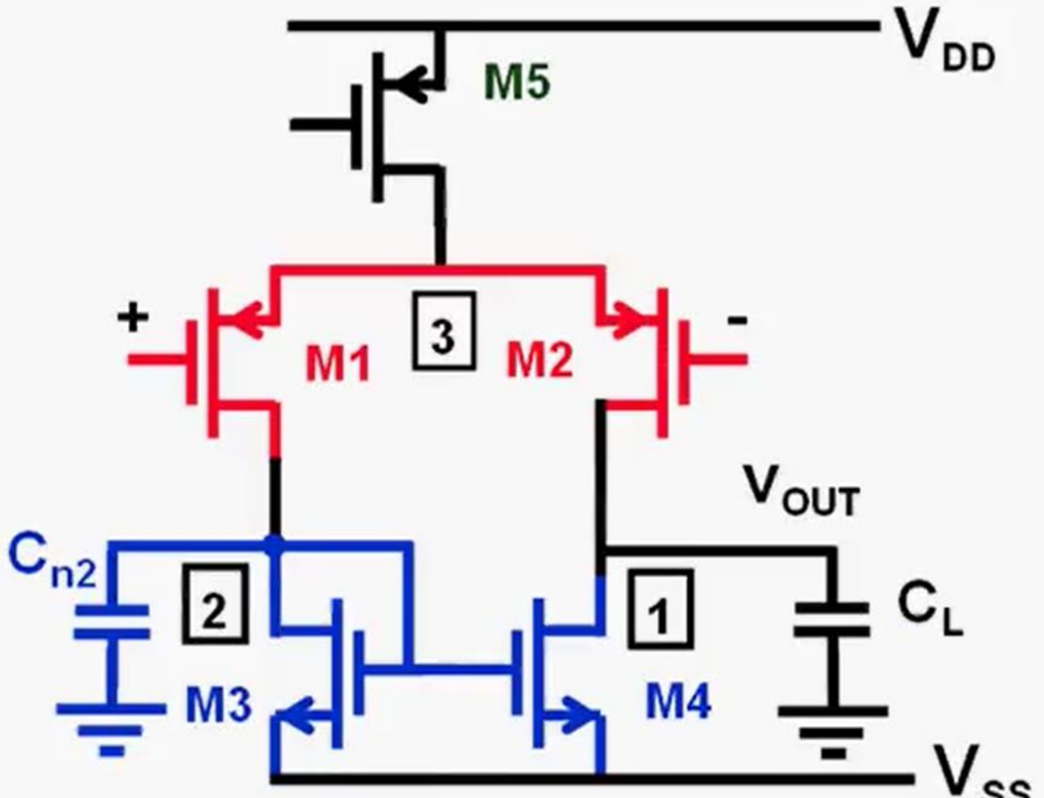
FOM of Two Stage AMP

$$GBW_{max} \approx \frac{f_{T6}}{16}$$

$$FOM = \frac{GBW.C_L}{B\ I_B} = \frac{1000}{1.5 * 2}$$

MHzpF/mA

FOM of Single Stage AMP



Differential pair

Current mirror

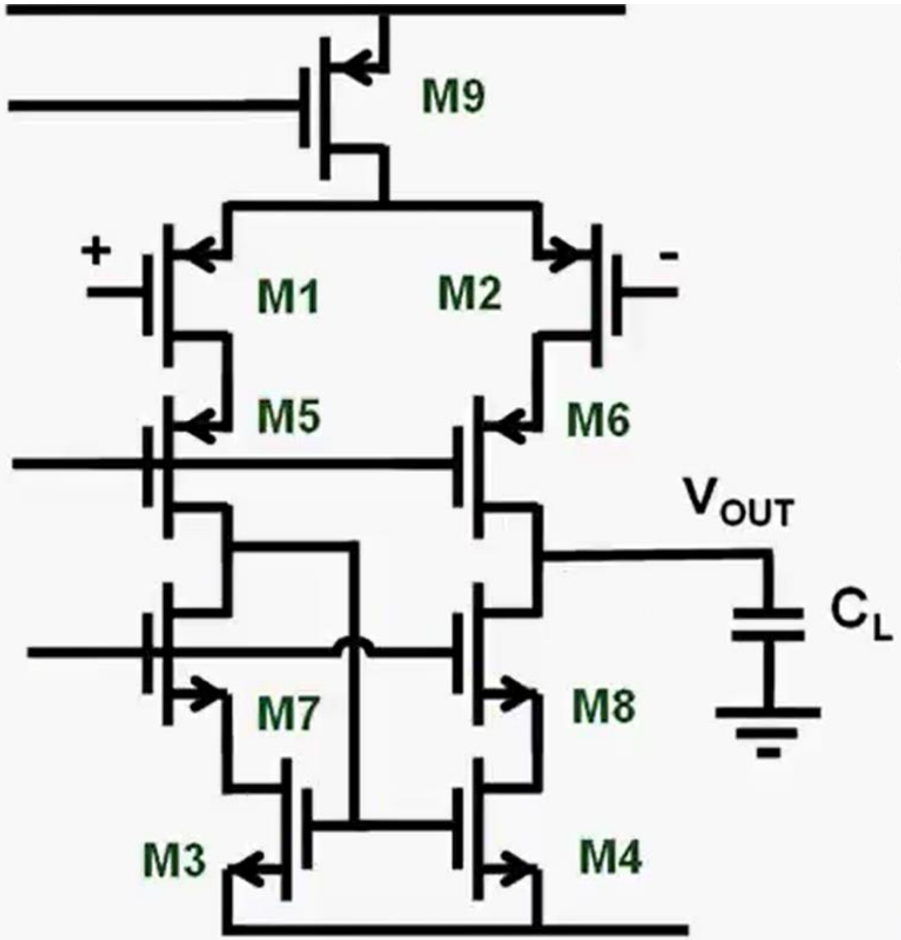
$$GBW = \frac{g_{m1}}{2\pi C_L}$$

$$f_{nd} = \frac{g_{m3}}{2\pi C_{n2}} \approx \frac{f_{T3}}{4}$$

$$\boxed{GBW \approx \frac{f_{T3}}{8}}$$

$$C_{n2} = 2C_{GS3} + C_{DB3} + C_{DB1} \approx 4C_{GS3}$$

FOM of Telescopic AMP



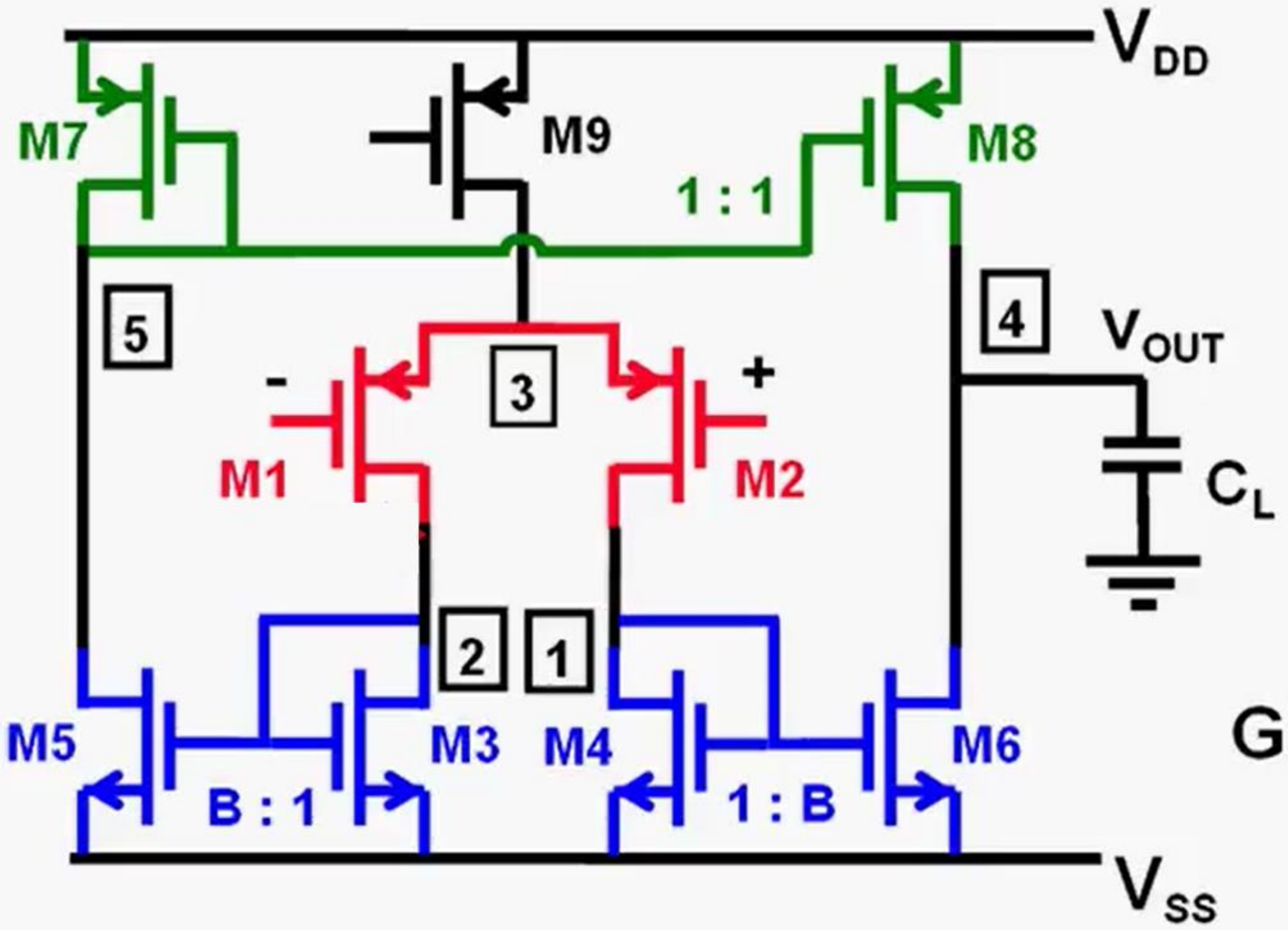
More gain
At low frequencies

$$GBW = \frac{g_{m1}}{2\pi C_L}$$

$$\boxed{GBW \approx \frac{f_{T3}}{8}}$$

Gulati, JSSC Dec.98, 2010-2019

Symmetric AMP



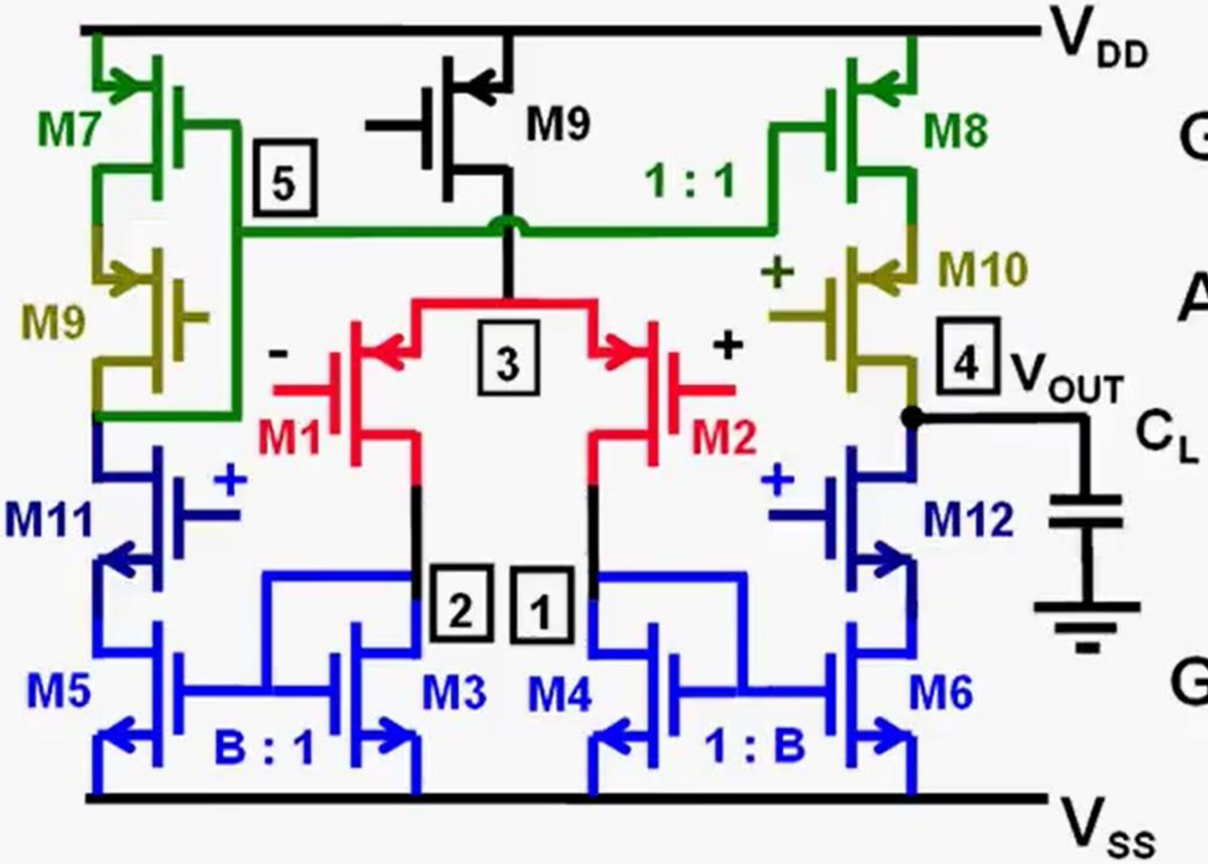
$$A_V = g_{m1} B R_{n4}$$

$$BW = \frac{1}{2\pi R_{n4} C_L}$$

$$GBW = B \frac{g_{m1}}{2\pi C_L}$$

B ?

FOM of Symmetric AMP



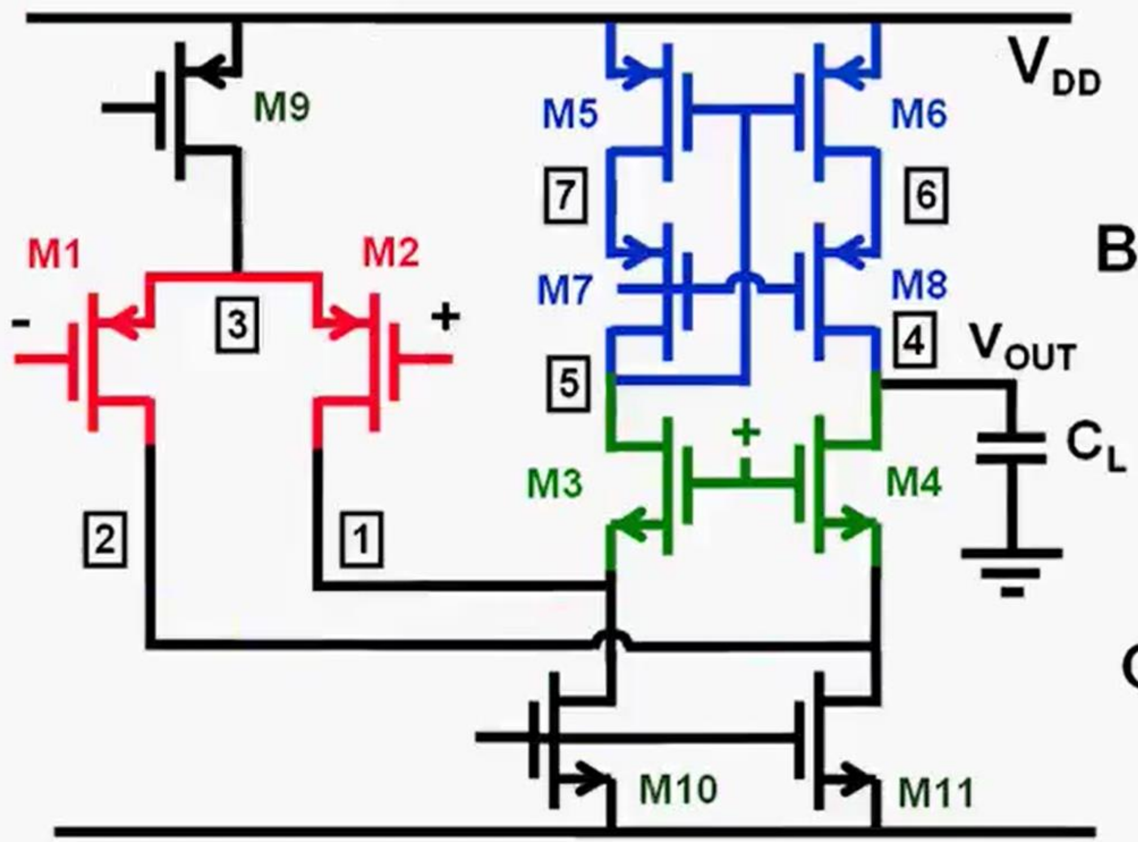
GBW same
but
A_V is x higher !!!

$$GBW = B \frac{g_{m1}}{2\pi C_L}$$

$$FOM = \frac{GBW.C_L}{B\ I_B} = \frac{1000}{3 * 2}$$

MHzpF/mA

FOM of FC



$$A_v = g_{m1} R_{n4}$$

$$BW = \frac{1}{2\pi R_{n4} C_L}$$

$$GBW = \frac{g_{m1}}{2\pi C_L}$$

$$FOM = \frac{GBW.C_L}{B\ I_B} = \frac{1000}{2}$$

MHzpF/mA

Comparison

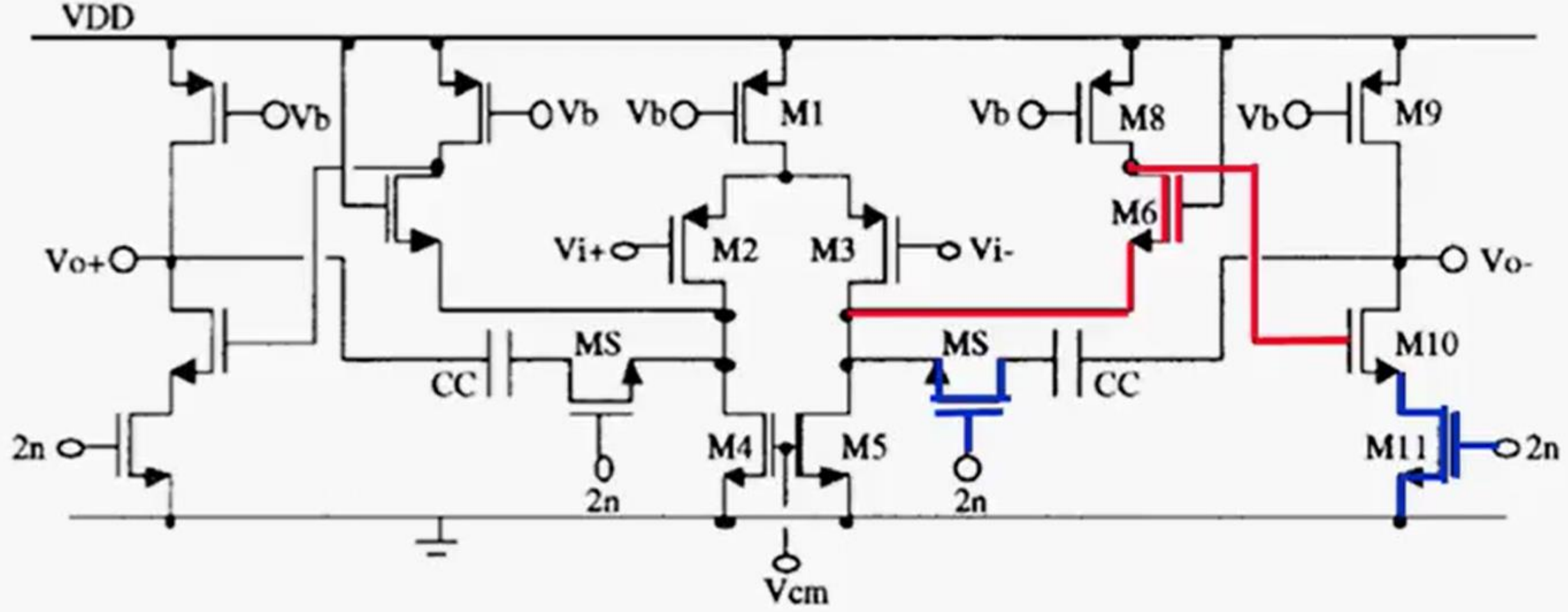
	I_{TOT} mA	$\frac{dv_{in,eq}^2}{8/3\ kT\ df / g_{m1}}$	Speed FOM	Swing
Telescopic casc.	0.25	4	1	small
Symmetrical (B= 3)	0.33	16	6	avg.
Folded cascode	0.5	4	2	avg.
Miller 2-stage*	1.1	4	3	max.

.

* $C_L/C_c = 2.5$

Sub 1V Configurations

JSSC'97

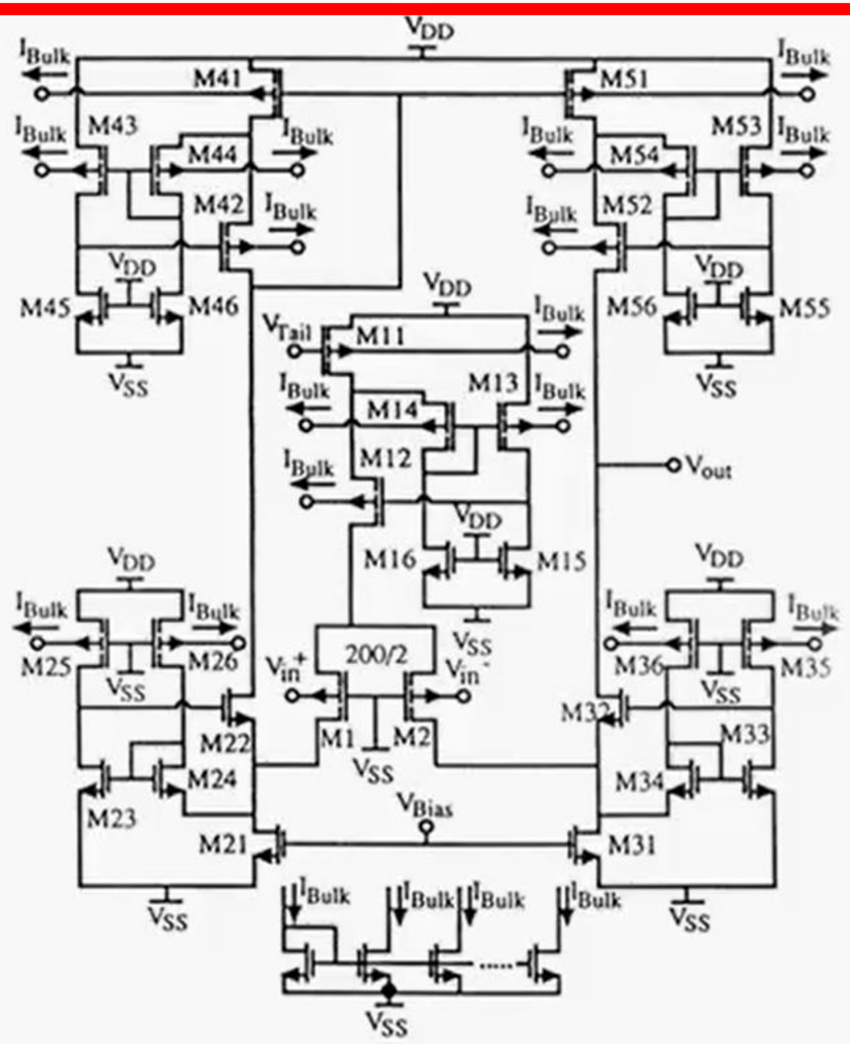


1 V 80 μW (min: $V_T + 2V_{DSsat}$)
Fully differential
75 dB 30 MHz

4 Switches 2n :
Only 2nd stage switched off !

Baschirotto, .. JSSC Dec.97,pp.1979-1986

ISSCC'95



$V_T \approx 0.7$ V (n-well CMOS)

Bulk input :

At 0.4 V only 0.1 pA

$$\frac{g_{mB}}{g_m} = n - 1 = \frac{C_D}{C_{ox}}$$

GBW = 0.6 MHz (15 pF)

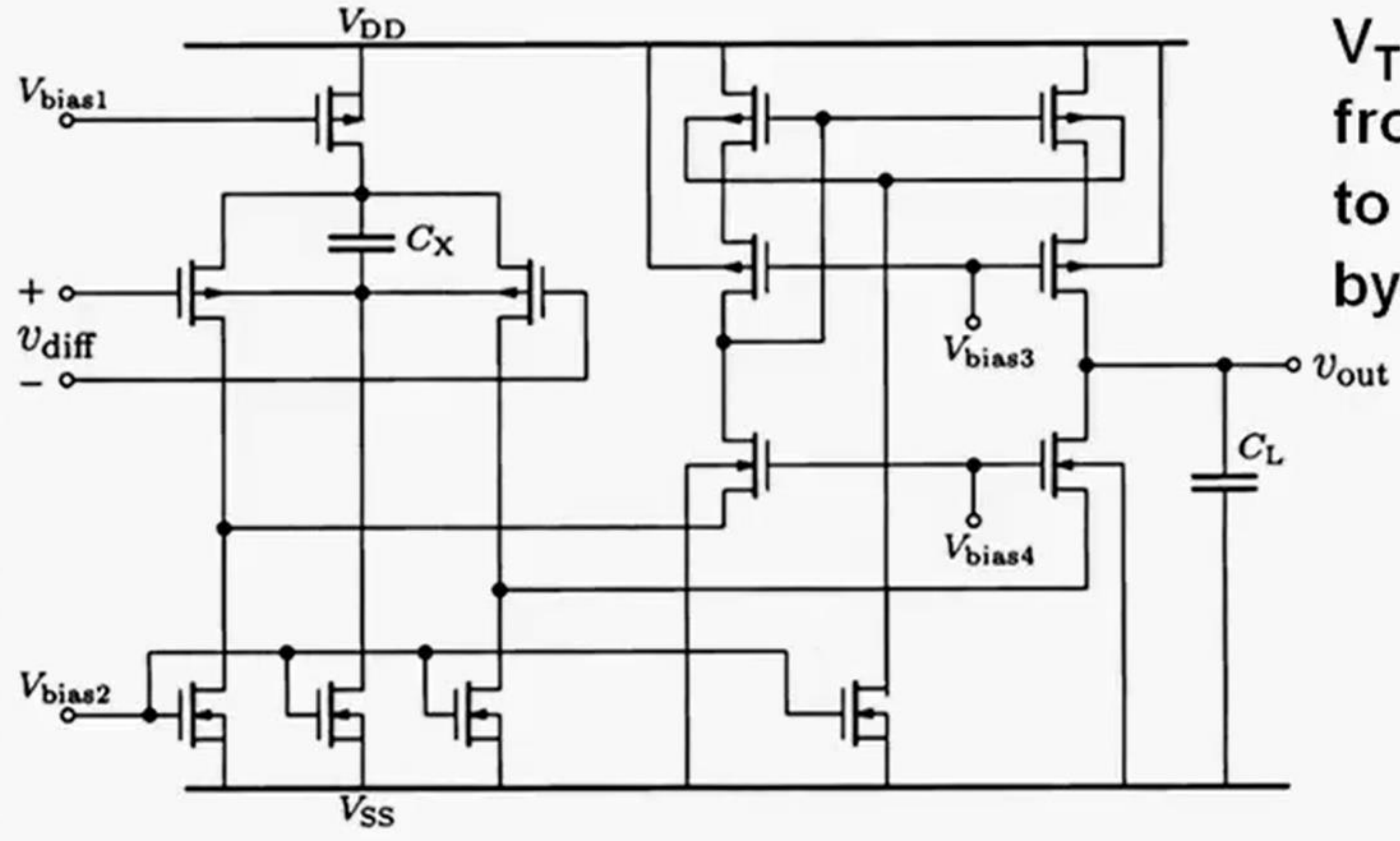
1 V 45 μA

In & Output range:

0.2 … 0.88 V

Allen, .. ISSCC Febr.95, 192-193

JSSC'00



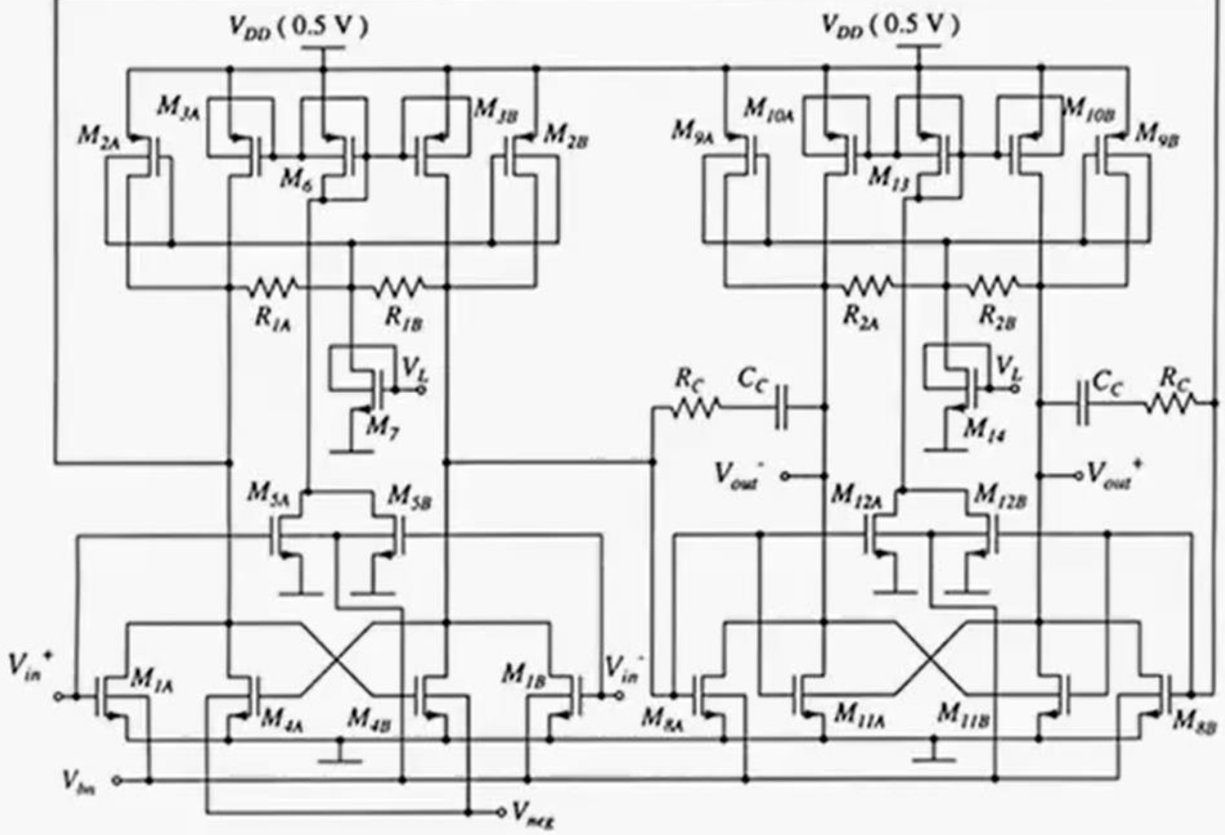
V_T reduced from 0.6 V to 0.4 V by bulk-bias !

$V_{DD,SS}$ = 1 V

$V_{in,\ CM}$ = 0 - 0.65 V

V_{out} = 0.1 - 0.9V

JSSC'05



$V_T \approx 0.5$ V (n-well 0.18 μm CMOS)

Pseudo-differential Cross-coupling for higher gain !

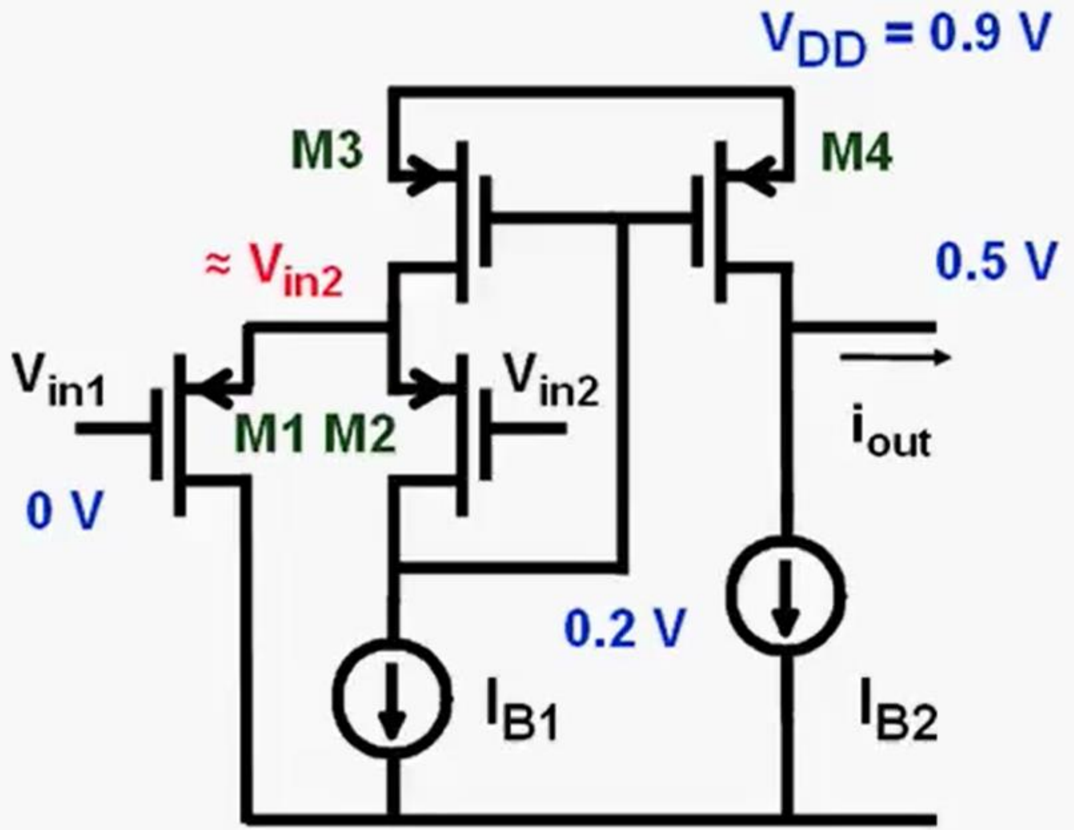
GBW = 15 MHz (10 pF)
0.5 V   0.4 mA

$V_{in,\,DC} = V_{GS1} = V_{GS8} \approx 0.4$ V
$V_{DS7} \approx 0.1$ V

V_{bn} sets $V_{outDC} \approx 0.25$ V
V_{neg} sets gain

Chatterjee, JSSC Dec.05, 2373-2387

JSSC'98



$V_T = 0.5$ V

$V_{GS} - V_T = 0.2$ V

$V_{GS} = 0.7$ V

$V_{DSsat} = 0.2$ V

M2 is source follower

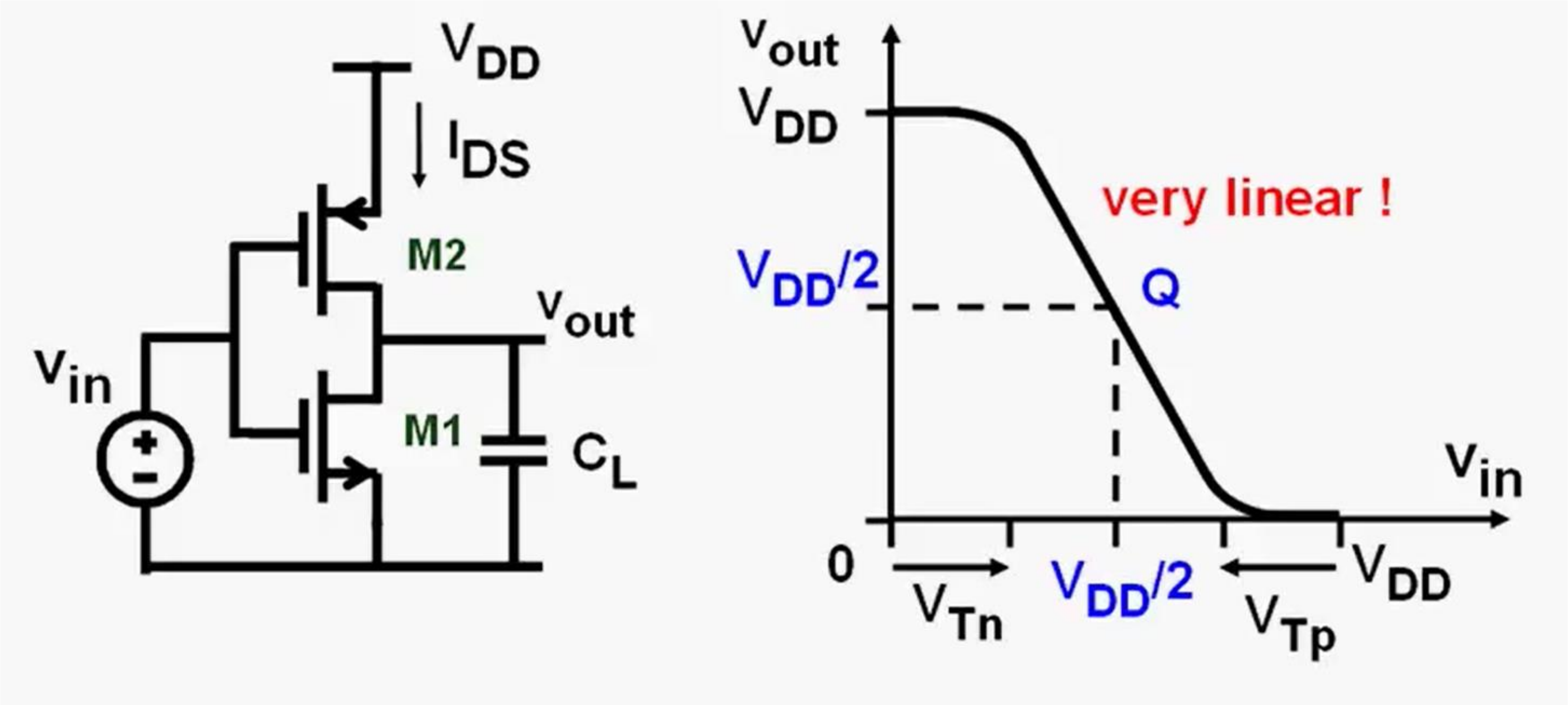
$V_{GS1} = V_{in1} - V_{in2}$

$i_{out} \sim (V_{in1} - V_{in2})^2$

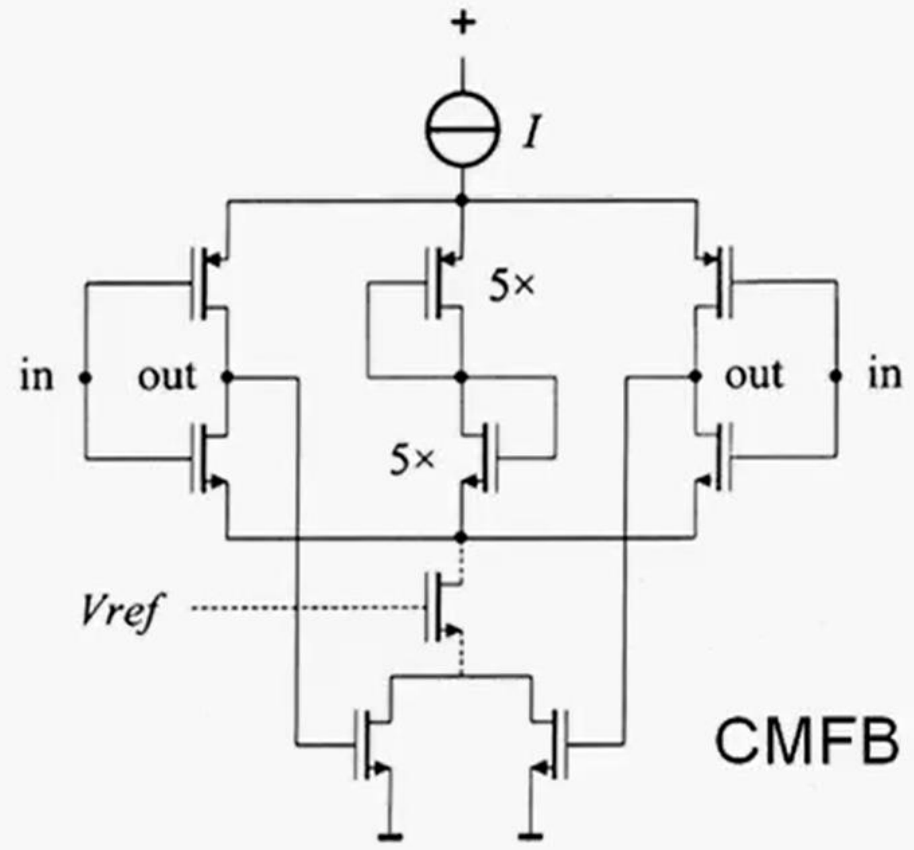
>>> Class AB

Peluso,..., JSSC Dec.98, pp.1887-1896

Inverter for Analog



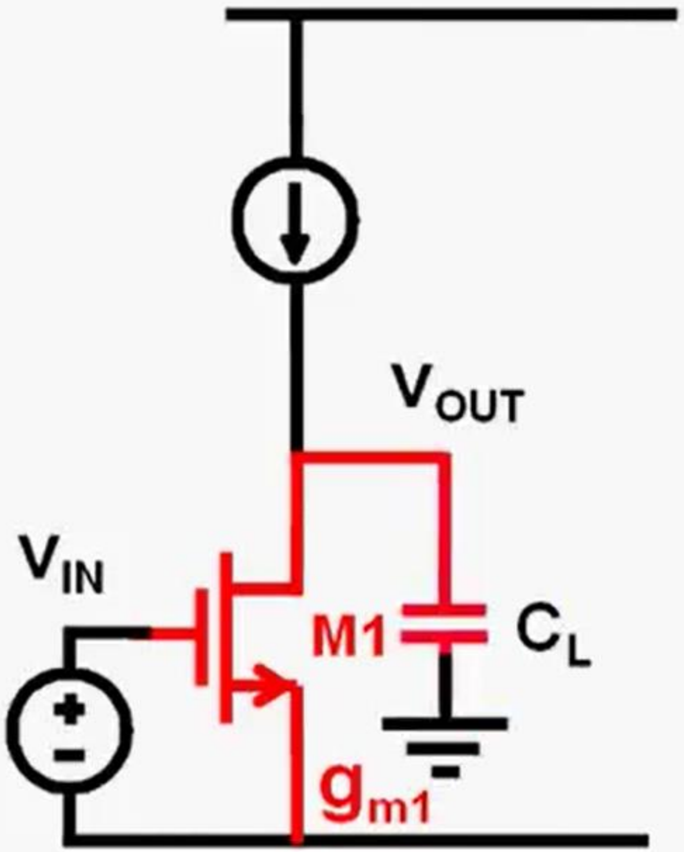
Inverter for Analog : JSSC'00



Ref. Voorman, JSSC, Aug.2000, 1097-1108

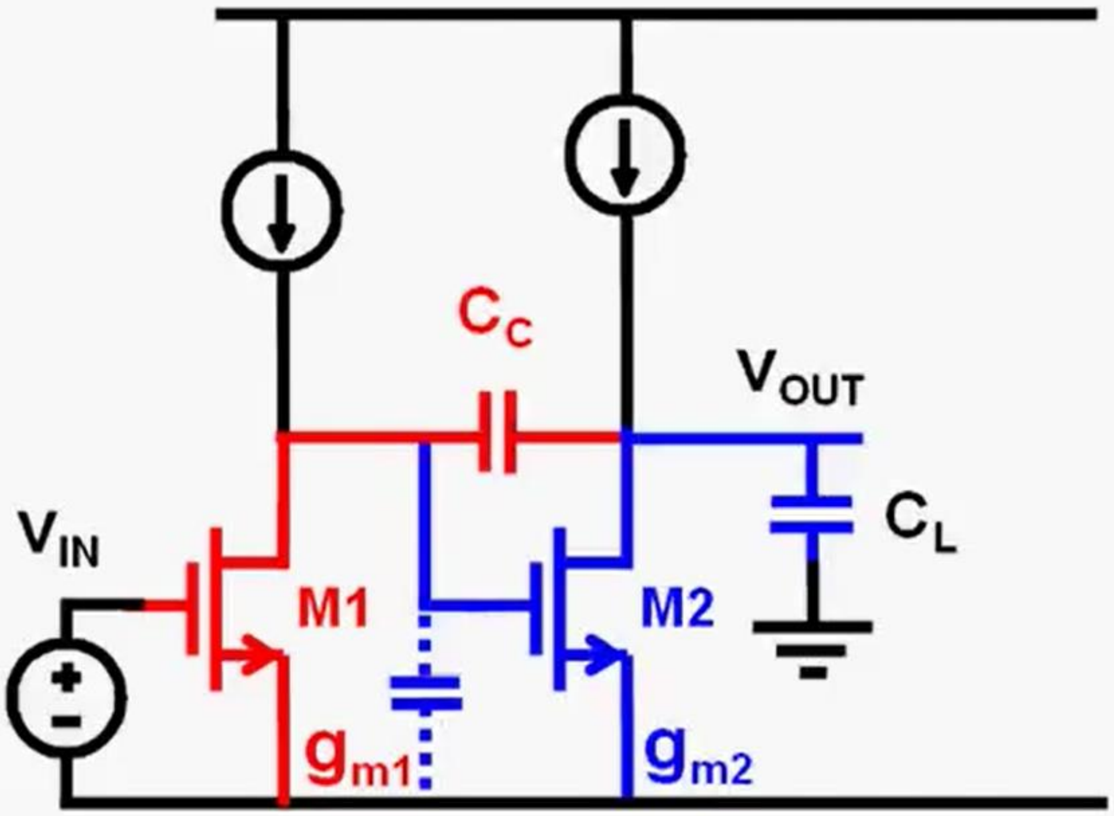
Multistage Amplifiers

Multistage Amplifier



$$GBW = \frac{g_{m1}}{2\pi\, C_L}$$

Multistage Amplifier

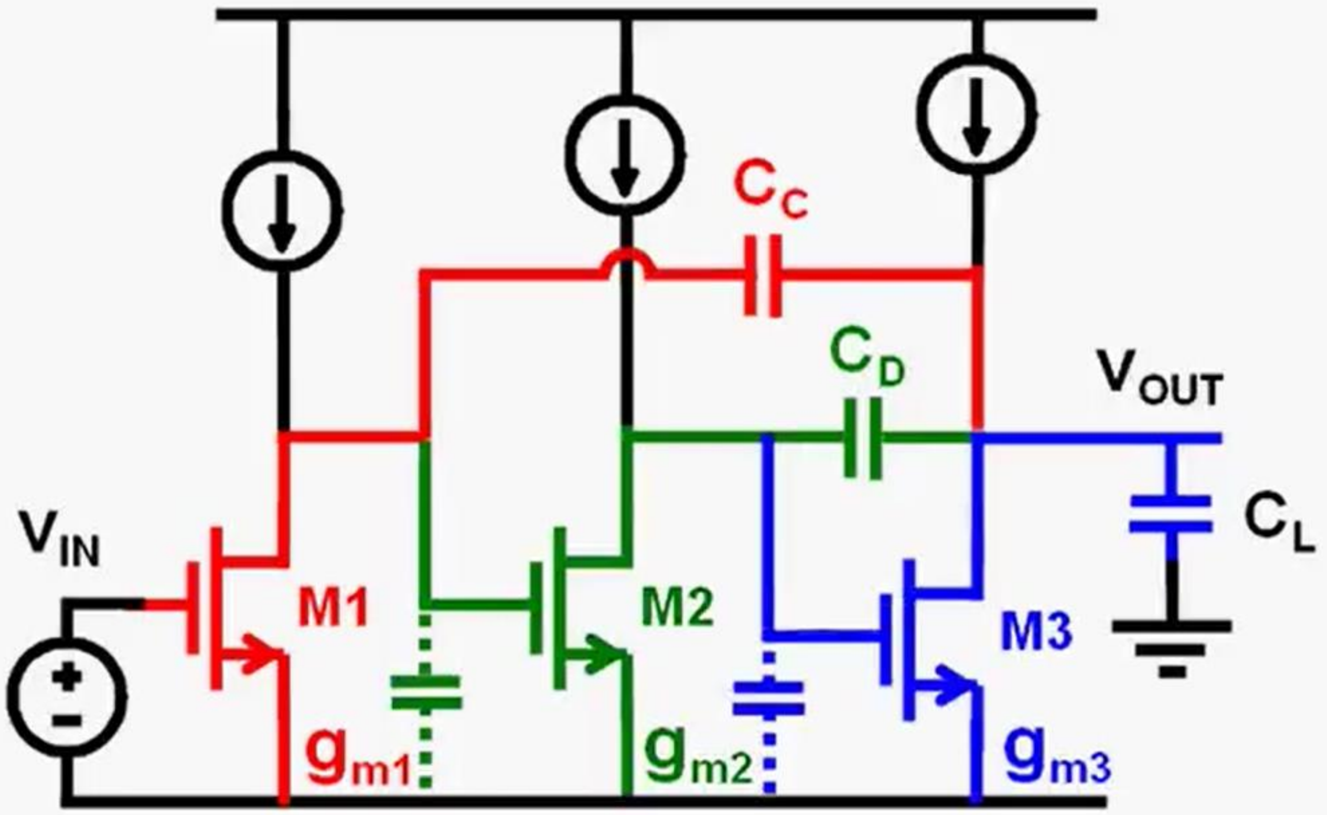


$$GBW = \frac{g_{m1}}{2\pi\, C_C}$$

$$f_{nd1} = \frac{g_{m2}}{2\pi\, C_L}$$

$$f_{nd1} = 3\ GBW$$

Multistage Amplifier (NMC)



$$GBW = \frac{g_{m1}}{2\pi C_C}$$

$$f_{nd1} = \frac{g_{m2}}{2\pi C_D}$$

$$f_{nd2} = \frac{g_{m3}}{2\pi C_L}$$

$$f_{nd1} = 3\ GBW$$

$$f_{nd2} = 5\ GBW$$

Multistage Amplifier

GBW = 50 MHz for $C_L = 2$ pF

Find I_{DS1}; I_{DS2} ; I_{DS3} ; C_C and C_D !

Choose $C_C = C_D = 1$ pF > $g_{m1} = 2\pi\, C_C GBW = 315\ uS$

$I_{DS1} = 31\ uA$

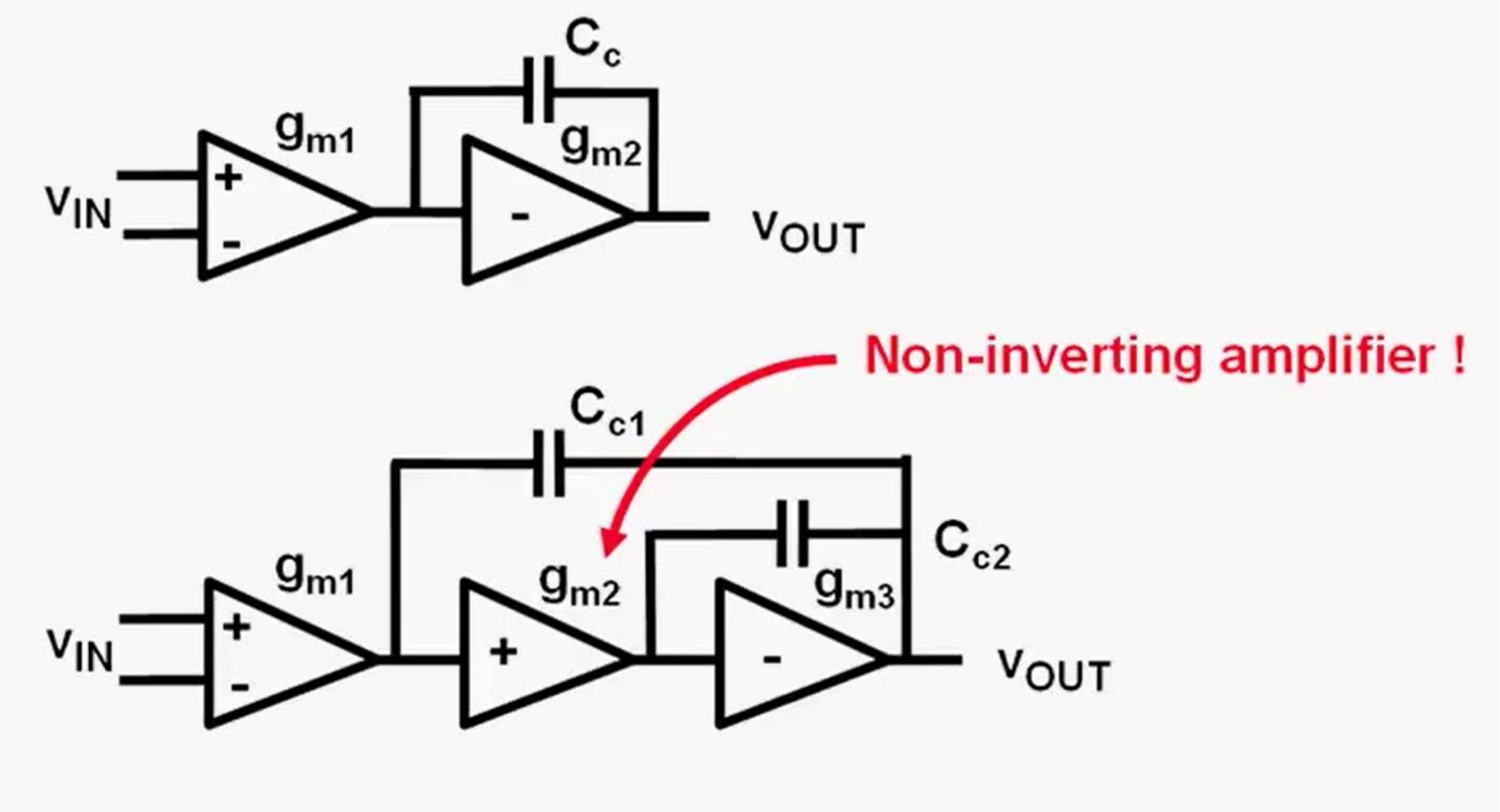
$f_{nd1} =$ 150 MHz > $g_{m2} = 2\pi\, C_D 3GBW = 3g_{m1} = 945\ uS$

$I_{DS2} = 95\ uA$

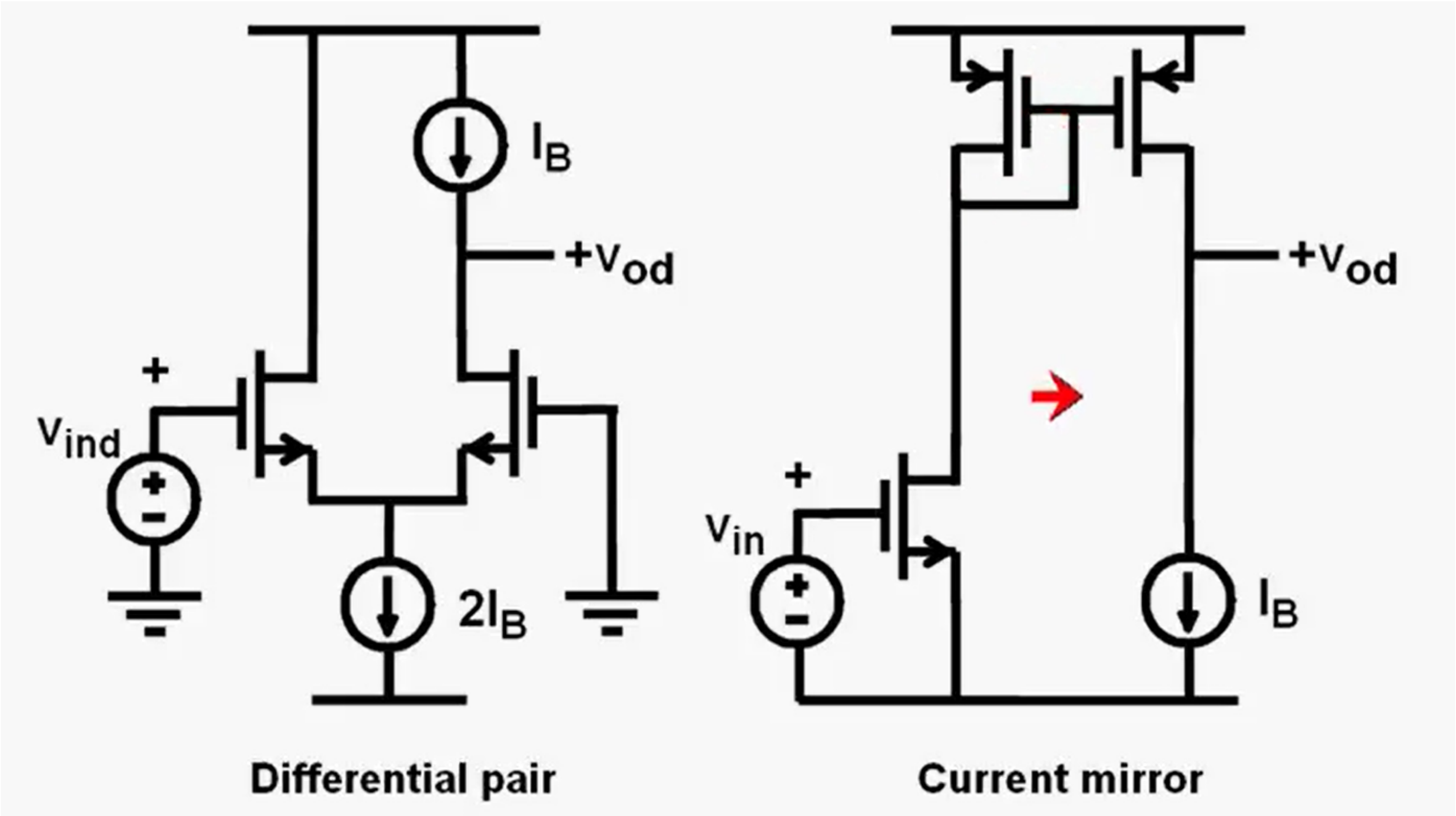
$f_{nd2} =$ 250 MHz > $g_{m3} = 2\pi\, C_L 5GBW = 10g_{m1} = 3150\ uS$

$I_{DS3} = 315\ uA$

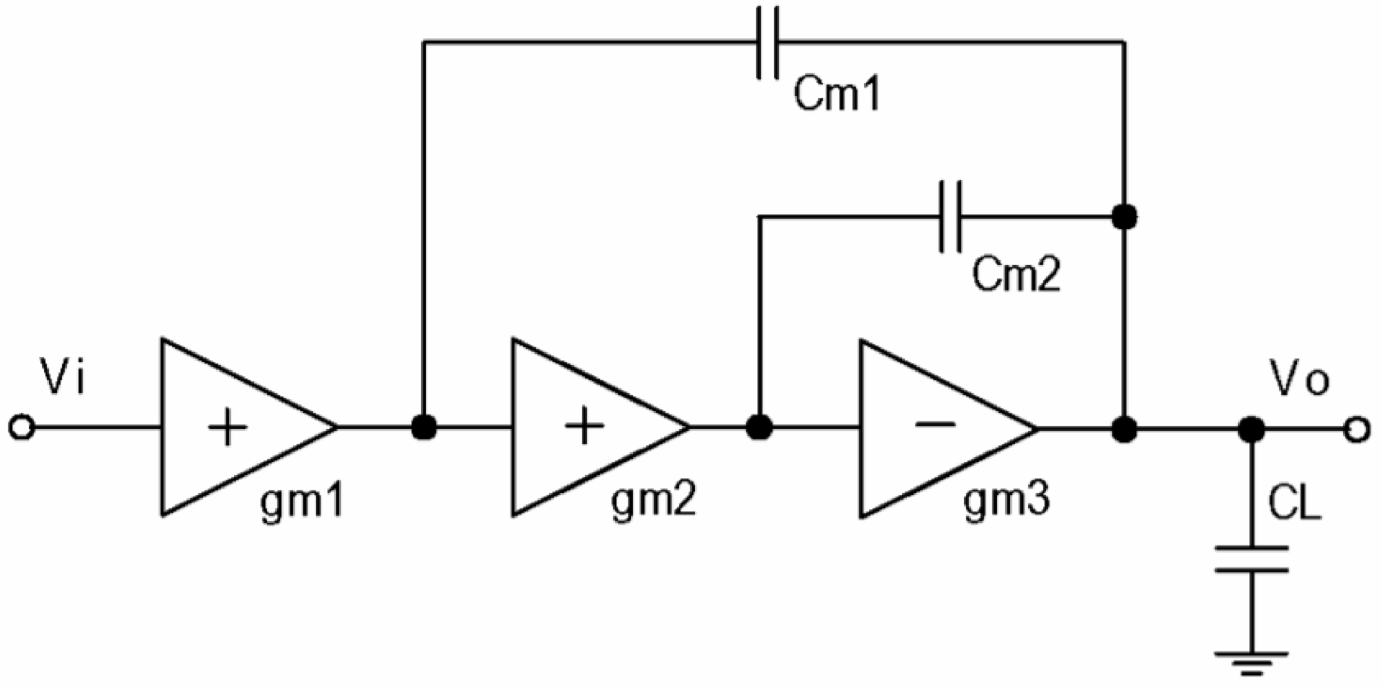
Multistage Amplifier



Multistage Amplifier : Non Inv AMP

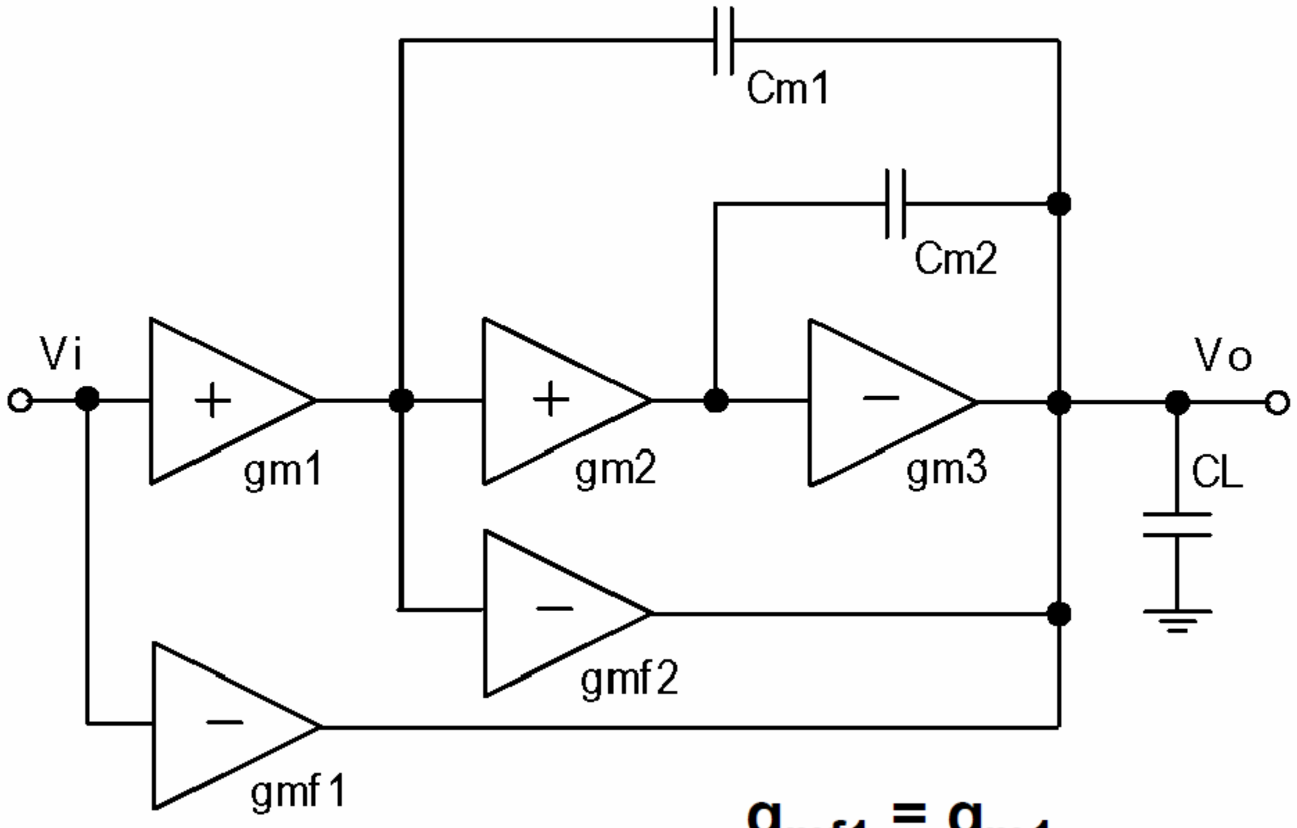


Multistage Amplifier : NMC



Huijsing, JSSC Dec.85, pp.1144-1150

Multistage Amplifier : NGCC



$$\omega_1 \approx \frac{g_{m2}}{C_{m2}}$$

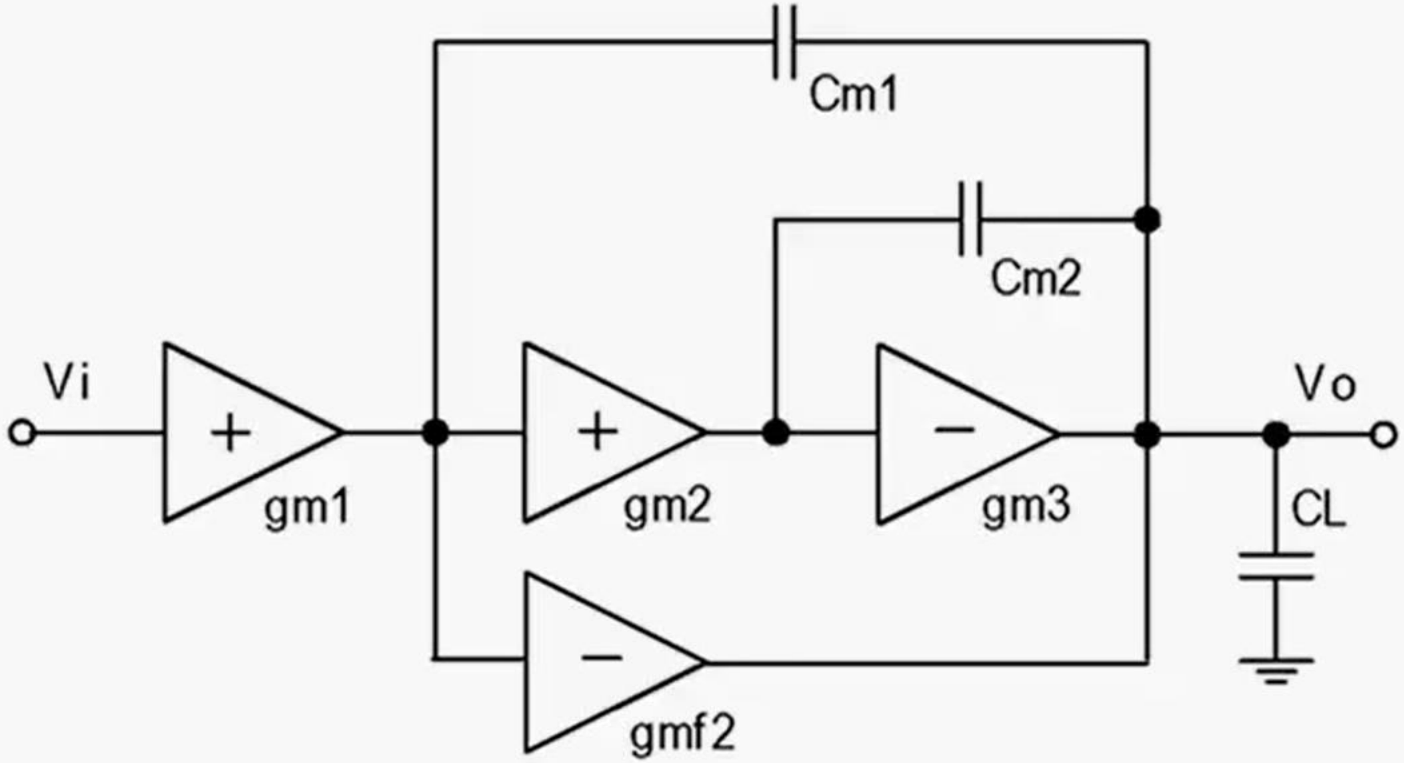
$$\omega_2 \approx \frac{g_{m3}}{C_L}$$

$g_{mf1} = g_{m1}$

$g_{mf2} = g_{m2}$

You, JSSC Dec.97, 2000-2011

Multistage Amplifier : NMCF



$$g_{mf2} = g_{m3}$$

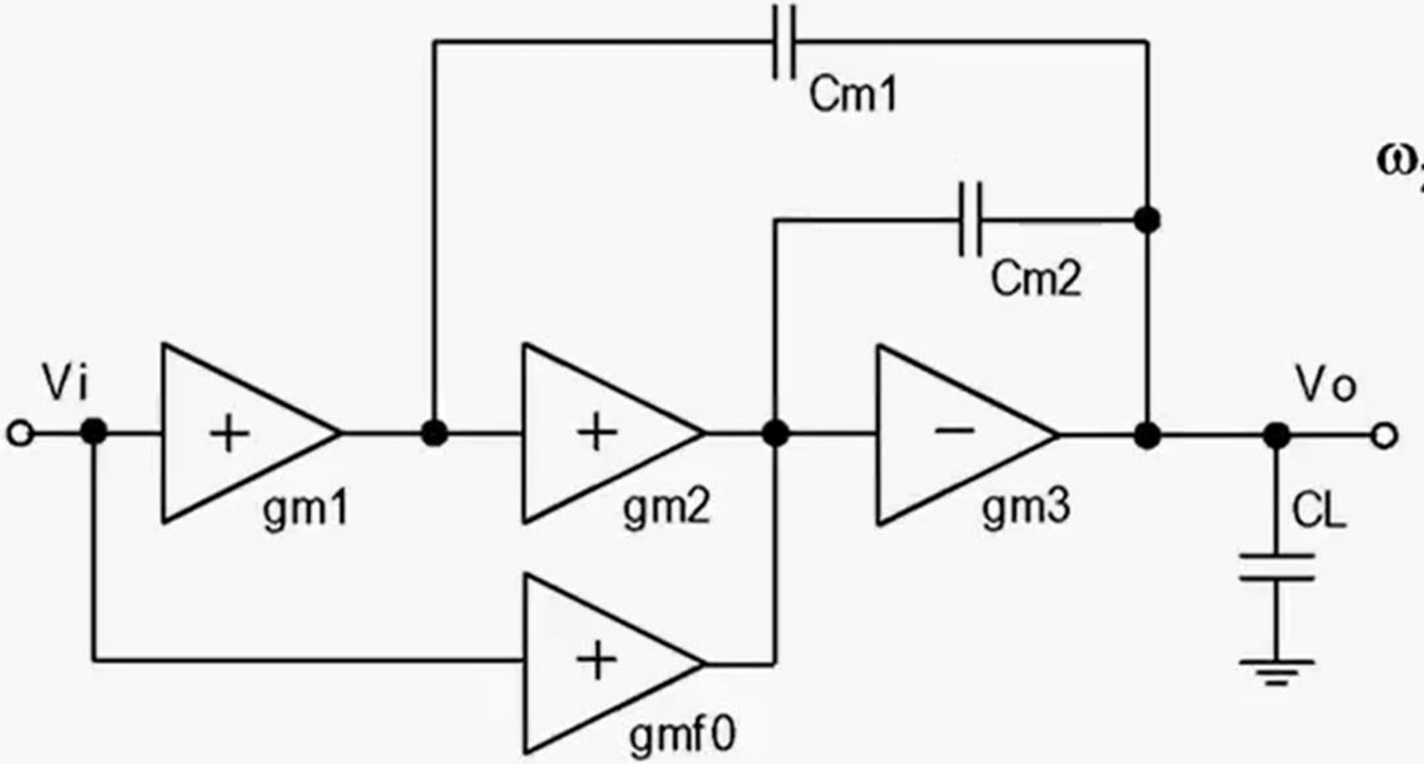
$$\omega_z \approx \frac{g_{m2}}{C_{m2}}$$

$$\omega_1 \approx \frac{g_{m2}}{2C_{m2}}$$

$$\omega_2 \approx \frac{2g_{m3}}{C_L}$$

Leung, CAS April 01, 388-394

Multistage Amplifier : MNMC



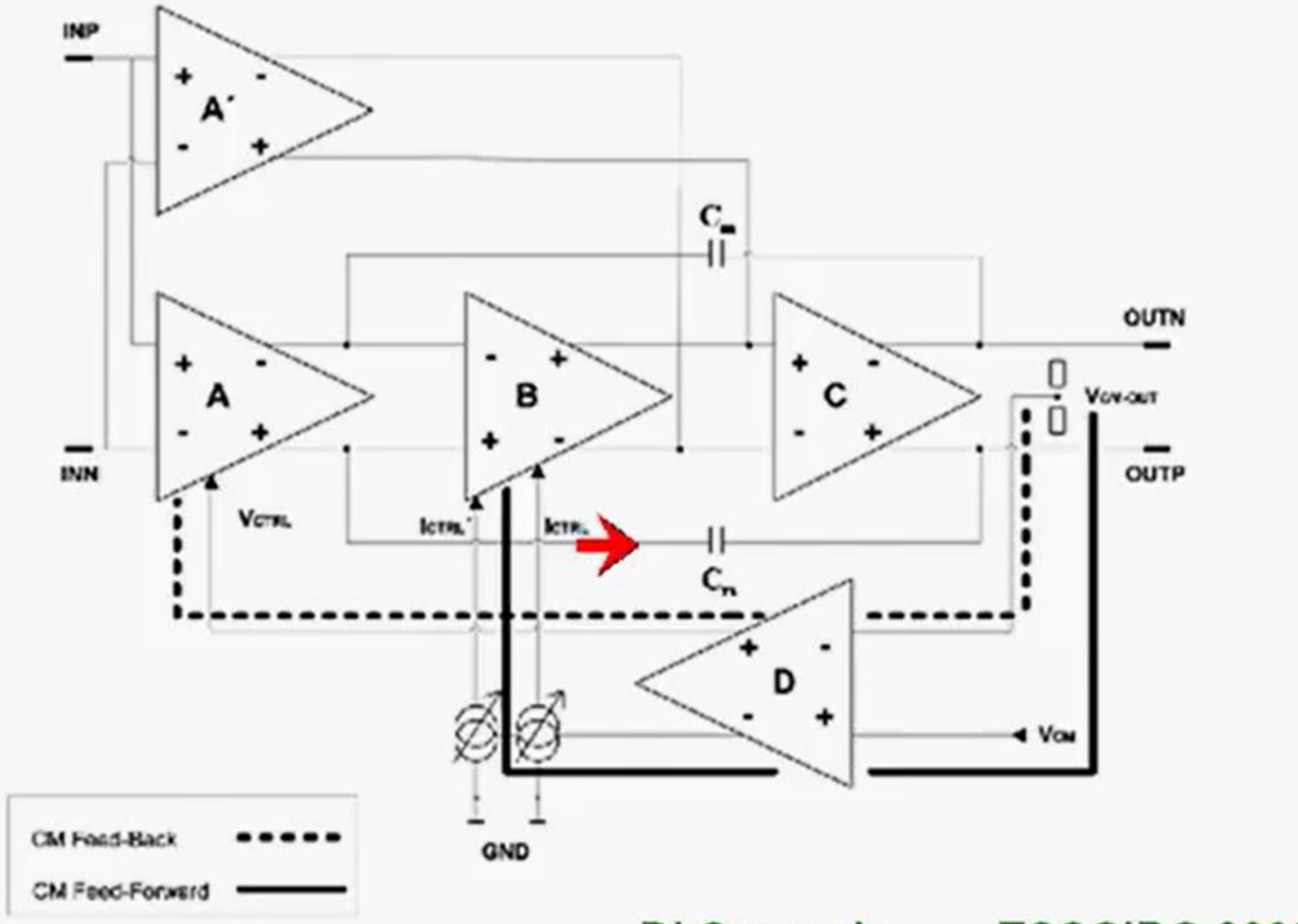
$$\omega_z \approx \frac{g_{m1}\, g_{m2}}{g_{mf0}\, C_{m1}}$$

$$\omega_1 \approx \frac{g_{m2}}{C_{m2}}$$

$$\omega_2 \approx \frac{g_{m3}}{C_L}$$

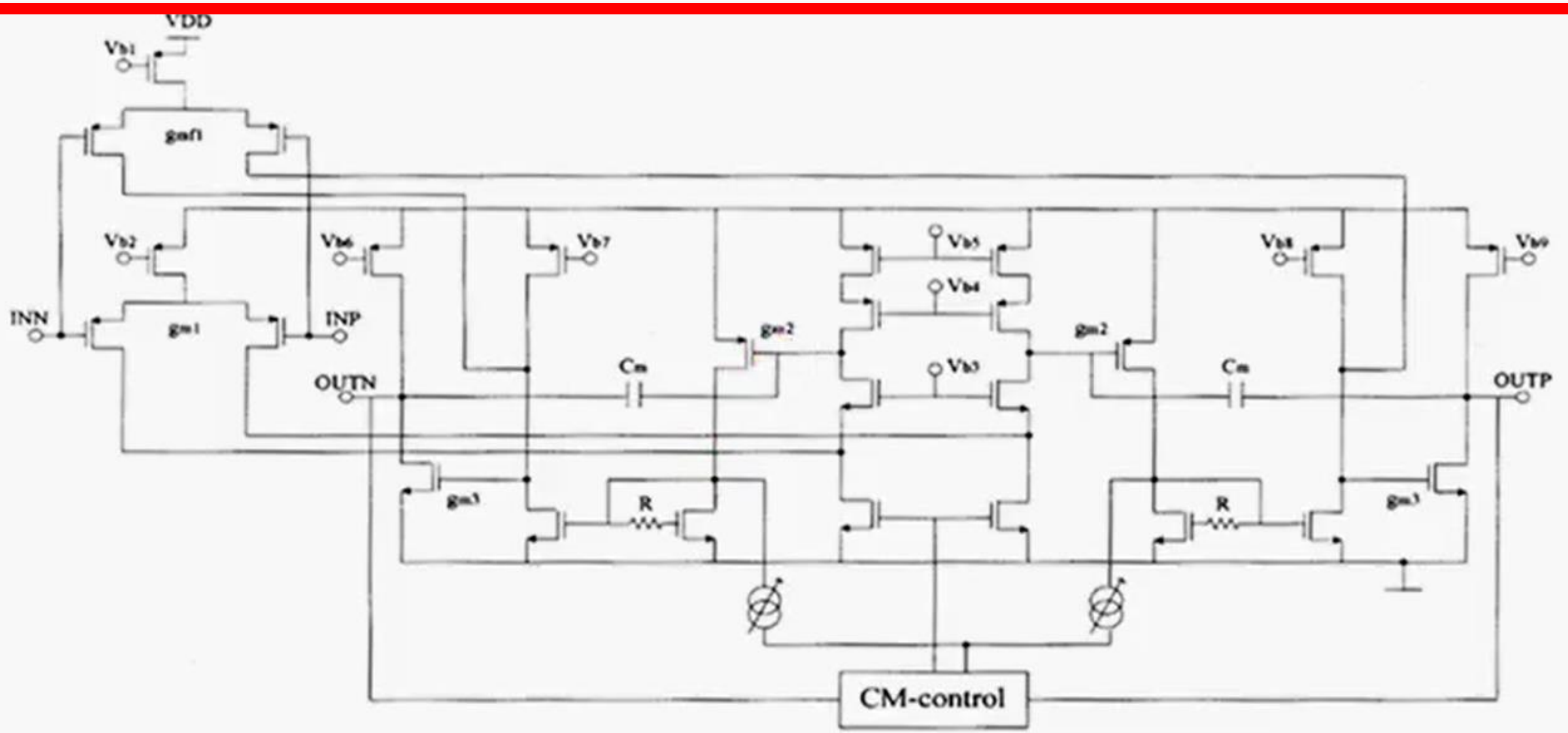
Eschauzier, JSSC Dec.92, pp.1709-1717

Multistage Amplifier : ESSCIRC' 08



Di Sancarlo, ... ESSCIRC 2008, pp. 314-317

Multistage Amplifier : ESSCIRC' 08



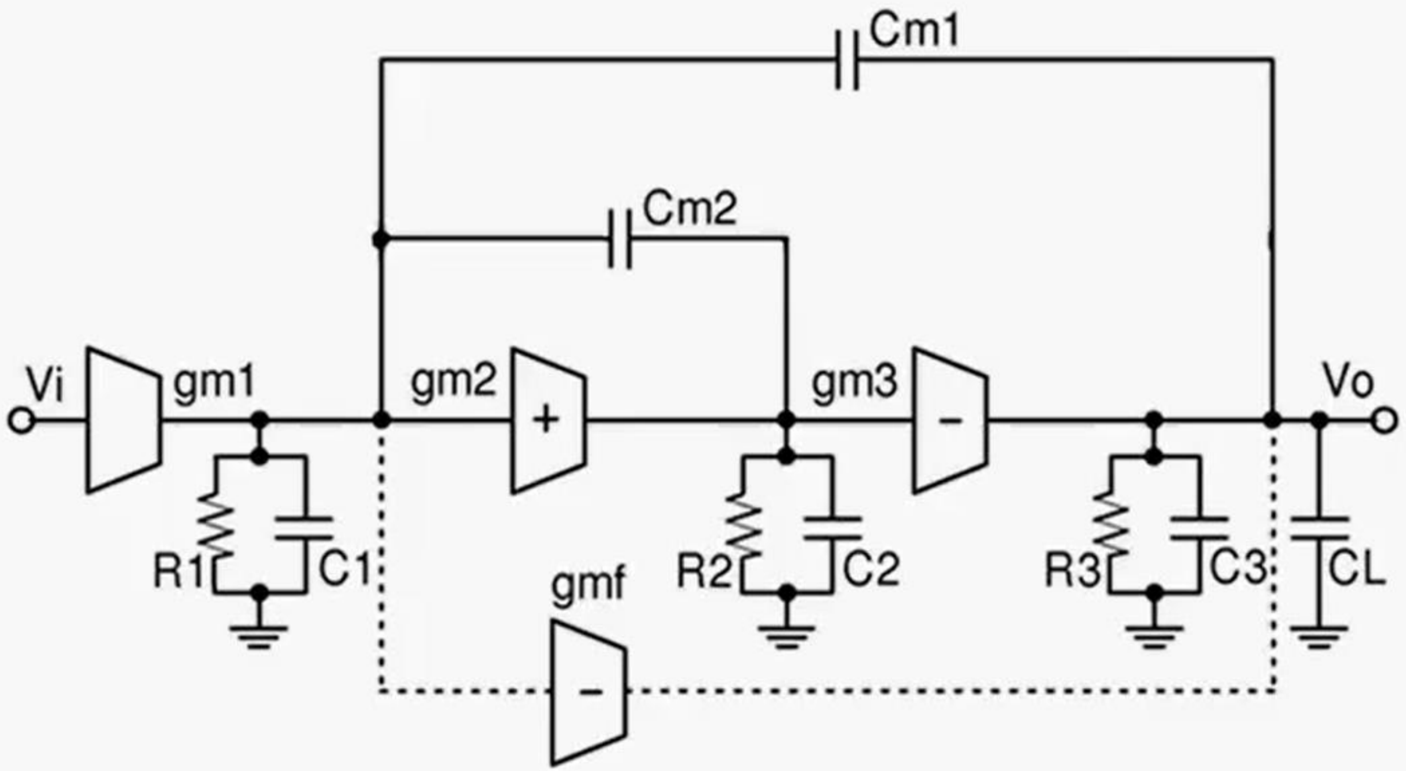
200 MHz 3 pF 10.2 mA 1 V

Di Sancarlo, ...
ESSCIRC 2008, pp. 314-317

Multistage Amplifier : Comparison

Topology	Stages	PM	GB=2πGBW
Single	One	$<90^\circ$	(g_m/C_L)
SMC	Two	$<63^\circ$	$0.5(g_{m2}/C_L)$
NMC	Three	$\approx 60^\circ$	$0.25(g_{m3}/C_L)$
NGCC	Three	$\approx 60^\circ$	$0.25(g_{m3}/C_L)$
NMCF	Three	$>60^\circ$	$<0.5(g_{m3}/C_L)$
MNMC	Three	$\approx 63^\circ$	$\approx 0.5(g_{m3}/C_L)$

CICC'02



$GBW / GBW_{NMC} \approx 6$

Ramos, CICC 2002, 333-336

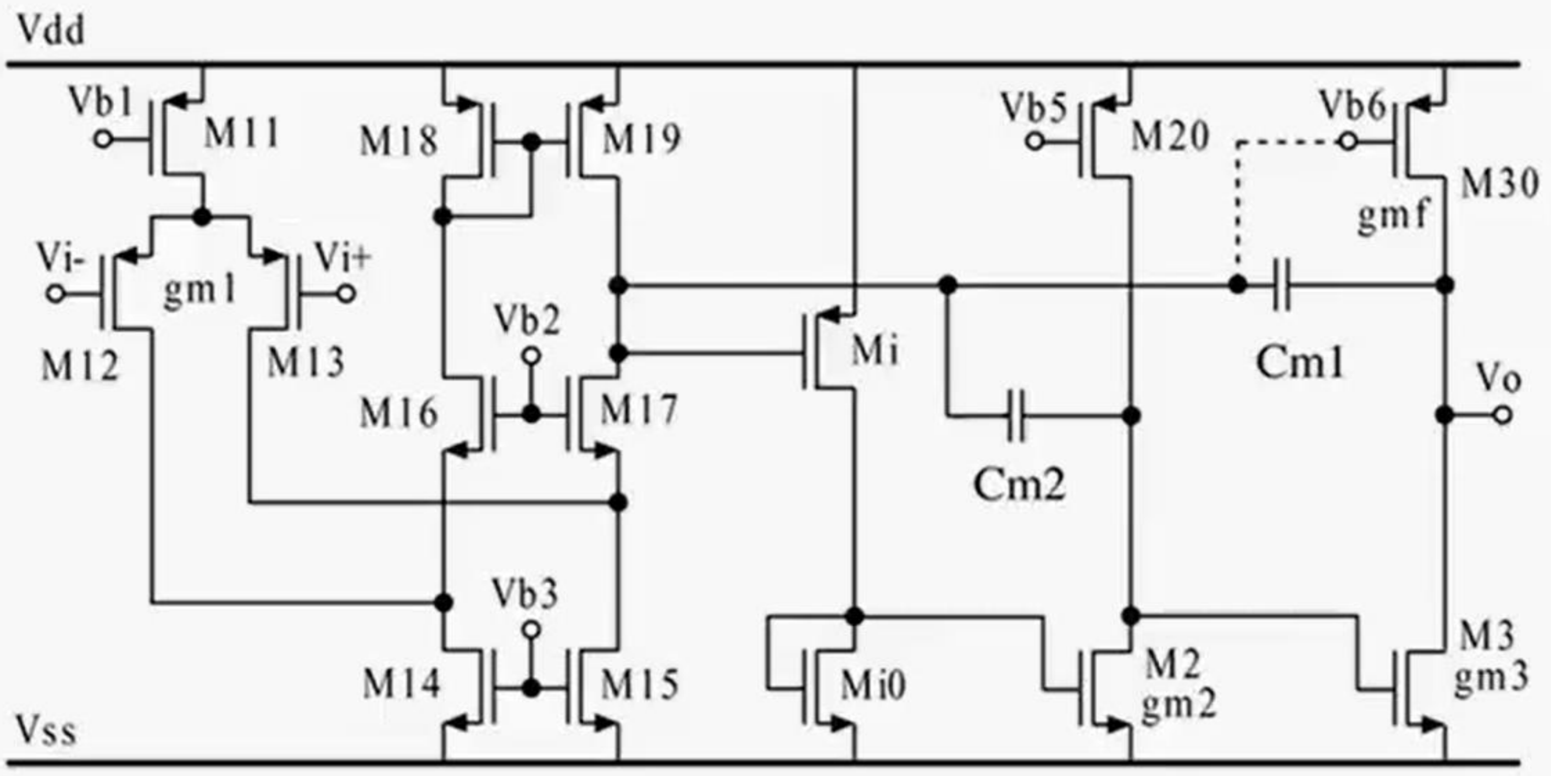
CICC'02

$$A_v(s) = \frac{A_{dc}\left(1 + \frac{s}{\omega_1} + \frac{s^2}{\omega_1\omega_3}\right)}{\left(1 + \frac{s}{\omega_d}\right)\left(1 + \frac{s}{\omega_1} + \frac{s^2}{\omega_1\omega_2}\right)}$$

$$A_{dc} = g_{m1}g_{m2}g_{m3}R_1R_2R_3$$

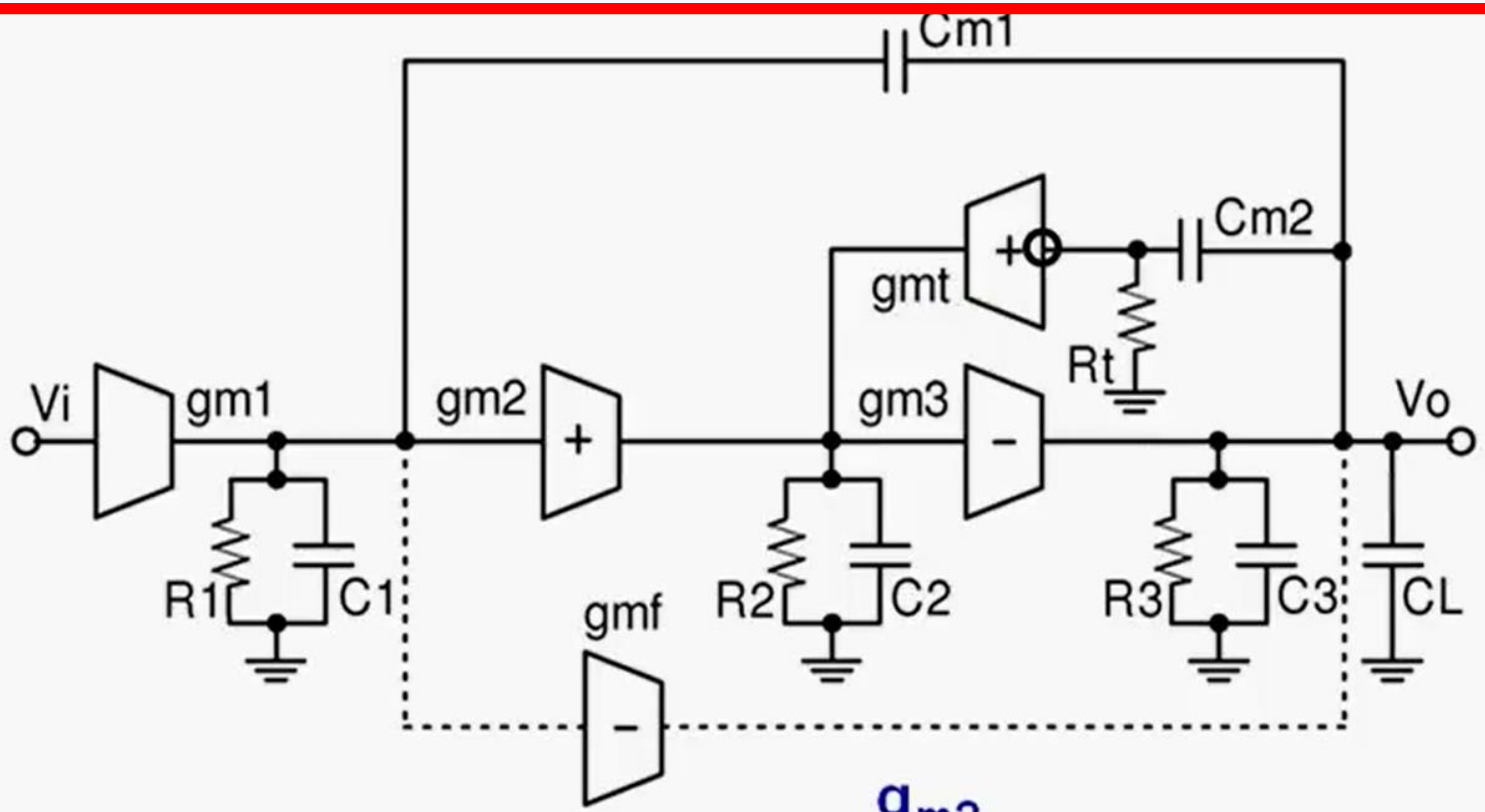
$$\omega_d = -\frac{1}{C_{m1}g_{m2}g_{m3}R_1R_2R_3}$$

CICC'02



Ramos, CICC 2002, 333-336

JSSC'05



$$k_t = \frac{g_{m2}}{g_{mt}}$$

$GBW / GBW_{NMC} \approx 41$

Peng, JSSC July 05,1514-1520

JSSC'05

$$A_v(s) = \frac{A_{dc}\left(1 + \frac{s}{\omega_2} + \frac{s^2}{\omega_2^2} + \frac{s^3}{\omega_2^2\omega_4}\right)}{\left(1 + \frac{s}{\omega_d}\right)\left(1 + \frac{s}{\omega_1} + \frac{s^2}{\omega_1\omega_2} + \frac{s^3}{\omega_1\omega_2\omega_3}\right)}$$

$$\omega_1 = \frac{1}{1+k_t}\frac{g_{m2}}{C_{m2}}$$

$$\omega_2 = \frac{1}{k_t}\frac{g_{m2}}{C_{m2}}$$

$$A_{dc} = g_{m1}g_{m2}g_{m3}R_1R_2R_3$$

$$\omega_d = -\frac{1}{C_{m1}g_{m2}g_{m3}R_1R_2R_3}$$

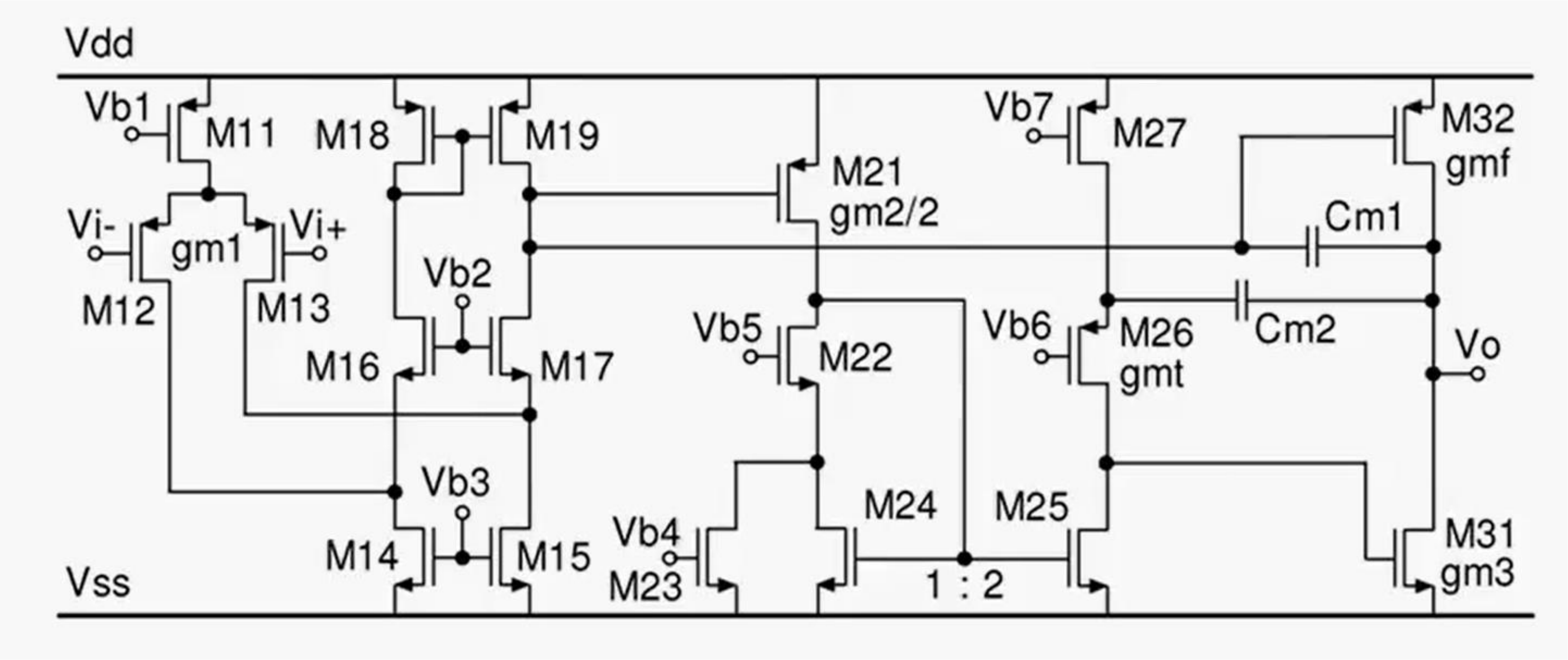
$$\omega_3 = (1+k_t)\frac{C_{m2}}{C_2}\frac{g_{m3}}{C_L}$$

$$\omega_{UG} = \frac{g_{m1}}{C_{m1}}$$

$$k_t = \frac{g_{m2}}{g_{mt}}$$

$$\omega_4 = -k_t\frac{C_{m2}}{C_2}\frac{g_{m3}}{g_{m1}}\omega_{UG}$$

JSSC'05

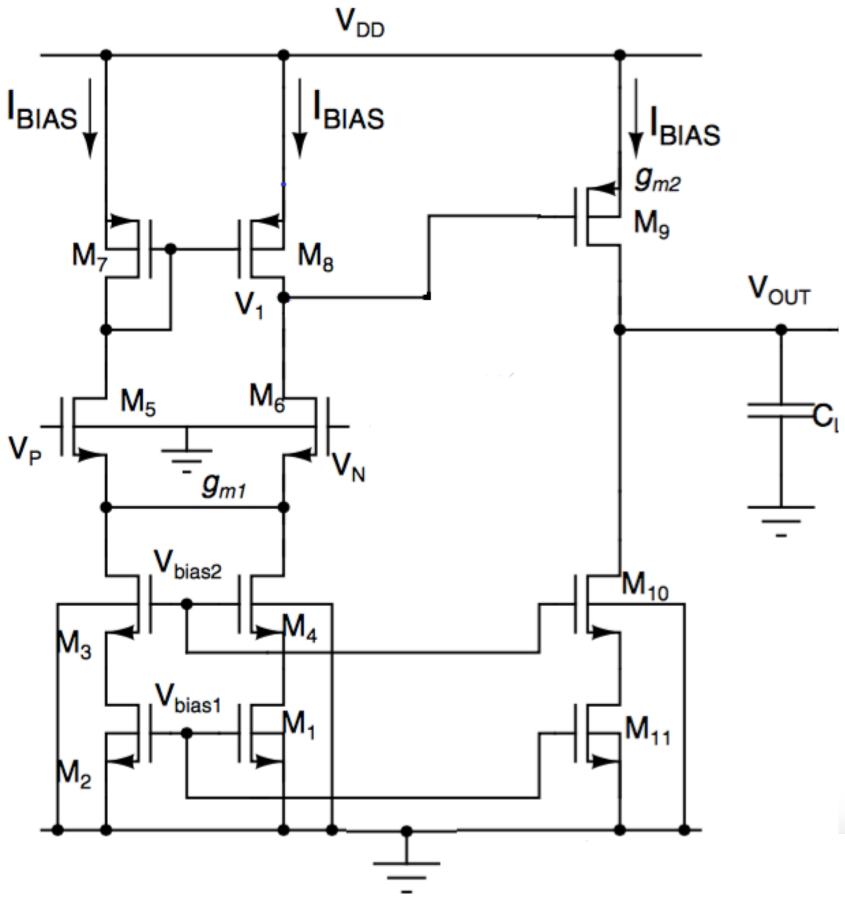

Vdd
Vb1
M11
M18
M19
M21
gm2/2
Vb7
M27
M32
gmf
Vi-
gm1
Vi+
Cm1
M12
M13
Vb2
Cm2
Vb5
M22
Vb6
M26
gmt
Vo
M16
M17
Vb3
M24
M25
M31
gm3
Vb4
M14
M15
M23
Vss
1 : 2

Case Study

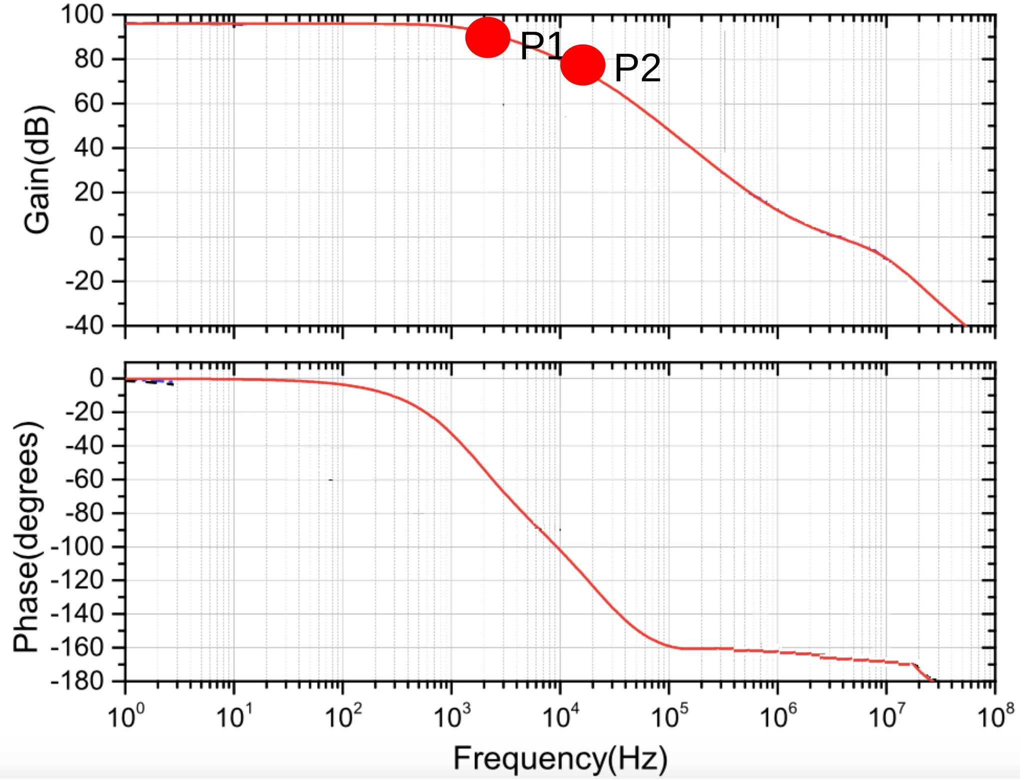
-3dB BW Enhancement Technique

(Hande et al, MWSCAS' 18)

-3dB BW Enhancement Technique



Two Stage Non-Compensated OTA



Phase Margin ~ 10

UNSTABLE !!

-3dB BW Enhancement Technique

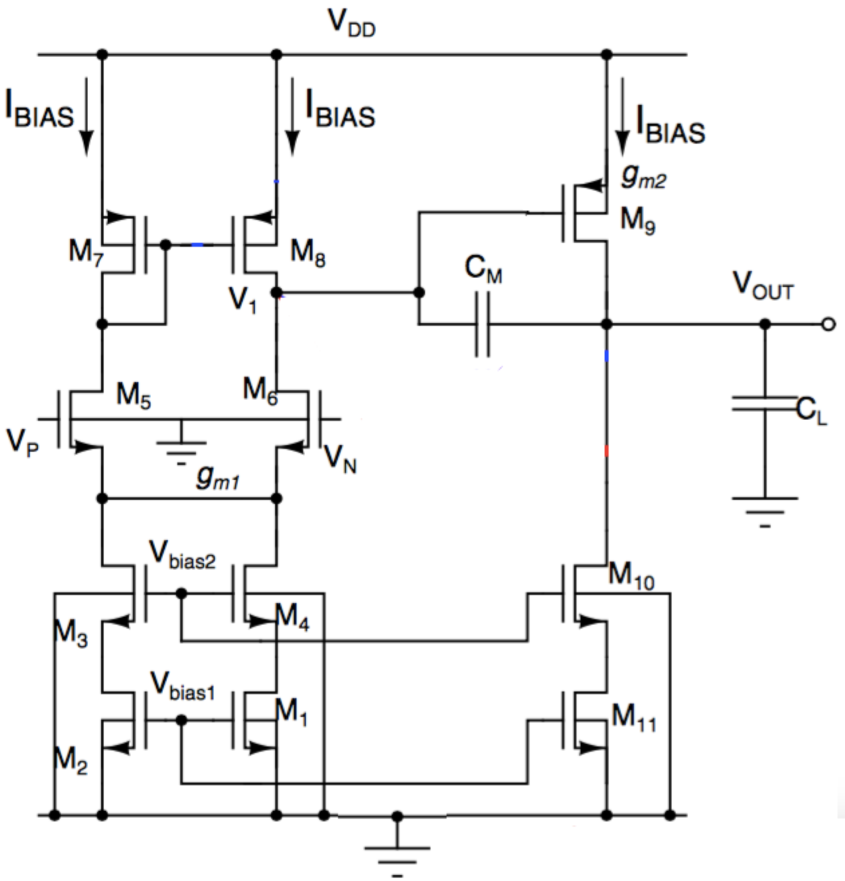
Solution :

Create a dominant pole in the system to make single pole system as it is inherently stable.

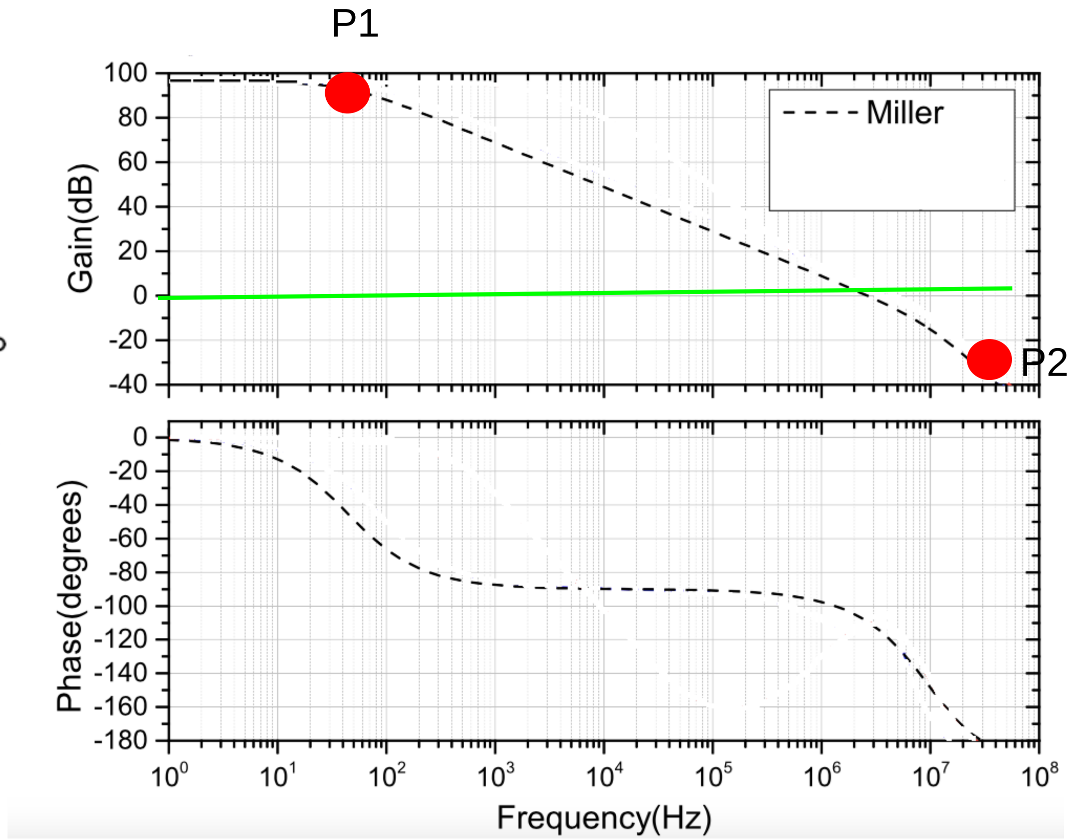
Push the P_1 to lower frequency

Disadvantage : *Reduction in -3dB bandwidth.*

-3dB BW Enhancement Technique

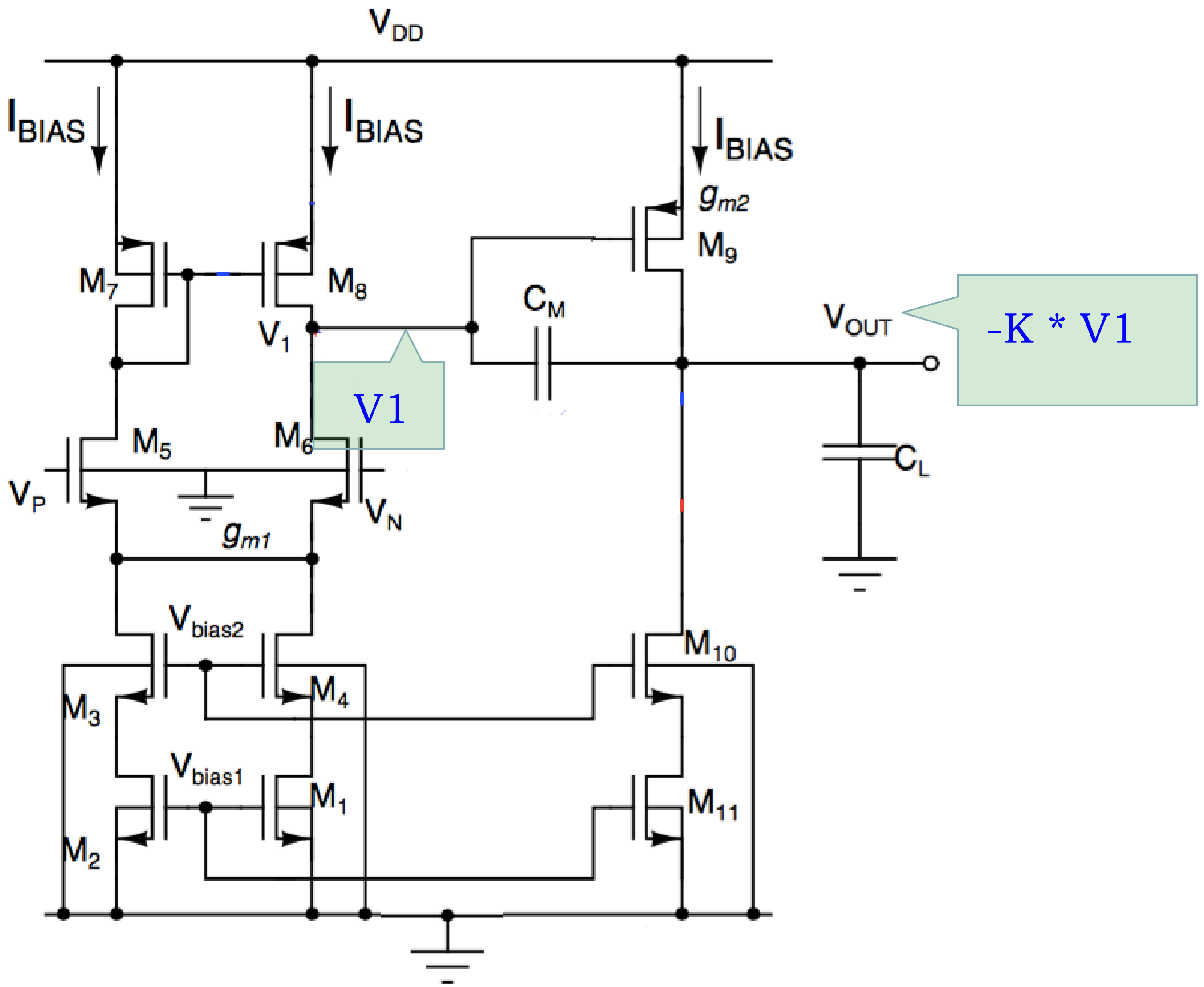


Two Stage Miller Compensated OTA

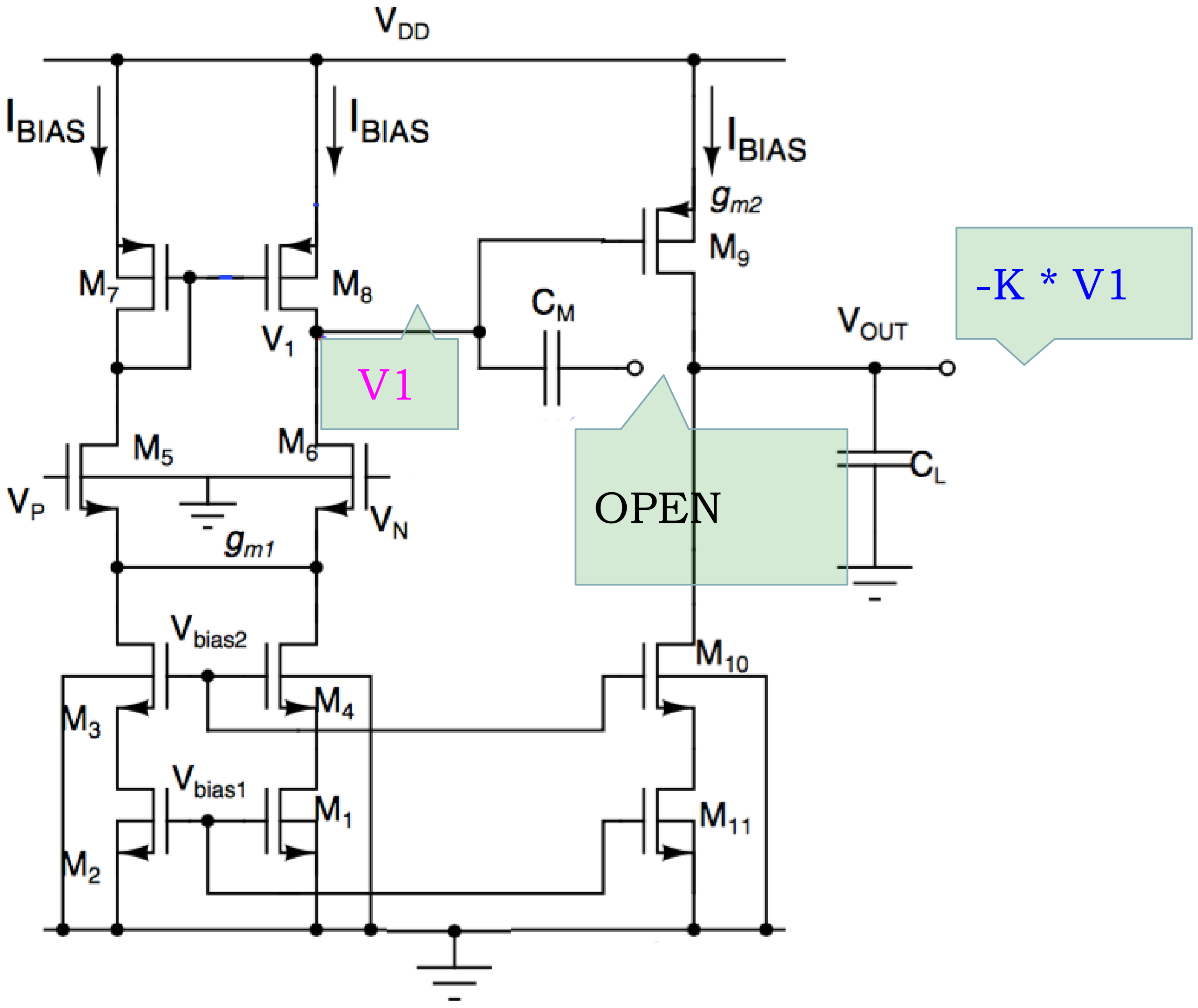


Phase Margin ~ 70

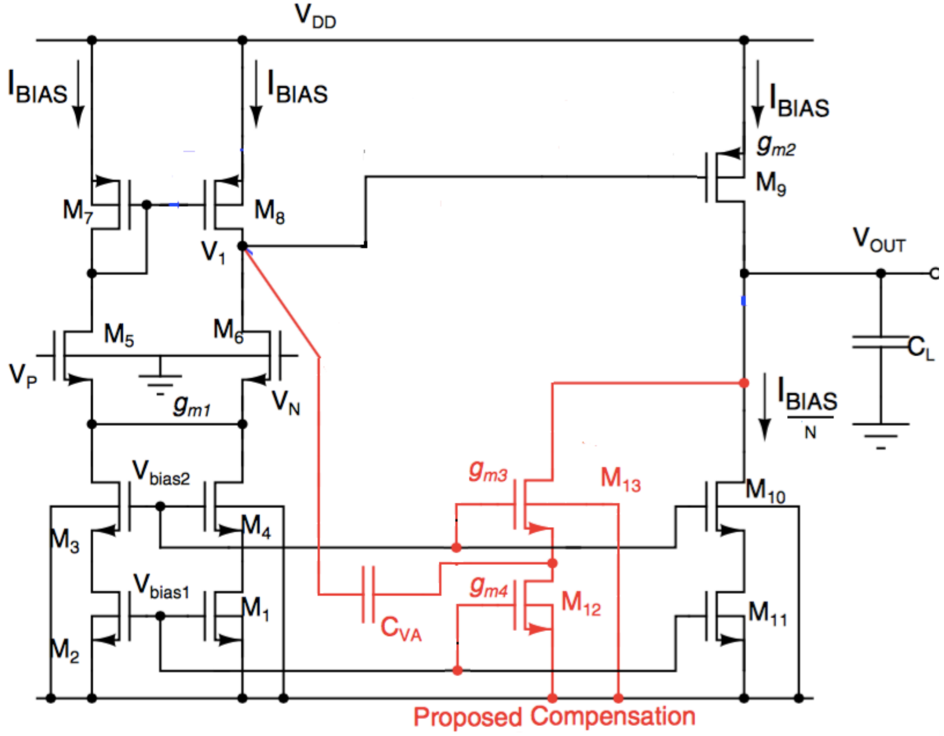
-3dB BW Enhancement Technique



-3dB BW Enhancement Technique



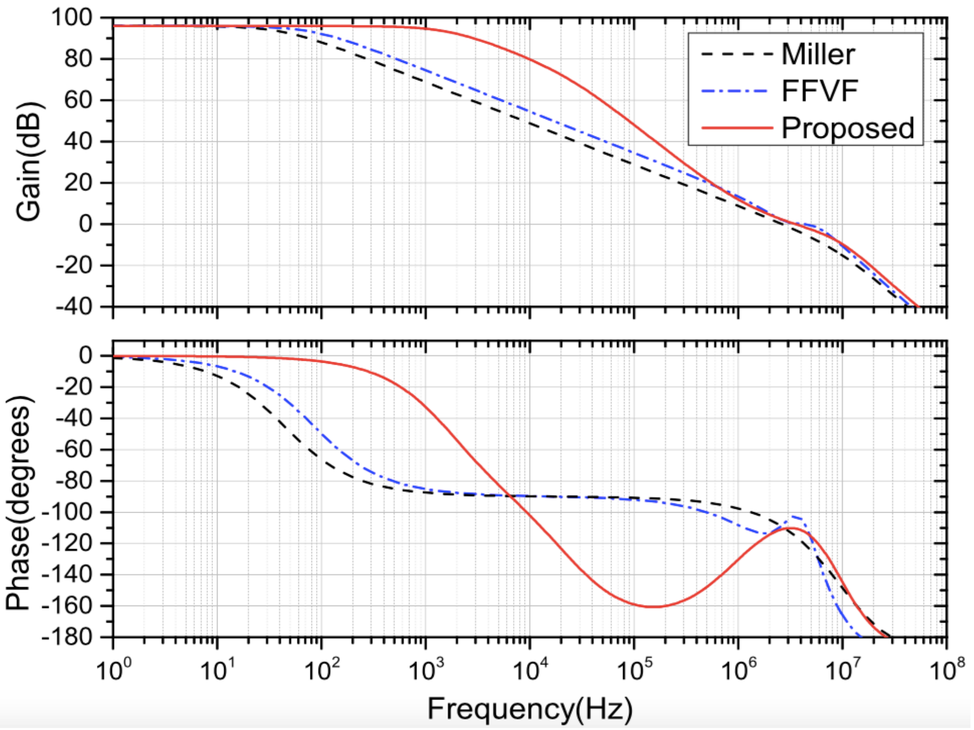
-3dB BW Enhancement Technique



Two Stage Miller Compensated OTA

LHP Zero introduced using $1/gm_3$.

-3dB BW enhanced by 20 times



Phase Margin almost 70

BW enhanced !!

Conclusion

Scaling in technology created opportunity to re-visit the classical aanalog design in terms of operational amplifier.

Reference :
Design of Analog Integrated Circuits and Systems by Kenneth R. Laker and Willy M. C. Sansen and letures provided by Willy Sansen